

Official Review of Submission598 by Reviewer 4Brc

Official Review by Reviewer 4Brc 📅 11 Aug 2025, 20:23 (modified: 15 Aug 2025, 14:47)

👁 Program Chairs, Area Chairs, Reviewers Submitted, Reviewer 4Brc, Authors

📄 Revisions (/revisions?id=Fq6m29sTcA)

Ethics: NO or VERY MINOR ethics concerns only

Position: Yes, the paper argues for or against a position related to machine learning.

Summary:

The paper argues that AI's climate impact must be assessed based on both its emissions and potential to reduce emissions, rather than focusing solely on its carbon footprint. The paper examines AI's climate impact by developing a framework that considers emissions and benefits. It introduces a Net Climate Impact Score equation to calculate impact. The authors show that AI applications create emissions, but strategic deployment could deliver benefits. They note that efficiency gains from AI create rebound effects that reduce benefits. The paper provides a framework for impact assessment that considers carbon time value and infrastructure, while offering recommendations to maximize benefits and minimize footprint.

Author Identification: No.

Support: 4: excellent

Significance: 4: excellent

Presentation: 3: good

Context: 4: excellent

Discussion: 4: very likely

Alternative Position: No

Strengths:

- The position has been well argued with data. Even early quantitative framework such as the proposed net climate impact score can be a valuable starting point
- The paper considers important aspects like rebound effect and time value of carbon in analyzing the impact on carbon
- The data points provided throughout are well grounded in literature
- The paper acknowledges the limitations of speculative nature of future benefit estimates and uses range vs point estimates

Weaknesses:

- The paper does not consider any alternative position. For example, focus on carbon does not consider impact on water use, mining of critical minerals, or human exploitation during annotation.
- The proposed equations do not consider regional impacts on grids, pollution, or economy.
- As much of the numbers included are speculative, it would be good to see a validation method before they are used for real-world decisions. How can AI-enabled reductions be measured at scale?
- While the time value of carbon and rebound effect have been discussed, the proposed equations do not incorporate it for estimating net impact
- Opportunity cost of investing in AI vs other climate solutions is not considered
- It is unclear why stakeholders will adopt policies that are based on net climate impact for AI related decision making

Questions:

1. What are the alternative positions to the one you have considered, and how do they fare compared to your arguments? This is a critical aspect missing in the paper.
2. How would you propose measuring and validating AI enabled emissions reductions?
3. How would you incorporate time value of carbon and rebound effects into your equations?
4. How would you incentivize stakeholders to adopt the proposed framework? Why would people stop generating cat videos regardless of what the equation says?
5. More of a comment than question -- impact of inference far exceeds that of training.

Agreement: 5: strongly agree

Rating: 6: Weak Accept: The paper presents a solid argument about an issue of moderate importance to at least one sub-area of the NeurIPS community.

Confidence: 5: You are absolutely certain about your assessment. You are very familiar with the related work.

Thoroughness: 5: You read the paper and appendices rigorously and checked all of the details carefully, including references and proofs (if present).

Code Of Conduct Acknowledgement: Yes

Add: **Official Comment**

Official Review of Submission598 by Reviewer 7Qe4

Official Review by Reviewer 7Qe4  30 Jul 2025, 13:58 (modified: 15 Aug 2025, 14:47)

 Program Chairs, Area Chairs, Reviewers Submitted, Reviewer 7Qe4, Authors

 Revisions (/revisions?id=Hu23wtZzwV)

Ethics: NO or VERY MINOR ethics concerns only

Position: Yes, the paper argues for or against a position related to machine learning.

Summary:

This paper argues that, instead of focusing on quantifying, reducing, and regulating the climate impact of AI, one must take a step further to also factor in the benefits from, for example, using AI tools to reduce carbon emissions of AI and other applications.

This work proposes to take a more holistic approach to consider the potential benefits of AI for climate mitigations and balance it with the carbon impact of AI. This is challenging because, first, the cost is incurred now while the reduction often comes later. Second, the scope is broad. Nonetheless, the paper describes three major factors to consider for the net carbon impact of AI: rebound effects, time value of carbon, energy and emission impact of AI.

While this work takes the first step describing what such framework can look like, there are many aspects that fall short. The model estimating the carbon cost of AI is preliminary and so is the methodology for potential benefits. It'd be great if the authors can continue the work to establish the described framework, refine and validate it with key use cases. The reviewer believes such a tool, by enabling what-if analysis, would be useful to many.

Author Identification: No.

Support: 2: fair

Significance: 2: fair

Presentation: 1: poor

Context: 2: fair

Discussion: 3: possibly

Alternative Position: No

Strengths:

The idea to make a framework assessing the net carbon impact of AI is commendable.

Weaknesses:

The framework for estimating the net carbon impact of AI needs significant improvement for it to be useful.

The paper is written with excessive direct quotes from other sources, such as news articles with eye-catching titles, hand-wavy projections. More scientific rigor is expected for a NeurIPS position paper.

Questions:

The idea to make a framework assessing the net carbon impact of AI is commendable. However, the scope is broad. Can you be more specific about the scope of carbon quantification across the "lifecycle" that will make it a useful tool? What are some key use cases you envision the tool enables? Can you please provide a few concrete examples?

Agreement: 4: agree

Rating: 4: Borderline reject: The paper presents a position, but the reasons to reject, e.g., unclear reasoning or limited support for the claims, outweigh reasons to accept.

Confidence: 4: You are confident in your assessment, but not absolutely certain. It is unlikely, but not impossible, that you did not understand some parts of the submission or that you are unfamiliar with some pieces of related work.

Thoroughness: 4: You read the paper and appendices and checked most of the details, including references..

Code Of Conduct Acknowledgement: Yes

Add: **Official Comment**

Official Review of Submission598 by Reviewer 26VL

Official Review by Reviewer 26VL (👤 Vivek Kumar Mishra (/profile?id=~Vivek_Kumar_Mishra1))

📅 23 Jul 2025, 11:49 (modified: 15 Aug 2025, 14:47)

👁 Program Chairs, Area Chairs, Reviewers Submitted, Reviewer 26VL, Authors

📄 Revisions (/revisions?id=Tb9WaRbPIT)

Ethics: NO or VERY MINOR ethics concerns only

Position: Yes, the paper argues for or against a position related to machine learning.

Summary:

This paper develops a framework to quantify AI's net climate impact, balancing its greenhouse gas emissions against potential benefits like optimized energy systems. It introduces the Net Climate Impact Score (NCIS), a novel equation factoring in emissions, benefits, and the time value of carbon. The paper highlights AI's growing energy demands (1.5-4% of global emissions) but argues strategic applications (e.g., grid management, electrification) could reduce emissions by 1.5-10% by 2030. It warns of rebound effects undermining benefits without policy support. Contributions include the NCIS framework, a mixed-methods approach, and policy recommendations to minimize AI's footprint while maximizing its climate potential.

Author Identification: No.

Support: 4: excellent

Significance: 2: fair

Presentation: 3: good

Context: 3: good

Discussion: 4: very likely

Alternative Position: Yes, and alternative positions are well-considered and named but not addressed

Strengths:

The NCIS framework for estimating AI's climate impact while weighing costs and benefits is expertly developed in this paper. It has a solid foundation thanks to its mixed-methods approach (literature, surveys, interviews) and 47 references (IEA, World Bank). It identifies pressing problems, such as data center energy consumption, and makes practical policy suggestions. It adds depth and is extremely pertinent to NeurIPS and climate research because of its emphasis on rebound effects and the time value of carbon.

Weaknesses:

Empirical case studies of AI applications (such as particular grid optimizations) could enhance the paper. Accessibility is diminished by its technical terminology and small OCR mistakes (such as "edationion"). Giving non-AI solutions (like expanding renewable energy) precedence over AI-driven ones is an alternate viewpoint that hasn't been thoroughly investigated. It would be strengthened by additional discussion on how to apply NCIS in environments with limited resources.

Questions:

How feasible is NCIS adoption in developing countries with limited data?

Agreement: 4: agree

Rating: 7: Accept: The paper presents a solid argument about an important issue that remains unresolved and is of importance to at least one sub-area of the NeurIPS community.

Confidence: 4: You are confident in your assessment, but not absolutely certain. It is unlikely, but not impossible, that you did not understand some parts of the submission or that you are unfamiliar with some pieces of related work.

Thoroughness: 4: You read the paper and appendices and checked most of the details, including references..

Code Of Conduct Acknowledgement: Yes

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Exhibit JR-29

NeurIPS 2025 Paper 598 - Official Review Posted Confirmation



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Paper title: The Net Climate Impact of AI: Balancing Current Costs with Future Climate Benefits

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Exhibit SR-1

Integrating Solar Photovoltaic Systems into the Grid: An Overview of AI
Applications

Integrating Solar Photovoltaic Systems into the Grid: An Overview of AI Application

Vivek Kumar Mishra

IEEE Senior Member

IEEE

Delaware, USA

vivekmishra@ieee.org

Abstract: The photovoltaic (PV) system business is seeing an increase in the number of applications for artificial intelligence (AI) as a result of the expansion of powerful computer resources, helpful tools, and an explosion of data. It has been shown that the methods that are currently used for a variety of jobs in the solar photovoltaic (PV) industry, such as design, forecasting, control, and maintenance, provide results that are not totally accurate. Additionally, artificial intelligence (AI) has enhanced the accuracy and precision of these professions, which has allowed them to become a topic of discussion at the present time. Within the context of this specific paradigm, the objective of this study is to investigate the impact that AI techniques have had on the PV value chain. In the course of the study, a cataloging of the current artificial intelligence systems, the compilation of a list of prospective applications of AI in the future, and an analysis of the benefits and drawbacks of these systems in contrast to more conventional approaches are all included.

Keywords: AI, solar systems, optimum size, irradiance prediction, condition tracking, transition management, dependability

I. INTRODUCTION

The capacity of artificial intelligence (AI) to automate processes for improved efficiency and performance has propelled it to the forefront of academic inquiry in recent decades [1]. Through the application of a set of intricate instructions, it empowers systems to mimic human learning, reasoning, and decision-making abilities. This procedure is widely employed in several sectors and is also used by consumers in their daily lives.

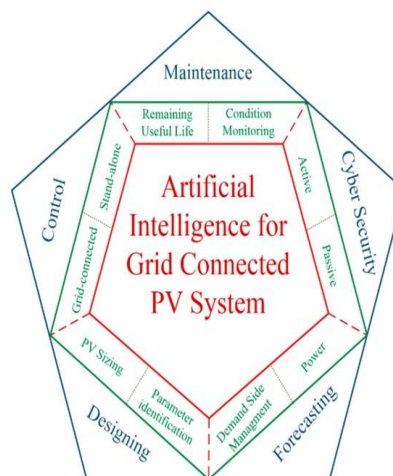


Fig. 1. The implementation of artificial intelligence for the power system industry.

The digital transformation of power systems, which includes the use of AI, has the potential to greatly enhance the network's stability, dependability, dynamic reaction, and other crucial features [2]. The power system is now being designed, forecasted, controlled, optimized, maintained, and protected using AI, as shown in Figure 1. There was hope

that emerging cyber security standards will lead to more innovative uses of artificial intelligence in PV power systems in the near future. Thanks to the abundance of data generated by PV power systems, artificial intelligence has made great strides in helping these systems learn how to improve efficiency and reaction time in design, control, and maintenance. In order to tackle the difficult and complicated issues with power systems, this method promoted data-driven research. Figure 2 shows a schematic that connects the methods used by AI in power systems to their respective functions and applications. Although several articles have shown positive outcomes from using AI in solar PV, it is important to remember that many of the suggested solutions were only tested in limited case studies. According to this research, the models have a low relative degree of generalizability and would not provide the same outcomes when used in a new setting.

The AI strategies shown in Figure 2 were used in the supplied situations, and in every relevant component, the majority of the findings demonstrated that they outperformed traditional methods. Data preparation and preprocessing are another critical factor to think about. Data preparation is the most crucial step in using AI since all techniques rely on gathered time-series data, which is prone to noise, missing datasets, and abnormalities.

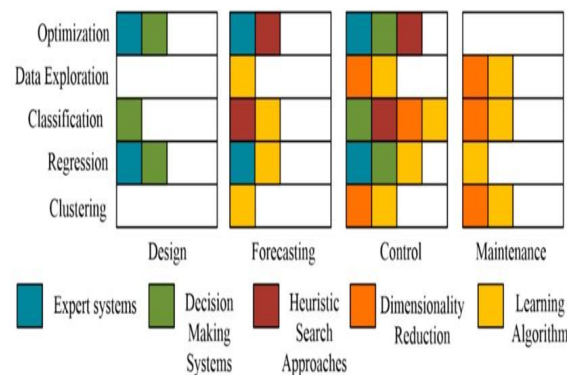


Fig. 2. Integration of various AI applications into grid-connected photovoltaic system design, forecasting, control, and maintenance processes.

According to a Crowd Flower assessment of 80 data scientists' time allocation practices, data preparation consumed about 80% of the total [3]. The training procedure ought to be simple given a clean and well-structured dataset. A thorough analysis of AI's role in power systems, with an emphasis on the following areas, is suggested to accomplish this goal: To accomplish an ideal power system design and appropriate resource usage, find the AI solutions that are suitable for PV system size. To determine how much longer certain parts of the system will last, examine how artificial intelligence methods tailored to condition monitoring and reliability analysis were used. To prevent unplanned power outages and maintain control and monitoring, it is important to anticipate emerging trends in artificial intelligence (AI) methods for digital twins and cyber security.

II. ARTIFICIAL INTELLIGENCE ARCHITECTURE FOR PHOTOVOLTAIC SYSTEMS CONNECTED TO THE GRID

A rundown of all the papers written on artificial intelligence in the electricity grid throughout the last few decades. The data was obtained by referring to the significant articles published in various publications. It is clear that the majority of researchers have been focusing on artificial intelligence methods for optimization and control applications in system design in the last several years. Intelligent photovoltaic (PV) plants are envisioned in [4], with system control achieved by model predictive control and array and energy storage system size optimized using linear programming. Here, AI-powered learning methodologies are made possible by the collected data, allowing the system to detect and respond to a wide range of problems and anomalies within the allotted time. Based on the findings, this procedure can be executed online using reinforcement learning techniques.

Data management and categorization allow for precise identification of the many operational phases and characteristics, which is crucial for power system operation, which relies heavily on processing massive volumes of data quickly. Various functioning phases and disruptions in the power system may be identified by monitoring its real-time

characteristics, as described in [5]. In addition, the power system employs an expert system analysis to differentiate between various voltage dips and disruptions in [6].

In addition, the data was used to its maximum potential by modifying regression methods for power system power flow analysis, demand side management, and forecasting. To find out how much power to scale up the heterogeneous virtual power plants, the authors of [7] use the K-clustering technique. To accommodate the power system's diverse ESS distribution, this section employs the distributed dynamic clustering technique. By taking power system uncertainty into account, a multi-cluster method is used to optimize ESS sizing for PV production in [8]. Similarly, different power grid linkages are investigated using an ordered clustering spectral approach [9]. The digital shift of in photovoltaic systems connected to the grid with AI is shown in Figure 3, considering all the purposes described before.

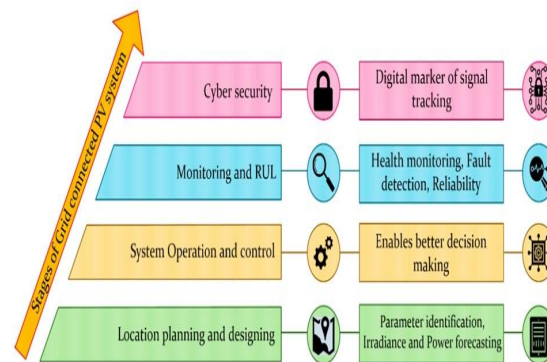


Fig. 3. AI-powered digital transformation of PV installations linked to the grid.

III. USING AI TO DESIGN POWER SYSTEMS

The existing status of artificial intelligence (AI) application in PV system design and optimization with respect to energy yield, costs, and permits is presented in this section. Solar panel equivalent circuit model-based numerical simulations for system operational performance description have long been the subject of discussion [10-11]. Through analytical or numerical methods, the parameters of these models are discovered. The problem with using analytical techniques is that they rely on a lot of assumptions and approximations, which might lead to inaccurate models. In contrast, numerical approaches have consistently shown superior results [12-13]. Despite its computational demands, these approaches include pattern search, non-linear least squares optimization, and the Newton-Raphson method. Also, by using Markov chains, parameter identification was successfully achieved [14].

Conventional approaches cannot be used if data covering a broad time range is not accessible, since these methods rely on this data. When modeling and simulating the PV system, as well as when diagnosing faults, parameter identification is crucial. A method for estimating currents and shunt resistance based on the temperature dependence of diode voltages has been devised. The top performers from the last generation are selected based on how closely they follow the experimental I-V curve with excellent convergence. When compared to other approaches found in the literature, the findings demonstrate that the created strategy performs the best in terms of RMSE. A similar approach is used in [15] to identify parameters of the solar cell's single and double diode models using an artificial bee colony. Compared to the previous algorithms, the one that was designed converges more quickly and with better accuracy (lower RMSE). In addition to heuristic search methods, other popular techniques include neural networks [16-17] and parameter identification methods based on the adaptive neuro fuzzy inference system (ANFIS) [18]. These methods have shown promising outcomes when used to solar panels whose parameters are not known. Figure 5 provides a high-level overview of how the ANFIS technique may be used to perform parametric identification. Table 1 also provides a high-level summary of the many traditional and AI-based parameter identification methods, with the goal of determining which methods are more accurate.

Table 1. Identification of parameters using traditional and advanced techniques.

S. No.	Algorithm	Diode Model	Accuracy
1	GA	Model with Two Diodes	Standard Reduced Mean Square Error
2	PSO	Models of Single and Double Diodes	High Reduced Mean Square Error
3	AIS	Model with Two Diodes	High Reduced Mean Square
4	ABC	Models of Single and Double Diodes	High Reduced Mean Square
5	PS	Models of Single and Double Diodes	Low Reduced Mean Square
6	ANN	Model for a Single Diode	Standard Reduced Mean Square Error

Solar photovoltaic (PV) systems must be appropriately sized to maximize economic life-cycle savings, provide a constant and high-quality power supply, and avoid power outages. There is a lack of sufficient precision in the findings produced by intuitive techniques, and both the non-AI and numerical approaches to system sizing in the literature need excessive quantities of data. Therefore, when there is a lack of necessary data for PV system size at a given location, other alternatives are investigated. Results showed that the created method achieved quicker optimization than the use of particle swarm optimization; the algorithm was trained using a database of current PV modules with technical specs. Technical and financial details of commercially available system devices were included in the algorithm database, together with climatic data for the suggested locations. The literature that was discussed is included in Table 2.

Table 2. Strategies for best PV system sizing.

S. No.	Algorithm	Advantage	Disadvantage
1	GA	Finds the worldwide best using the provided data efficiently Workable with complicated, not-well-defined datasets as well as discrete and continuous models When steps in the solution fail, the final product is unaffected.	It takes more time for the method to handle data when fitness functions are evaluated many times. This method will provide erroneous outcomes if it becomes stuck in local optima.
2	PSO	Simple to set up and tweak with few settings Capable of enduring simultaneous computation Locates global optima efficiently and reaches convergence quickly	Unsuitable for parameters that are distributed Loss of local minimum due to convergence occurring too soon. Identifying basic design parameters is a challenging task.
3	AIS	Effectiveness in determining the global optimum, even when faced with difficult optimization challenges	Implementation method that is quite complicated
4	BA	Does not need many input parameters. The construction is straightforward. Strong performance overall.	The pace of convergence is slow, and the accuracy of optimization is poor.
5	GRNN	Easily maps the intricate relationship that exists between the independent factors and the dependent variables, Handles the noise in the dataset in an effective manner	It becomes stuck in local minima, which leads to over-fitting of the distribution. A significant amount of time is required for processing massive neural network architectures.
6	ANN	The computing time efficiency, ease of implementation, and high precision factor	Has limited scalability; it can only handle a single goal and a single distributed renewable energy source simultaneously.

IV. USING AI FOR GRID FORECASTING IN PV SYSTEMS

Having precise predictions for the power generation delivered into the grid has grown in importance due to the recent uptick in grid-connected photovoltaic (PV) installations. The decline in investment costs—10–20% between 2019 and

2021—is the main driver of the growth, while other variables including incentives, rules governing technical requirements for construction projects, and other directives have also contributed. There will be more fluctuations in the power grid and the possibility of instability owing to unexpected weather changes as a result of grid-connected PV systems, and this trend is projected to persist for the foreseeable future [19]. In addition, spot markets for electricity were established as a result of energy market deregulation; these markets were crucial in regulating the supply and demand for power. This means that communities, merchants, generators, and major end users all need to have precise demand and production estimates. This has been made possible by the heavy use of forecasting techniques by these market participants. Figure 4 provides a high-level picture of the energy market mechanics, which in turn necessitates predictions of power output from intermittent renewable sources.

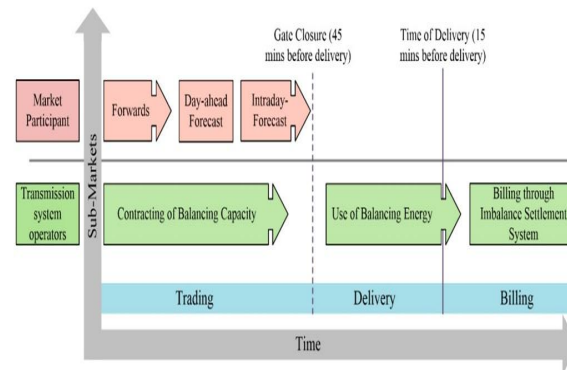


Fig. 4. An overview of the criteria for forecasting in the process energy marketing industry.

This synopsis is useful for controlling reserves, creating energy markets, and balancing power output and consumption [20]. Market participants often struggle to predict solar irradiance and PV outputs, which has made system management more challenging as PV has increased [21-23]. In addition, the literature identifies the balancing market as a place where merchants and generators pay hefty charges for their imbalances when they fail to fulfill their predicted demand or production. Because of these problems, improved market processes rely heavily on accurate forecasting models.

On the other hand, distributed energy resource outputs have not received nearly as much attention as load projections [24]. Improved system functioning has been achieved via the use of load projections in several studies. In order to improve the dependability of forecasts, there is research that looks at both the methods and accuracy of power grid forecasting, in addition to studies that just use forecast outputs [25-26]. The literature has shown that weather, a chaotic system, is very dependent on forecasting. So, looking far into the future, like into the following season, is completely out of the question [27].

In addition, the use of AI approaches offers a solution to the constraints of conventional methods and enables the resolution of intricate issues that are difficult to model and evaluate. The research indicates that artificial neural networks (ANNs) are the predominant machine-learning approaches used in solar power forecasting due to their shown efficacy in many scenarios and with multiple input variables. Support vector machines are the second most often utilized methodology, using supervised modeling methods. They possess robust generalization capabilities and excel at solving non-linear issues. Due to the fluctuating weather conditions over multiple seasons, using data from just one season for a model would need a substantial quantity of information. As a result, it is recommended that the model's performance might be improved by using normalized radiation.

V. ANTICIPATED DEVELOPMENTS AND PROSPECTS

A. Virtual Twin

By integrating AI methods with grid-connected PV systems' many functionalities, we have shown that digitization is a widely accepted process. Even more so, this prompted the creation of a plethora of novel methods, which in turn enhanced grid integration with contemporary energy systems. The advent of the digital twin (DT) has dramatically altered the trajectory of scientific inquiry by greatly expanding the interconnections among real-world objects, their

associated data, and digital simulations. By creating a model of the energy system in a digital and virtual environment, this method paved the way for real-time synchronization, monitoring, and other services [28-29]. In order to learn about the system's status in real time, this method employs intelligent analytics in conjunction with visualization software [30] and real-time data [31-32]. The voltage source inverter's sensing and actuation capabilities were used to create this strategy, which aims to improve system performance. The DT framework typically receives as contribution the fallout of the insight period and the data-gathering modus operandi at the PV system. Furthermore, energy forecasting, the energy internet, the internet of things (IoT) platform, and sophisticated data analytics are all crucial technologies linked to the DT framework that ensure the system runs smoothly. Connected devices, the Internet of Things (IoT), and artificial intelligence (AI) data mining capabilities are laying the groundwork for a data-driven future in product design, production, and service via the DT framework [33-34]. In order to build the DT framework, the IoT platforms additionally integrate data produced by the physical system with historical datasets acquired from the grid-connected PV systems' prior operations [35]. A more efficient and consumer-oriented DT framework may be achieved by enhancing data processing skills using AI. In addition, most of the PV system's functions are incompatible with the raw data due to its high stochasticity. Data preprocessing using the DT framework is crucial for feature extraction in datasets [36]. These characteristics shouldn't impede the uniqueness of the measured data, but they should take into account the elements impacting the dynamics of the PV system's operation.

B. Internet safety

As digitalization incentives have spread across all industries and ICTs have proliferated, cyber security has become an increasingly pressing issue. Adaptive control via digital twins, condition monitoring, predictive maintenance, and fault diagnostics are only a few of the activities made possible by data centric methods, as described in the preceding part. Nevertheless, they simultaneously increase susceptibility to cyber-attacks. Changes to PV systems are likely to be substantial when new cyber security requirements for DERs are developed. Figure 5 provides a bird's-eye view of the situation by comparing data-driven [37-39] and physics-informed [40] solutions in terms of speed, selectivity, and accuracy.

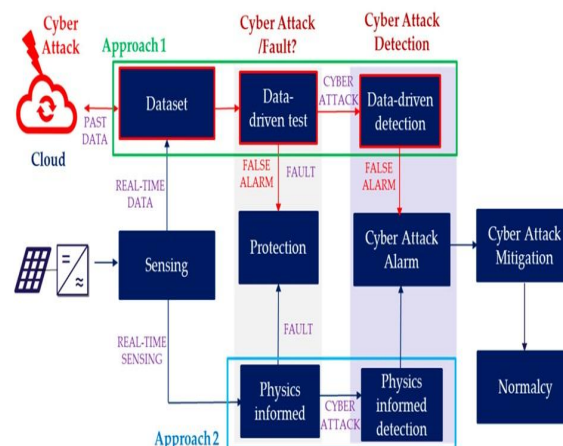


Fig. 5. presents a viewpoint on the cybersecurity methods used for photovoltaic (PV) systems. It provides a comparative assessment of artificial intelligence (AI) techniques and physics-informed approaches.

Figure 5 shows that in Approach 1, grid-tied PV systems' historical data can be used to build cyber security technologies that are both fast and accurate. Because of this, anomaly detection may end up going off more often than it should. The precision and discrimination of data-driven cyber security systems may be readily undermined in this manner. But with method 2, you can only influence the physics-guided tools' decision-making from the real-time sensing stage, and that's already taken into consideration. The outcome is that method 2's decision-making process is more confident than method 1's. To effectively create data-driven cyber security solutions, this calls for further study into this area to account for the opaque character of AI tools.

VI. CONCLUSIONS

According to this paper's review of the literature, numerous studies use different AI methods for different purposes at the structure level of the solar power plant value chain. While specific AI methods may differ depending on the task at hand, artificial neural networks and related sub-architectures are often used. Optimizing generally makes use of adaptive algorithms like PSO and GA, even though artificial neural networks ANN have shown to be highly effective when working with historical data, such in sunlight projection or electrical prediction. Despite their prevalence for management and upkeep, methods of learning, based on data methods, and reasoning frameworks have shown to be too problem-or condition-specific to be useful in other contexts or with different datasets.

Even though most of the publications claim to have achieved considerable results, it should be pointed out that they only tweaked one model while leaving the others with their default values. On top of that, it's likely that some of the suggested models aren't generalizable given the input data. There will very certainly be improvements to the existing AI methods in the not-too-distant future. The industry is currently experiencing a data discrepancy, but this is expected to change as a result of advancements in internet of things solutions, widespread sensor deployment, drone video streams for maintenance purposes, and natural language processing techniques.

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Exhibit SR-2

The Role of Generative AI

The Role of Generative AI in Revolutionizing Healthcare, Education, and Finance: A Mini Review

Vivek Kumar Mishra¹, Gaston O. Adoyo²,
Aayush Bharat Mandavia³, Subhash Kumar Chand⁴

IEEE Senior Member DE, USA¹

Jomo Kenyatta University of Agriculture and Technology Nairobi, Kenya²

IEEE, CA, USA³

IEEE, DE, USA⁴

vivekmishra@ieee.org, mandaviaaayush@gmail.com, skchand@buffalo.edu

Abstract: Today, generative artificial intelligence is enabling industries to transform at an unprecedented pace. In this paper, illustrated case studies of the revolutionary role of GenAI in three sectors are explored: healthcare, finance, and education. GenAI accelerates the domain of healthcare by making its way into drug discovery, medical imaging and diagnostics, and virtual healthcare care assistance. In the financial sector, GenAI applies to crime detection, risk management, and prescriptive financial advisory services. GenAI helps educators in the education sector with personalized learning, automates teaching tasks, and creates an interactive learning environment. In this review, we check out exactly how these various kinds of GenAI devices have been made use of in the form of components such as GANs, VAEs, or transformer-based versions through these sector applications. These case studies are analyzed, pointing out realworld developments and their effect on outcomes. The paper ends by discussing the inevitable pitfalls, ethical concerns, and regulatory barriers to GenAI adoption, such as data privacy, algorithmic discrimination, and cybersecurity issues. In addition, it discusses future research directions and opportunities for responsible innovation, focusing on AI transparency, ethical frameworks, and the critical role of human oversight

Keywords: Generative AI, Healthcare, Education, Finance, AI Ethics

I. INTRODUCTION

Generative artificial intelligence is recognized as one of the most disruptive trends of the 21st century, with great potential to revolutionize multiple industries, ranging from healthcare and finance to education. Unlike conventional AI, which analyzes and classifies the prevailing data, generative artificial intelligence (GenAI) can generate human-like images, text, audio, and code [1, 2]. This unique ability to create humanlike outputs as guided by large datasets makes GenAI valuable to address complex challenges in critical sectors.

In healthcare, using GenAI helps strengthen aspects such as drug development, big data, and patient profiling, among other roles. Machine learning algorithms can work through various large datasets and find intricate relationships much more complicated for a human mind to discern [3]. As an example, through analysis of patient information and clinical trial data, GenAI models can identify how patients are likely to respond to certain treatments to perfect the fitting of treatments to patients [4]. In addition, GenAI can optimize the design of new drugs by using molecular interaction simulations, the results of which would usually take time and laboratory procedures to establish [5].

GenAI offers great utility in the finance sector, especially in fraud detection, credit risk assessment, financial modeling, and customer service automation. GenAI can assist financial companies in improving security as it relates to such transactions because it can detect possible fraudulent activities and suspicious transactions in real time [6]. Also, GenAI creates credible financial reports and analyzes them to minimize the time it would take for human intervention [6]. This means that finance professionals can utilize their time more efficiently to make tactical evaluations instead of drowning in numerous calculations.

Education is another domain where GenAI has brought significant transformational change. GenAI can develop concepts that suit any learning style and pace for students [7]. With such an adaptation, GenAI helps increase benefits and engagement so the educators have more time for the methods that engross the learners more. Educators also benefit from GenAI in terms of generating teaching content, building lesson plans as well as quizzes, and generating knowledge to supplement the process of teaching [7]. An overview of the GenAI application in these three sectors is presented in Figure 1.

Despite these immense gains from GenAI, a myriad of challenges remain, including ethical issues, regulatory hurdles, and technical inadequacies. To this end, a comprehensive review of GenAI applications across education, finance, and healthcare is critical in understanding its transformative potential, limits, and future trajectory. While the present research discusses, comprehensively, the general role of AI in the health, education, and financial fields, this review particularly tackles the knowledge gap concerning the distinct impact of GenAI in terms of automation, decision-making, and ethics [6, 4, 8].

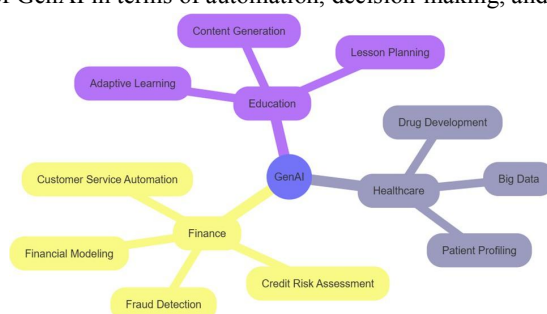


Figure 1: GenAI applications across health, education, and finance sectors

Systematically, this paper provides a review of the applications, case studies, challenges, and future directions of GenAI in education, healthcare, and finance. Through analysis of the latest research and trends in the industry, this review provides insights into how GenAI is transforming these sectors, the presented opportunities, and the challenges to be tackled to ensure sustainable and responsible adoption. On the other hand, the exclusion criteria involved studies generally discussing AI applications and not focusing on GenAI, published papers or industry reports beyond 2017 in exclusion of those providing foundational insights, and non-English-published articles or industry reports. Finally, after carefully identifying relevant and credible sources, an extraction of key themes followed and included applications of GenAI in education, healthcare, and finance; case studies including real-world applications of GenAI in these sectors; challenges, regulatory, and ethical concerns on GenAI applications in these sectors; and future direction and gaps in research.

II. METHODOLOGY

The present paper presents an extensive review of the existing literature and trends on GenAI applications in education, healthcare, and finance. To this end, the methodology involves a systematic literature review of relevant information on the application of GenAI in the education, health, and financial sectors. To begin with, relevant literature was identified and pinpointed from academic databases, including Google Scholar, Web of Science, Scopus, IEEE Xplore, and PubMed. The search was specifically focused on peer-reviewed journal articles, industry reports, and conference papers published recently, i.e., from 2017 to 2024. The keywords included "Generative artificial intelligence in healthcare," "Generative artificial intelligence in education," and "Generative artificial intelligence in finance." The inclusion criteria were studies delving specifically into the applications of GenAI in education, healthcare, and finance, peer-reviewed published papers or credible industry-based reports, and research on case studies, practical implementation, or empirical results as far as applications of GenAI in education, healthcare, and finance sectors are concerned. A summary of the methodology is presented in Figure 2.

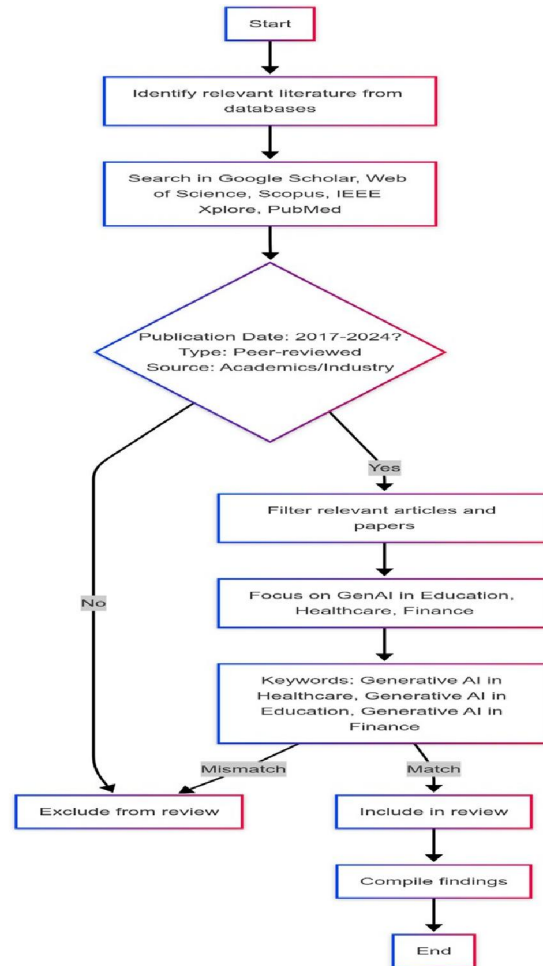


Figure 2: A summary of the methodology used in the review

III. OVERVIEW OF GENAI TOOLS AND APPLICATIONS IN EDUCATION, HEALTHCARE, AND FINANCE

GenAI operates via advanced deep learning models, enabling the synthesis of knowledge, the creation of new content, and the autonomous making of decisions. These AI tools have the capability of processing gigantic amounts of data, learning intricate patterns, and generating human-like images, audio, texts, and molecular structures. As such, they are highly valuable in data-intensive industries like education, healthcare, and finance. GenAI technologies operate under three primary architectures, and they include generative adversarial networks, variational autoencoders, and transformer-based models.

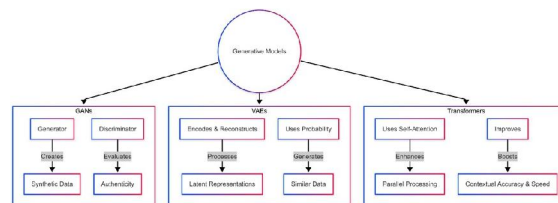


Figure 3: Overview of GenAI technologies

To start with, generative adversarial networks (GANs) comprise two neural networks, the generating system (generator) and the discriminating system (discriminator), and as their names suggest, the duo works antagonistically in creating highly realistic-synthetic data [9]. Principally, the generator yields new samples of data, whereas the discriminator evaluates the authenticity of the generated data by the generator against the real data, prompting the former (generator) to improve its output. Variational autoencoders (VAEs) are deep learning models working on the principle of probability to generate new samples of data with similar characteristics to that of the training set [9]. VAEs execute this by encoding and reconstructing latent representations of space. VAEs, unlike GANs, do not need adversarial training. Transformer-based models are advanced in nature and have played a key role in revolutionizing content generation and the processing of natural language. Unlike VAEs and GANs, transformer models employ the use of self-attention systems in the parallel processing of data to improve contextual accuracy and speed [9]. A summary of these tools and their applications in education, healthcare, and finance is represented in Table 1

IV. APPLICATIONS OF GENERATIVE AI IN HEALTHCARE

GenAI is a force to reckon with in healthcare, especially in the wake of COVID19, which crippled the worldwide healthcare system. Since GenAI is integrated with new technologies such as the Internet of Things (IoT), it has made it easier for patients to adopt digital health technologies [9]. Indeed, the digital collection of patient data and AI-empowered intelligent medical technology have been met with enthusiasm among patients, making GenAI accentuate the 4Ps of medicine (i.e., predictive, preventive, personalized, and participatory) [10]. To this end, integrating GenAI into healthcare is effective in terms of being better, faster, and less costly [11].

Table 1: Summary of GenAI applications across the three sectors

Domain	Key Applications
Education	Personalized learning, automated grading, adaptive learning, generating teaching materials, and curriculum design.
Healthcare	Drug discovery, medical imaging, clinical decision support, and virtual healthcare assistance.
Finance	Fraud detection, credit risk assessment, financial reporting, personalized financial advising, and algorithmic trading.

4.1 Drug Discovery and Medical Research

Drug discovery methods demand an enormous financial investment and multiple years of development that exceed ten years before delivering new medications to the market. The field of drug discovery benefits from GenAI technology, which speeds up molecular design practices while forecasting pharmaceutical couplings and improving selection outcomes, thus lowering expenses and duration [5]. It enables rapid analysis of biochemical datasets, molecular structure simulation, and identification of promising drug conjugates to improve precision medicine and treatment conservation outcomes. Figure 4 below showcases how GENAI reduces the timeframe for drug discovery. GenAI is also being employed to repurpose existing drugs for new therapeutic applications, which has previously been very effective in leveraging existing treatments to combat new diseases like COVID19 in a short amount of time [5].



Figure 4: Application of GenAI in drug application

Predicting protein structure and understanding drug-target interactions with high accuracy is how GENAI transforms molecular simulation. AlphaFold, a GenAI developed by DeepMind, has drastically shortened the amount of time needed to predict protein folding and, in turn, allowed researchers to design efficient targeted drugs for complex diseases like cancer and Alzheimer's [12]. GenAI simulations enable pharmaceutical firms to simulate chemical characteristics while detecting toxic compounds and finding the best drug stability levels that minimize first-trial

failures [13]. Pharmaceutical companies utilize GenAI systems to create new compounds suitable for rare disease treatment, so the medicinal discovery process becomes faster [12].

Drug repurposing using GenAI accelerates the identification of new therapeutic uses for existing drugs without the need to perform early-stage clinical trials and expedite regulatory approval. This was the case during the COVID-19 pandemic, when thousands of drugs from the FDA's approved drug list were screened by the models to find possible treatments [5]. By applying analytical methods to molecular interactions and clinical trial data, pharmaceutical companies cut down drug discovery schedules [5]. Additionally, GenAI repurposing is improving personalized medicine (i.e., drugs are tailor-created based on genomic and biomarker profiles to improve patient-specific treatment results) [5].

4.2 Medical Imaging and Diagnostics

Medical imaging is used in the diagnosis and monitoring of diseases, providing health care professionals with information about the internal structures and allowing early detection of diseases, in particular cancer, neurological disorders, and heart diseases. Despite the accuracy of human interpretation in medical imaging, the accuracy of human interpretation is subject to variable human radiologist experience or workload and the limits of humanly realizable medical imaging technology [9, 14, 3]. Additionally, low-resolution scans, poor contrast, and movement artifacts can decrease the quality of the image to the point of not being able to detect subtle abnormalities [15]. The applications of GenAI are to boost image resolution, automate anomaly detection, and generate synthetic medical data to train the AI model. GenAI-based imaging systems combine deep learning models such as GANs and transformers that eventually help radiologists detect diseases earlier and make fewer interpretation errors [16, 17]. In oncology, neurology, and cardiology, most notably where early intervention can make an impact on patient outcomes, the impact of AI-enhanced imaging is particularly significant.

Super-resolution imaging is one of the most transformative applications of GenAI in medical imaging by improving the clarity, contrast, and resolution of MRI, CT, and ultrasound scans. Such AI image enhancement techniques can bring forth high-resolution medical images from low resolution inputs so that radiologists can see minute tumors, early-stage neurological changes, and vascular abnormalities that otherwise would go unobserved [18, 19]. Furthermore, deep learning algorithms applied in automated anomaly detection systems such as Qure.ai and Aidoc evaluate Xrays, MRIs, and CT scans and predict with higher accuracy and efficiency the presence of suspicious lesions, fractures, and pulmonary abnormalities, as well as signs of stroke [20, 21, 22]. AI-driven systems eliminate diagnostic errors, reduce radiologist workload, and allow instant clinical decision support as it prevents critical situations from being missed out on and patients receive timely, lifesaving treatments. Current research has demonstrated that GenAI-supported diagnostic tools can augment radiological efficiency by up to 30 percent, reduce false negatives, and, most importantly, facilitate early diagnosis for patients [3, 18].

Due to the lack of high-quality, annotated datasets, medical imaging is a major challenge, especially for rarer diseases, pediatric conditions, and less common cancer types. The solution offered by generative AI is the ability to make synthetic medical images with GAN and use these synthetic images as the dataset for training AI models with these generative images without having much real-world imaging data [23, 24]. In medical specialties where realworld datasets are scarce, this technique proves valuable because it allows AI models to learn and generalize from synthetic training data and improve diagnostic accuracy [25, 26]. In addition to that, GenAI-generated medical images are important in the preservation of the patient's privacy because they remove the need for such real patient data, allowing the AI models to learn the disease patterns and variations. It is important, especially for compliance with data protection regulations as well as in terms of ethical and legal standards, that AI-driven medical research should comply with [27]. Augmenting AI training with synthetic medical images allows for more robust, more reliable, and less biased medical diagnostic and medical imaging systems that are more accessible and more accurate for patients around the world.

4.3 Virtual Healthcare Assistant

Virtual healthcare assistants powered by GenAI are changing the game regarding how practitioners can engage with patients, how they can access providers, and how they can be more efficient. NLP and machine learning algorithms are

leveraged by these assistants to engage with patients and provide personalized recommendations in healthcare [4, 28]. The integration of AI into chatbots as mental health support and telemedicine platforms can reduce the workload in hospitals, improve medical outcomes, and provide a better experience for patients. In particular, AI-driven solutions are extremely beneficial in rural and underserved regions where immediate medical assistance and mental health support may be difficult to achieve and maintain [29, 28].

Out of all the applications of AI in virtual healthcare, chatbots are employed to assist with answering patient inquiries, assessing symptoms, and making preliminary diagnoses. Several AI-powered platforms interact with users to answer patients' medical questions, guide them, or provide a preliminary diagnosis on how their symptoms relate [29, 30]. In addition, these GenAI chatters also analyze the symptoms and the history of a patient to find out if a patient needs emergency care, must be treated himself, or must speak with a doctor. These tools can cut down unnecessary hospital visits, thus optimizing healthcare resources by prioritizing critical case admissions [31, 30]. AI chatbots also perform well in terms of patient satisfaction, especially when used in primary care settings such as triage, followup consultations, and chronic disease management [31].

Mental health AI assistants are becoming a more important tool for people with anxiety, depression, and stress-related disorders. Platforms that are driven by GenAI provide real-time text-based conversations that mimic emotional support and delivery of cognitive behavioral therapy (CBT) to the user in a continuous, judgment-free, and instantaneous manner [32, 31]. AI-powered tools have been particularly effective at helping individuals who may be socially, financially, or spatially limited in our traditional mode of seeking out therapy using a licensed mental health professional [32]. Mental health results from research imply that mental health AI assistance can enhance well-being, decrease symptoms of depression and anxiety, and aid in the supplementation of traditional therapy [31]. AI mental health tools are scalable and affordable and provide important coverage in public health efforts to address the increasingly serious global mental health problem [31, 30].

Since telemedicine has risen, AI-powered virtual assistants have been integrated into video consultations, and that has improved efficiency in terms of remote healthcare delivery. AI assistants assist physicians in retrieving patient history, transcribing consultations in real time, and generating predictive analytics of personalized treatment recommendations [33]. These tools facilitate improved physician workflow by freeing doctors from administration instead of interacting with patients [34]. Speech-to-text technologies driven by AI allow for the use of machine transcription of medical documentation to lessen physician burnout directly related to inpatient electronic health record entry [35, 34]. AI assistants in telemedicine can also analyze patient speech, facial expressions, and behavioral cues to identify early cognitive decline, stress, or other medical conditions to improve early intervention and preventive care [34]. With telehealth adoption rising, AI-based assistants play a vital role in ensuring easy, efficient, and high-quality virtual care.

V. APPLICATIONS OF GENAI IN EDUCATION

The integration of GenAI in educational settings is a transformative force that changes how students learn, teachers engage in education, and learning institutions operate. Through personalized learning experiences, automated administrative responsibilities, and real-time feedback delivery, GenAI is taking the educational landscape by storm while bridging gaps and promoting a more inclusive and efficient learning architecture [36].

5.1 Personalized Learning and Tutoring

GenAI has been able to personalize the learning experience for students, which is one of the most significant contributions that GenAI can make to education. The traditional education system works in a one-size-fits-all manner, which is not suitable for all students. As generated AI analyzes student performance, learning styles, and knowledge gaps, it adapts lesson plans to cater to these students' needs by delivering tailored content that is both engaging and comprehensible [36].

Adaptive learning algorithms are also used by AI tutoring systems and allow real-time feedback, personalized exercises, and targeted interventions. AI-powered tutors set up exercises with different grades of challenges to ensure that a student gets a particular grade of difficulty to exercise order and avoid getting frustrated with too hard-to-digest material [36, 37]. The results of the research have also indicated that students who interact with AI tutors are more successful, with a stronger retention rate and deeper insights regarding the subject matter [38, 39].

5.2 Teaching Materials and Assignment Automation

Even more, outside of personalized learning, GenAI is transforming the way we develop curricula, plan lessons, and generate content. AI-based platforms can automatically create textbooks, interactive quizzes, and practice tests for learning materials. [37, 39]. These are up-to-date, contextually relevant, and cater to the needs of education. AI-powered grading systems help save time in marking assignments and give instant feedback and consistency in grading so that teachers can spend more time with students on engagement and mentoring [36]. Additionally, AI feedback on essays, code assignments, and problem-solving exercises provides students with real-time feedback and helps them understand what they are doing wrong [36].

5.3 Virtual Classroom

In recent days, as virtual learning environments have become a trend, virtual AI classrooms have also emerged, which provide real-time interaction, instant feedback, and personalized learning paths. These AI tutors that are AI-driven are ChatGPT, Claude AI, and IBM Watson Tutor, which help students answer questions, provide a summary of complex topics, and generate interactive study materials [40, 39]. In addition, these intelligent tutoring systems can simulate real-world classroom experiences, enabling students to participate in virtual experiments and historical reenactments and react to AI debates. For example, AI platforms such as Querium and Cognii employ AI to teach students critical thinking and analytical skills through dialogue learning. [37, 40]. Empirical evidence indicates that students who use AI-powered tutoring systems tend to get better results in standardized tests, and they also have higher levels of engagement than those who use traditional e-learning systems [36].

VI. APPLICATIONS OF GENAI IN FINANCE

The financial sector has witnessed a profound transformation due to GenAI solutions. Financial institutions rely heavily on process automation and big data and, therefore, are in a "unique" position to lead the adoption of AI. To begin with, automation is possible with GenAI, leading to increased efficiency and productivity in the financial sector. In addition, GenAI-like machines are not prone to human erroneous tendencies and psychological factors, ensuring accuracy and unbiasedness in predictive analytics and trading strategies [41]. GenAI also leads to radical changes in customer relationships through the promotion of customized digital finance, cost-effectiveness, and service efficiency [42]. Moreover, GenAI is capable of overseeing the financial conduct of employees and assists supervisors in identifying potential violations and unscrupulous activities while also helping regulators anticipate the effects of regulatory changes with precision [41]. Fintech leaders can reap from GenAI in terms of making faster credit decisions with beneficial effects to both lenders and customers.

6.1 Fraud Detection and Risk Management

Increased use of digital transactions, online banking, and decentralized finance paved the perfect ground for financial fraud to grow. Typically, traditional fraud detection methods, such as rule-based systems, are ineffective at catching emerging fraudulent patterns. This generates fraud detection with the real-time generation of anomalies in transaction data to detect suspicious activities and prevent financial losses before they occur [6, 43]. Anomaly detection algorithms, behavioral analysis, and risk scoring models form AI-driven fraud detection models based on which unusual patterns in credit card transactions, online payments, and stock market activities are found.

For example, Darktrace AI or Feedzai use AI-based fraud analytics to detect phishing attacks, identity theft, and money laundering schemes [43]. In addition, AI-based credit risk assessment models employ customer credit history, spending behavior, and real-time financial data to assess the risk of lending and make investment decisions and, therefore, determine the approval of loan applications for banks and lending institutions [43]. In addition, AI can also use NLP and predictive analytics to watch financial news, regulatory updates, and events on social media for indications of market risks. This method enables hedge funds, central banks, and financial regulators to act judiciously against economic changes and cybersecurity risks [6, 43].

6.2 Financial Advisory

Gen AI-powered financial advisory services have transformed investment strategies, wealth management, and financial literacy in important ways. The key difference between a traditional financial advisor and an AI-based advisory platform is that an AI-based advisory platform analyzes market trends, historical data, and individual financial goals to provide appropriate investment recommendations [44]. GenAI employs machine learning algorithms to establish market risks, assemble asset allocation, and rebalance AI-powered investment portfolios. In particular, these systems do not rely on human bias or emotional trading and present objective financial strategies using a data-driven approach [45]. GenAI helps new investors learn more about the topic by simplifying complex investment concepts for novices. Financial markets become more accessible and understandable to people with the use of AI chatbots, such as real-time insight into market trends, stock performance, and economic indicators [44]. At the retirement planning level, AI-powered financial advisory services are equally important in tax optimization and risk assessment. Using historical spending patterns and economic forecasts, AI can calculate future needs and suggest specific saving plans for individuals and businesses [44]

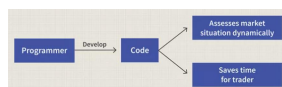


Figure 5: The algorithmic trading process

6.3 Algorithmic Trading

Using AI-powered software for investing in financial markets, the use of algorithmic AI algorithms has been used to perform stock or futures trading. The algorithmic trading program is also termed non-programmatic/quantitative trading and aims to analyze the financial markets, find profitable trading opportunities, and perform trading at high speed (Figure 5). By removing human bias, curing human decision-making, and enhancing market efficiency, AI-driven trading systems can all achieve these [46]. Buy/sell strategies in these trading models are optimized in real time using deep reinforcement learning, neural networks, and statistical arbitrage techniques. For example, Goldman Sachs and Morgan Stanley have introduced AI-driven trading systems that increase the accuracy of market forecasting by 35 percent, implying improved profitability and reduced financial risks.

6.4 Algorithmic trading and high-frequency

trading (HFT) are instrumental in AI, for thousands of trades are executed in milliseconds. AI models can help process realtime financial data, news sentiment analysis, and geopolitical events to figure out how to predict market movements and take advantage of short-term price fluctuations [46]. It also incorporates AI-powered sentiment analysis that helps to deal with financial news, earnings announcements, and social media trends to collect the market sentiment, which is directly proportional to the stock price. For instance, sentiment analytics and natural language processing (NLP) apply to determining investor emotions, determining market volatility, and revising trading strategies based on them [46].



Figure 6: Carnegie Learning's AI empowered interface for learning math

VII. CASE STUDIES

7.1 Education: Personalized Learning at Carnegie Learning

As an educational technology company, Carnegie Learning has implemented AI-driven personalized learning systems at K12 and higher education institutions with success. The platform uses machine learning and NLP to analyze students' patterns of learning and adjust lessons based on specific needs, providing personal instruction. MATHia software is based on AI and customizes students' mathematics learning paths by providing targeted feedback focused on each student's progress [47]. In Pennsylvania school districts, students who used MATHia excelled 28 percent more on measures of algebra proficiency than other students subjected to traditional teaching methods. Results from this case

demonstrate the capability of GenAI to improve student engagement, reduce learning gaps, and improve student learning outcomes through adaptive educational technologies [47].

7.2 Healthcare: Medical Imaging and Diagnostics at Stanford Health Care

GenAI-driven diagnostic tools have been integrated by Stanford Health Care in radiology and medical imaging analysis. A most impactful application is the AI-trained CheXNet deep learning model, which is trained to detect pneumonia, tuberculosis, and lung cancer from chest X-rays at rates better than human radiologists. Aiming to enhance services and reduce wait times, this AI model was developed at Stanford University, was trained on 112,120 chest X-rays, and has been proven to perform 4 percent better than human radiologists in identifying pneumonia [48]. Stanford, too, allows patients to benefit from AI-powered MRI enhancement tools to improve scan clarity, which reduces the need for repeated imaging and radiation exposure to patients. GenAI has been used in medical imaging to speed up diagnostics, reduce human error, and enable early detection of disease, which has benefited the patient [48].

7.3 Finance: Algorithmic Trading at Goldman Sachs

A global financial heavyweight, Goldman Sachs, has employed GenAI algorithmic trading models to drive better market forecasting and execution. It is done by the firm by utilizing deep learning techniques, reinforcement learning, and real-time financial analytics to boost trade profitability as well as risk reduction. The 'SecDB' (Securities Database) system, using realtime market data analysis to optimize highfrequency trading (HFT) strategy, has been one of their most successful AI applications [49, 50]. Goldman Sachs said in 2022 that the use of its AI-driven trading algorithms increased market forecasting accuracy by 35 percent and resulted in a greater return on investments and fewer losses. The AI system also monitors economic indicators, social media sentiment, and global news trends to dynamically adjust trading positions, thus giving an edge in dynamic markets. The results of this case study show that Generative AI can improve financial decision-making, reduce risks, and improve trading efficiency in the markets [49, 50].

VIII. CHALLENGES TO AI APPLICATIONS

8.1 Healthcare

As the field moves towards integrating GenAI in medical research, diagnostics, and patient care, these pose significant ethical and privacy concerns. In health care, AI systems lead to the processing of huge quantities of sensitive patient data that may lead to incidents of data breaches, hacking, and misuse [4]. As medical data handling is subject to the strict compliance of the Health Insurance Portability and Accountability Act (HIPAA) and the General Data Protection Regulation (GDPR), a lack of full AI transparency and security remains a challenge. Moreover, biased artificial intelligence models trained on nondiverse datasets also lead to misdiagnosis, disparities in healthcare outcomes, and reduced trust in AI-powered medical tools [4]. There is another major issue about the lack of accountability in AI-driven healthcare decisions. Interpretation of the decision making process of the GenAI models is a difficult task since they are functioning as "black boxes." The lack of transparency in AI-generated diagnoses and treatment recommendations leaves medical professionals to not justify the diagnoses and treatment recommendations, thus raising legal and ethical questions [51, 52].

8.1.1 Education

AI-driven personalized learning and automated grading in education raise bias and fairness concerns along with accessibility. Likewise, if AI models are trained on biased datasets or few training samples, they might reinforce stereotypes or disadvantage students from underrepresented groups [53]. With the emergence of AI-generated teaching materials and assessments, such materials and assessments could contain misinformation or even errors that may negatively affect academic integrity and overall education quality [54].

Another challenge is the digital divide that comes with the lack of access to artificial intelligence-powered learning platforms and stable internet connections as well as necessary devices, for example, students coming from low-income areas or rural backgrounds. There is also the risk of using AI tools ineffectively without AI literacy training for

educators [54]. Finally, there are still worries over student privacy due to AI-powered educational tools that collect and analyze a lot of data from students, which could be used for wrongful use by other firms [55].

8.2 Finance

Historical financial data is essential for fraud detection, algorithmic trading, and risk management using GenAI. Despite that, biased datasets can have an impact on inaccurate credit risk assessment, unfair loan rejections, and discriminatory investment strategies [56]. The bad or misuse of algorithmic trading systems leads to market volatility and financial instability. To illustrate, high-frequency trading algorithms have launched AI-driven flash crashes by rapidly trading erroneously [57].

Additionally, hackers are taking advantage of AI that can neutralize fraud prevention systems. By using deepfake technology and AI-generated phishing scams, attackers attack financial markets and steal sensitive data. As AI systems improve, the enhancements required in cybersecurity protocols by the financial institutions to avert AI-powered fraud. [58]. It must be robust and in real time.

IX. FUTURE DIRECTIONS

AI Transparency and Explainability

The explanations of AI should be as transparent and understandable as possible, as they are one of the most important areas for improvement in healthcare, finance, and education. Future research should aim to develop “white box” AI models with clear and interpretable decision-making processes [52]. Explainable AI can assist physicians in validating AI-generated diagnoses and treatments, increasing the trust that AI-aided decision-making will be better accepted [59]. By using AI-driven auditing tools, finance regulators need to ensure fairness and compliance in financial prediction and risk assessment [43].

AI Regulations and Ethical Frameworks

Despite this, we'll need stronger AI policies to regulate the ethical use of AI in the industry. To ensure AI complies with security compliance, algorithmic fairness, and data privacy, there need to be clear AI guidelines for policymakers [59]. AI fairness frameworks can be introduced to mitigate the possibility of healthcare treatments, financial services, and education access discriminating against individuals [56]. Similarly, companies at the stage of developing AI models should also build mechanisms of bias detection to minimize such unwanted discriminatory outcomes.

Cybersecurity

As AI becomes ensconced in more of our lives, many more cybersecurity challenges will arise. Future cycles of AI development should target AI-based security solutions that use the appropriate AI to identify and stop AI-generated cyberattacks in real time [56]. In the finance industry, AI-based fraud detection models need to be updated continuously to cope with growing fraud techniques and deepfake scams [56]. Blockchain-based AI security could also prove useful for financial institutions to enhance the security of sensitive transactions and customer data.

Accessibility and Digital Equity

Ensuring equitable access to AI-powered learning and healthcare solutions is crucial for bridging the digital divide. Governments and organizations should invest in AI infrastructure for low-income communities, ensuring fair access to AI-driven education, healthcare, and financial services [57, 56]. AI literacy programs for teachers, doctors, and financial professionals will also be vital in maximizing AI's potential while minimizing risks [56].

Human Decision-Making

Rather than replacing human professionals, the future of GenAI should focus on augmenting human expertise. AI should act as an assistive tool for doctors, financial analysts, and educators, helping them make informed decisions while maintaining human oversight [56]. Future AI models should emphasize collaborative intelligence, where AI and humans work together to enhance efficiency, accuracy, and fairness across industries.

X. CONCLUSION

GenAI is a critical force to reckon with across healthcare, education, and finance. For instance, Carnegie Learning Center is a successful story of how personalized learning can be augmented by GenAI to enhance student learning outcomes. Stanford Health Center employs the use of GenAI-driven diagnostics to improve medical imaging and, therefore, drastically reduce waiting time. In the realm of finance, Goldman Sachs uses algorithmic trading to optimize the financial decision-making process. Indeed, GenAI accelerates drug discovery, diagnostic imaging, and the revolution of patient care in healthcare. In the financial sector, GenAI enhances risk management, security, and access to financial advice. In education, administrative automation and the creation of engaging learning material, in addition to personalized learning, have been achieved through GenAI. Nonetheless, care should be taken in terms of ethical implications, technical limitations, and regulatory frameworks. Matters related to data privacy, cybersecurity vulnerabilities, algorithmic bias, and human job displacement must be addressed for sustainable and responsible adoption. Future directions should prioritize transparency, robust regulations and ethical frameworks, digital equity, and accessibility. With such developments, the power of GenAI can be harnessed to create more accessible healthcare, an equitable financial sector, and a more effective and personalized education.

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Exhibit SR-3

MDPI Sustainability Journal Publication

Article

Framework for Rail Transport Inequality Assessment: A Case Study of the Indian Railway Zones with Superfast Express (SE) Trains

Shailesh Chandra *  and Vivek Mishra

Department of Civil Engineering and Construction Engineering Management, California State University, Long Beach, CA 90840, USA

* Correspondence: shailesh.chandra@csulb.edu

Abstract: The paper presents a framework for assessing inequality in passenger rail services by examining connectivity and accessibility across two different travel frames: favorable (daytime travel) and unfavorable (nighttime travel). The unfavorable timeframe is often characterized by a lack of adequate first-/last-mile transport availability to train stations, impacting passenger access. The study employs a newly developed inequality-to-investment ratio to classify rail transport entities like stations, routes, or zones as either winners or losers, using a performance persistence analysis. A case study was conducted using data on the superfast express (SE) train schedule and investments from 2016 to 2020 across sixteen zones of the Indian Railway, assessing service inequalities during the hours of favorable (7 a.m. to 9 p.m.) and unfavorable (9 p.m. to 7 a.m.) timeframes. Results revealed inconsistencies in inequalities among zones. Moreover, the research demonstrates that the choice between connectivity and accessibility metrics significantly influences the identification of better or worse performing zones as winners or losers, respectively, throughout the 2016–2020 period and during both timeframes. The study underscores the importance of selecting appropriate measures and suggests re-evaluating or increasing investments in certain zones of Indian Railways based on these insights. The framework proposed in this paper can be used to assess inequalities within any transportation system receiving investments or government funds, contributing to the broader goal of ensuring equitable access to transportation, which is essential for a sustainable transportation system.



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Keywords: passenger rail; superfast express trains; connectivity; accessibility; inequality; Indian Railway; sustainable transport

1. Introduction

Passenger rail is critical to the economic growth of both developed and developing economies [1–3]. For example, in India, the intercity passenger rail is vital to transportation needs, having large-scale impacts on the country's economy [4]. In other countries like China, connectivity by train is so vital that 19 of the 20 top-ranking domestic transportation routes are by rail [5,6]. The dependency on rail in other countries like the USA is similar, which is critical in its economic sustenance [7].

Passenger rail facilitates access to opportunities, and the lack of it and, in general, any public transportation facility translates to social exclusion emerging due to a lack of social opportunities [8]. Several scholars agree that rail's influence on the economy rests on its ability to address inequalities, fulfilling the need to provide a means of transport for justice and fairness between cities experiencing differential economic activities [9,10]. Thus, in essence, the role of rail in the economic development of a nation is evident in its meeting the equity goals of connecting places and attenuating regional disparities of spatially far-apart places [11–13].

Some studies, however, have highlighted the disadvantages of rail access [14–17]. They have noted that rail transportation has caused wealthy cities to enjoy higher accessibility benefits than hinterland cities [18]. Similar observations were made for cities with rail transportation in Europe [19].

Inequalities stemming from transit connectivity and accessibility are contingent upon essential factors and amenities in the vicinity of the stations utilized by train passengers. The popularity and ridership of a rail system hinge on the effective connection of its stations with first-/last-mile transportation. This connection significantly contributes to the overall service quality, which is particularly important for low-income shift workers, especially those commuting during nighttime when first-/last-mile connectivity is often limited [20]. Research shows discomfort and insecurities arise in accessing transit stations during often unconducive or odd times of the day. For example, Jehle et al. [21] have pointed out the challenges passengers often face in being unable to access railway stations due to a lack of safety or first-/last-mile connectivity at night. Other scholars have also emphasized these findings on the need for nighttime transport, whether in a developed [22] or a developing nation [23].

During nighttime or unconventional hours of travel, passengers may face safety challenges at a station due to insufficient infrastructure facilities, such as inadequate street lighting and patrolling. These hours are particularly ‘unfavorable’ for vulnerable populations undertaking travel [24]. Lack of first-/last-mile transport connectivity at night signifies increased inequalities [25]. Hence, it is imperative to implement policies that mitigate inequality, with a specific focus on enhancing equity in rail services. This approach is essential to ensure fair access to resources, a prerequisite for both economic growth and upliftment. To formulate effective policies, it is crucial to first understand the extent of existing inequalities.

Often, investments (especially capital) in rail are made to eliminate regional inequalities and provide equal opportunities for transport access to all. However, this has not always been the case. Constantin et al. [17] showed that despite substantial investments, the Romanian Railway did not positively contribute to reducing regional inequalities. This occurred due to uneven growth and low accessibility.

In general, evaluating the effects of inequality can be a complex task in most research findings related to rail connectivity and accessibility. Recommendations have been proposed to employ spatial impact analysis based on changes in accessibility distribution within the particular urban clusters affected by rail projects [10]. Proximity to population centers, the presence of rail stations, and the service quality of the overall transportation system impact the inequality implications [20], which not many studies have addressed comprehensively in the past for semi-high-speed, premium, or superfast rail transportation in developing countries.

Therefore, this research is focused on evaluating inequalities among service zones of rail on its performance measures of connectivity and accessibility. Further, this paper enriches the knowledge of the performance of entities such as zones or regions of operations of rail services, identifying them as either beneficiaries (as winners) or disadvantaged (as losers), focusing on the investment patterns and shifts in inequality over time.

The framework presented in this paper is specifically studied with an application for India, a developing nation, with the impacts of the superfast express (SE) trains of the Indian Railway. The SE train impacts are studied on the inequalities of the sixteen zones of the railways utilizing data on train arrivals and departures from 2016 to 2020. The results are evaluated for two passenger travel timeframes: (i) between 7 a.m. and 9 p.m. (during daytime and favorable), and (ii) between 9 p.m. and (next day) 7 a.m. during nighttime as unfavorable. While other boundary times could have been chosen, our focus was to identify when first-/last-mile connectivity to stations is typically at its lowest in a developing country, which usually occurs after dark. We assumed this period to be from 9 p.m. to 7 a.m. Therefore, the 9 p.m.–7 a.m. timeframe was considered ‘unfavorable’ for passengers (in contrast to the 7 a.m.–9 p.m. timeframe, which was considered favorable),

mainly due to the presumed lack of first-/last-mile connectivity and safety concerns during nighttime hours. These timeframes were selected to highlight the differences in inequalities between daytime and nighttime travel. The findings shed light on any policy-related interventions that might be needed to minimize the inequalities that were observed, which can be brought about by reviewing existing train schedules, ensuring first-/last-mile access to stations at night, etc.

2. Research Contributions

Although the importance of passenger rail in achieving equity goals has been frequently highlighted in the literature, evaluating its impact due to investments in transport infrastructure remains complex. Past studies on economic agglomeration at a spatial level indicate that increased infrastructure investments and connectivity can exacerbate regional disparities. This occurs when larger cities draw more resources from newly connected smaller cities [12]. Connectivity between wealthy and poor regions does not always yield mutual benefits and can result in a ‘siphon effect’ where resources flow from small cities to large cities due to low transportation costs [26,27]. Therefore, public spending on rail transportation infrastructure aimed at benefiting all regions, especially those needing development, might backfire and lead to further marginalization due to this siphoning effect. Our research contributes to the literature on inequality and passenger rail transportation in the following four ways:

First, given the challenge of a potential siphoning effect, it is crucial to investigate to what extent passenger rail can meet equitable connectivity and accessibility goals for regions that consistently receive public funding. Our study complements the efforts of other studies that emphasize addressing transportation inequality, which, if unmet, can hinder sustainable economic growth in a region. Numerous proponents stress that integrating equity into transportation decision-making should be a priority for public agencies and policymakers [28,29]. There are multiple instances where public agencies have sought to reduce inequality by considering it a performance measure for transportation infrastructure investments, both in developed nations like the United States [30] and in developing countries such as India [22] and China [31]. Our research provides a framework to assess how effectively public investments in passenger rail can fulfill inequality goals when these efforts are prioritized.

Second, our research framework and methodology can inform regional planning initiatives by identifying areas needing focused attention through targeted transportation interventions. This will support the inequality efforts of passenger rail systems globally. For instance, this is relevant for Europe as it strives to meet its 2030 and 2050 plans and investments in the Trans-European Transport Network (TEN-T), which is proposed to ensure connectivity and accessibility of the European Railway network. Similarly, the framework for evaluation presented in this paper could be applicable to India, as the Indian Railways plans to invest USD 715 billion by 2030 in the passenger rail infrastructure [31]. Thus, our research can help ensure that these investments are effective in addressing regional disparities and achieving equitable outcomes.

Third, as we have found in our extensive review of the literature, none of the prominent studies have successfully integrated both connectivity- and accessibility-based formulations into inequality assessments that are directly linked to investments. For instance, Liu et al. [26] focus solely on incorporating centrality into inequality formulations, and Zhou et al. [32] employ a connectivity–accessibility index followed by a performance persistence analysis. However, both studies fall short in incorporating investment data or other investment proxies to thoroughly explore the relationship between institutional efforts (investments) and their outcomes (inequality). Our research stands out with a unique contribution: a newly developed inequality-to-investment ratio, complemented by a robust performance persistence analysis to identify and test high-performing (winner) and low-performing (loser) zones of interest for consistency.

Fourth, investments in rail infrastructure and operations often do not seem to adequately address inequalities. It remains uncertain whether past or future investments will effectively bridge the gap between high-performing and lower-performing operational regions of the rail over a period. This inquiry aligns with the broader goal of public transit systems, which is to understand the impact of connectivity and accessibility between different entities over time, whether they are cities [18] or zones. We showcase our framework's capability to link investments with inequality that could yield winners and losers through the example of the superfast express (SE) trains of the Indian Railways.

3. Materials and Methods

We first adopt a conventional formulation for connectivity and accessibility (two important performance measures in transit studies) by incorporating them into the inequality formulation for evaluating the performance of a typical passenger rail system. Centrality, a key topological feature in network analysis, is employed for the connectivity aspect, a concept extensively utilized in various fields like physical networks, computer science, and epidemiology [33]. While connectivity captures a network's topological characteristics, accessibility serves as another frequently employed metric for assessing passenger flow [26].

Using the two measures of connectivity and accessibility, a network's two important features, namely, the topology and the population (as a surrogate of passenger flow), were simultaneously considered. Note that past studies have carried out similar accessibility calculations with population data surrounding stations or stops rather than considering actual passenger flow or ridership [34,35]. Though various forms of these two measures exist in the literature, none we came across were sensitive to the train's scheduled arrival or departure times at a station within a network. Therefore, both connectivity and accessibility measures were first developed to be time-sensitive, reflecting the train's schedule effects.

3.1. Connectivity

The 'degree centrality' formula has been modified to develop the connectivity measure. The degree centrality of a node is the number of edges it has [36]. It is usually deployed to identify the most important node in a network and indicates the number of neighboring stations directly close to a station [37]. This means that the higher the centrality of a node, the more central and connected it is to the other nodes. In the formulation for centrality presented here, we use the centrality measure as a station's connectivity, which should be sensitive to the favorable (or unfavorable) travel times. Our connectivity formula uses the number of trains instead of the number of edges, with each train arriving at or departing from a station representing an edge.

The connectivity formula (denoted using $C_{i,z}^y$) for a node or a station i in zone z in the year y is expressed as follows:

$$C_{i,z}^y = \sum_{k=1}^K \left(\sum_{i \neq j \in N} \tau_{j,i,a}^{k,z,y} w_{ji}^{k,z,y} + \sum_{i \neq j \in N} \tau_{i,j,d}^{k,z,y} w_{ij}^{k,z,y} \right) \quad (1)$$

where

i, j = a station in a zone;

k = a train running between stations;

y = year of analysis;

z = a zone with stations;

a = train arrival;

d = train departure;

K = the number of trains;

N = number of stations being considered in the network;

$\tau_{j,i,a}^{k,z,y} = 1$ if a train k departs station j to arrive at station i (in zone z) during favorable travel timeframe in a given year y ; otherwise, it is 0 for unfavorable timeframe;

$\tau_{i,j,d}^{k,z,y} = 1$ when in a given year y a train k departs station i (in zone z) to arrive at station j during favorable travel timeframe; otherwise, it is 0 for unfavorable timeframe;
 $w_{ij}^{k,z,y} = 1$ if in a given year y there exists a train k from station i (in zone z) to station j , 0 otherwise;
 $w_{ji}^{k,z,y} = 1$, if there exists a train k from station j to station i (in zone z), 0 otherwise.

3.2. Accessibility

The accessibility used in the paper is a potential accessibility measure that has found a wide use and strong representation in the economic geography of a region, as evidenced by its use in various studies by Rotoli et al. [38] and Zhou et al. [32]. The potential accessibility measure is the state-of-the-art application of an empirically constructed measure and represents a direct relationship between population (often called ‘opportunity’) and is inversely proportional to the impedance between studied stations or zones. Accessibility derived from a transportation facility (such as rail) is often used as a representation of a region’s economic potential and attractiveness [39].

With a focus on the accessibility (or attraction) of a station in a zone, a formula is proposed for accessibility using a dummy variable $\delta_{i,j}^k$ and considering the year of analysis. The accessibility $A_{i,z}^y$ of station i in zone z in the year y is defined as follows:

$$A_{i,z}^y = \sum_{k=1}^K \sum_{j=1}^N \frac{\delta_{i,j,k}^y P_j^y}{I_{i,j,k}^y} \quad (2)$$

where

$\delta_{i,j,k}^y = 1$ if in a given year y , the arrival time of train k from station i (in zone z) to station j is within the favorable travel timeframe; otherwise, it is 0 if it is to be in the unfavorable travel timeframe;

P_j^y = the population of the city/town with station j for the year y ;

$I_{i,j,k}^y$ = the impedance in the year y between station i and station j with train $k \in K$ (total number of trains). The impedance used is the travel time between stations, with additional elaboration on its application as demonstrated in the case study;

N = the number of stations in the network.

The other common variables are defined as noted under Equation (1).

3.3. Generalized Connectivity–Accessibility Index

For a passenger, although a station with the largest connectivity would be attractive, there is a possibility that this may not be true when travel times from that station to the other stations are high since connectivity expressed in Equation (1) does not take into consideration the travel time between stations. On the other hand, accessibility—involving station/city population and travel times between stations—presents a more holistic measure of the attractiveness for rail passengers [39].

As evidenced by research, both connectivity and accessibility are equally important for enjoying the benefits of rail transportation [32]. To account for both centrality or connectivity and accessibility measures, we develop a generalized measure ($\Omega_{i,z}^y$) for a station i in zone z for the year y , expressed as follows:

$$\Omega_{i,z}^y = \begin{cases} C_{i,z}^y & \text{if connectivity is sought} \\ A_{i,z}^y & \text{if accessibility is sought} \end{cases} \quad (3)$$

Therefore, the generalized connectivity and accessibility index, Ω_z^y , for a zone (with stations) is the average of $\Omega_{i,z}^y$ of all the stations contained in that zone.

3.4. Inequality Measurement

Public transportation projects should consider equity during the planning stages, and appropriate inequality measuring instruments must be used [40]. In this paper, the inequality is measured at the regional or zonal level based on the connectivity and accessibility of all the stations in a zone. Analyzing inequality or disparity among zones at the spatial scale can reveal gaps in rail connectivity and accessibility, which can further justify and guide future zonal-level investments. Thus, historically, zones with stations that have benefited or not benefited from train service during favorable (or unfavorable) travel timeframes are identified with the disparity analysis. A disparity analysis can reflect any modifications, expansions, or adjustments needed in operating the train and expanding service connectivity among the cities within regions or zones.

In this paper, we use a modified form of the entropy index (Theil's T index) as a decomposable measure of inequality that can be disintegrated into population groups, income sources, or other dimensions [41,42]. Although there are some commonly available inequality measuring indicators (such as Gini and Lorenz curves) that are widely used in policy-related issues in transportation, the use of Theil's index is appropriate for our case as the index is much sensitive (due to its logarithmic structure) to even small variations in connectivity (and accessibility) of several regions or zones such as deployed for the case study later on the Indian Railways. While the use of other indices would also be appropriate—indeed, the Gini index produced relatively similar results for the zones—our decision to use Theil's index was driven by the need to enhance the sensitivity of the inequality outputs, particularly when two or more zones exhibited inequalities that were numerically very close.

In practice, Theil's index has been successfully used for measuring accessibility-based inequalities [43] and for measuring regional inequalities [26,44].

The modified Theil T index (T_z^y) for a zone z in year y incorporating the generalized connectivity–accessibility index of a station is expressed as follows:

$$T_z^y = \frac{1}{n_z} \sum_{i=1}^{n_z} \frac{\Omega_{i,z}^y}{\Omega_z^y} \ln \left(\frac{\Omega_{i,z}^y}{\Omega_z^y} \right) \quad (4)$$

where

n_z = number of stations in a zone z (assumption is that the total number of stations remains constant during each analyzed year y);

$\Omega_{i,z}^y$ = measure of the connectivity or accessibility of station i in zone z in year y ;

Ω_z^y = mean connectivity or accessibility for all the stations in zone z during year y .

We use the inequality formulation above to evaluate the performance of zones for connectivity and accessibility, as described in the next section.

3.5. Performance Persistence Analysis

We utilize a nonparametric method to conduct a performance assessment of a zone for inequality with connectivity and accessibility over a specified investment period. The performance analysis guides stakeholders to achieve regional and social inequality by directing investments in public transport [45–47]. The performance persistence analysis is an established approach that has been used to track the evolution of winner and loser status of analyzed entities such as zones, allowing for the evaluation of the relationship between past and future performances [33]. Zonal-level performance analysis should be conducted to understand how zones have improved in inequality over the years through policy interventions, particularly from investments. Therefore, we classify zones into winners and losers based on their performance, assessed through inequality improvement for the investment they received.

Utilizing the fundamentals of the performance persistence analysis by Agarwal and Naik [48], an inequality-to-investment ratio is developed. The inequality used here can be derived interchangeably from connectivity or accessibility using Equations (3) and (4).

The inequality-to-investment ratio ($E_z^{y-1,y}$) is defined as the percentage change in inequality of a zone z divided by the percentage change in investment from year $y - 1$ to y and is expressed as,

$$E_z^{y-1,y} = \left(\frac{T_z^{y-1} - T_z^y}{T_z^{y-1}} \right) / \left(\frac{I_z^y - I_z^{y-1}}{I_z^{y-1}} \right) \quad (5)$$

where I_z^y is the investment in zone z at the beginning of fiscal year y and T_z^y is the inequality at the end of fiscal year y .

The purpose of our assumption in providing linearly comparable improvements due to investment in Equation (5) is to simplify the decision-making process for public transport managers and stakeholders, enabling them to gather insights quickly without needing to calibrate parameters that would arise from a complex model. Such a model would precisely describe the relationship between zone inequality and investments.

The assumption with the ratio $E_z^{y-1,y}$ is that it reflects the investments at the beginning of a fiscal year y that bring about changes for inequality (which is connectivity- and accessibility-based) during the year y itself. This suggests the effectiveness of promptly utilizing rail investments to reduce disparities in specified regions or study zones.

Decision-makers typically finalize significant monetary investments in infrastructure within a zone well in advance of the start of a fiscal year. If, at the conclusion of the fiscal year, the inequality in that zone remains notably high despite the investment, it is regarded as an ineffective allocation of funds, especially in zones where reducing inequality poses significant challenges. Hence, it is crucial to evaluate zones that receive investments by identifying those that succeed and those that struggle based on this principle. We, therefore, classify zones as winners (that are high-performing and succeed) and loser zones (that fail or struggle) based on this evaluation scheme utilizing the proposed ratio $E_z^{y-1,y}$ in Equation (5) and as further illustrated in the next section.

Determining Winners and Losers

We assume that there must have been a direct or an indirect impact of investment in causing a decrease or an increase in inequality. Although it can be noted that other causal factors might impact the inequality and investment relationship, we excluded this investigation since it will digress from our current focus.

As evident from the formulation of the ratio $E_z^{y-1,y}$ in Equation (5), two cases can arise, as described below:

The first case occurs when $E_z^{y-1,y}$ is positive, which can happen under two scenarios:

1. When both $\frac{I_z^y - I_z^{y-1}}{I_z^{y-1}}$ and $\frac{T_z^{y-1} - T_z^y}{T_z^{y-1}}$ are positive. This implies that $I_z^y > I_z^{y-1}$ and $T_z^{y-1} > T_z^y$, indicating that an increase in investment has occurred while inequality has decreased from $y - 1$ to y , which is desirable by stakeholders and investors;
2. When both $\frac{I_z^y - I_z^{y-1}}{I_z^{y-1}}$ and $\frac{T_z^{y-1} - T_z^y}{T_z^{y-1}}$ are negative, meaning $I_z^y < I_z^{y-1}$ and $T_z^{y-1} < T_z^y$. This reflects a decrease in investment from $y - 1$ to y , which led to an increase in inequality, an undesirable outcome.

The second case is when $E_z^{y-1,y}$ is negative for a zone z . This can happen under two conditions:

1. $\frac{I_z^y - I_z^{y-1}}{I_z^{y-1}}$ is positive and $\frac{T_z^{y-1} - T_z^y}{T_z^{y-1}}$ is negative, indicating that investment increased from year $y - 1$ to y ($I_z^y > I_z^{y-1}$) but inequality also increased ($T_z^{y-1} < T_z^y$), which is undesirable since investment is expected to reduce inequality;
2. $\frac{I_z^y - I_z^{y-1}}{I_z^{y-1}}$ is negative while $\frac{T_z^{y-1} - T_z^y}{T_z^{y-1}}$ is positive, meaning investment decreased from $y - 1$ to y ($I_z^y < I_z^{y-1}$), yet inequality also decreased ($T_z^{y-1} > T_z^y$), an unusual outcome as reduced investment typically leads to increased inequality.

A zone is considered a ‘winner’ when the ratio $E_z^{y-1,y}$ is positive, based on the values of $\frac{I_z^y - I_z^{y-1}}{I_z^{y-1}}$ and $\frac{T_z^{y-1} - T_z^y}{T_z^{y-1}}$, as indicated by the green-shaded cells in Table 1. A winner zone is characterized by $E_z^{y-1,y} > 0$ (i.e., $T_z^{y-1} > T_z^y$ and $I_z^y > I_z^{y-1}$). Conversely, the red-shaded cells represent loser zones, where inequality increased despite changes in investment from $y - 1$ to y . A loser zone z is defined by $E_z^{y-1,y} < 0$ with $I_z^y > I_z^{y-1}$ and $T_z^{y-1} < T_z^y$ or $I_z^y < I_z^{y-1}$ and $T_z^{y-1} > T_z^y$.

Table 1. Scenarios for winner or loser zones (green shade indicating winner, red shade indicating loser, and gray shade indicating neutral).

	$\frac{I_z^y - I_z^{y-1}}{I_z^{y-1}} > 0$	$\frac{I_z^y - I_z^{y-1}}{I_z^{y-1}} < 0$
$\frac{T_z^{y-1} - T_z^y}{T_z^{y-1}} > 0$	(+)	
$\frac{T_z^{y-1} - T_z^y}{T_z^{y-1}} < 0$	(-)	(+)

In the cases presented above, a zone experiencing an increase in inequality or decrease in investment from $y - 1$ to y would indicate diminished connectivity and accessibility.

The grayed-out cell in Table 1 indicates an anomaly, as the investment has decreased from year $y - 1$ to y but has caused a decrease in inequality for the same period. Any zone falling in this classification is a neutral zone.

Once the winner/loser zones are determined, the cross-product ratio is computed using the odds ratio of the number of repeat performers, whether winners or losers, in two consecutive years. The formula for the odds ratio, $O_{y-1,y}$, is expressed as follows:

$$O_{y-1,y} = \left(\frac{WW_{y-1,y} \times LL_{y-1,y}}{WL_{y-1,y} \times LW_{y-1,y}} \right) \quad (6)$$

where

$WW_{y-1,y}$ = number of zones that are categorized as winners for two consecutive years ($y - 1$ and y);

$LL_{y-1,y}$ = number of zones that are categorized as losers for two consecutive years ($y - 1$ and y);

$WL_{y-1,y}$ = number of zones that are categorized as winners in year $y - 1$ and losers in the year y ;

$LW_{y-1,y}$ = number of zones that are categorized as losers in $y - 1$ and winners in y .

An odds ratio above 1 indicates a positive association between an outcome (e.g., being a winner or loser) in one time period and the same outcome in the subsequent period. On the other hand, an odds ratio of less than 1 indicates a negative association, which means that the outcome in the following period is more likely to be the opposite of the outcome in the previous period.

3.6. Example Calculation

A summarized version of the proposed framework is presented in the flowchart shown in Figure 1 below. Additionally, an example calculation with assumed data for two zone examples is provided in Figure 2, illustrating the steps for implementing the inequality assessment to identify winner and loser zones.

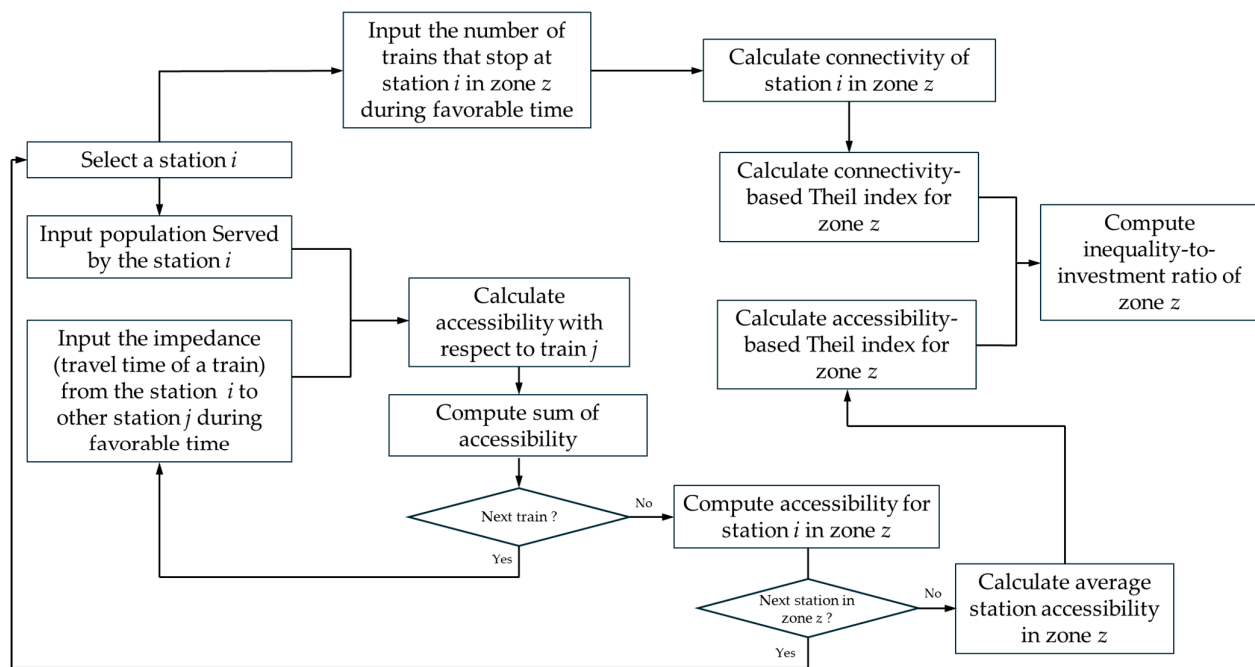


Figure 1. Flowchart for inequality calculation of a zone.

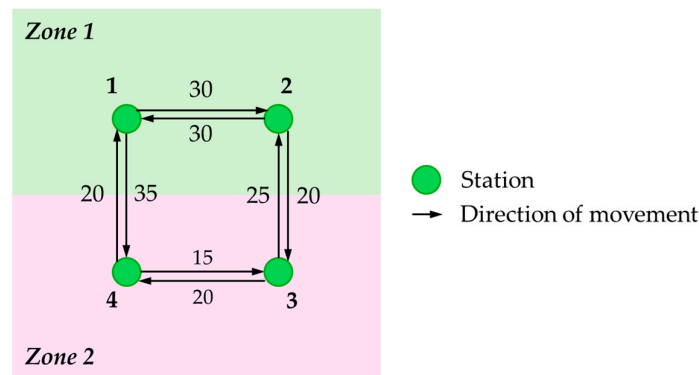


Figure 2. Setup example of inequality calculation with four stations and two zones.

Referring to the setup in Figure 2, with impedance (travel time) indicated for each direction of movement, let us consider data from two years: Year 1 and Year 2. For zones 1 and 2 during these two years, the data required to determine the winner and loser zones are as follows:

- Origin–destination travel time data in Table 2, as referenced in Figure 2;
- Frequency of trains arriving and departing from each station in Table 3;
- Population served by each station in Table 4.

Table 2. Origin–destination (OD) travel time between stations for Year 1 and Year 2.

Station	1	2	3	4
1		30	50	35
2	30		20	40
3	40	25		20
4	20	40	15	

Table 3. Frequency of trains arriving and departing for Year 1 (with data for Year 2 presented in parentheses).

Station	1	2	3	4
1		2 (3)	3 (5)	4 (6)
2	3 (5)		4 (6)	5 (7)
3	4 (6)	5 (7)		6 (8)
4	7 (9)	8 (10)	9 (11)	

Table 4. Population served by stations of trains for Year 1 and Year 2.

Station	1	2	3	4
1		2000	2000	2000
2	3000		3000	3000
3	3000	3000		3000
4	4000	4000	4000	

The subsequent calculations for the connectivity, accessibility, and inequality for the two years for the two zones are shown in Tables 5–7.

Table 5. Calculation output for Year 1.

Station	Connectivity	Normalized Connectivity	Mean Connectivity of Zone	Theil's Index with Connectivity	Accessibility	Normalized Accessibility	Mean Accessibility of Zone	Theil's Index with Accessibility
1	23	0.19			424	0.17		
2	27	0.23	0.21 (Zone 1)	0.0032	470	0.19	0.18 (Zone 1)	0.0014
3	31	0.26			732	0.30		
4	39	0.33	0.29 (Zone 2)	0.0065	795	0.33	0.32 (Zone 2)	0.0009

Table 6. Calculation output for Year 2.

Station	Connectivity	Normalized Connectivity	Mean Connectivity of Zone	Theil's Index with Connectivity	Accessibility	Normalized Accessibility	Mean Accessibility of Zone	Theil's Index with Accessibility
1	34	0.20			539	0.19		
2	38	0.23	0.22 (Zone 1)	0.0015	612	0.22	0.21 (Zone 1)	0.0020
3	43	0.26			802	0.29		
4	51	0.31	0.28 (Zone 2)	0.0036	849	0.30	0.29 (Zone 2)	0.0004

Table 7. Output with assumed investment data for zones.

	Investment (Assumed in Crore INR)		Ratio of Change in Investment	Ratio of Change in Theil's Index (Connectivity)	Ratio of Change in Theil's Index (Accessibility)	Inequality-to-Investment Ratio (Connectivity)	Inequality-to-Investment Ratio (Accessibility)
	Year 1	Year 2					
Zone 1	100	300	$(300 - 100)/100 = 2$	$(0.0032 - 0.0015)/0.0032 = 0.52$	$(0.0014 - 0.0020)/0.0014 = -0.49$	$=0.52/2 = 0.26$ (Winner)	$-0.49/2 = -0.25$ (Loser)
Zone 2	200	400	$(400 - 200)/200 = 1$	$(0.0065 - 0.0036)/0.0065 = 0.45$	$(0.0009 - 0.0004)/0.0009 = 0.53$	$=0.45/1 = 0.45$ (Winner)	$-0.53/1 = -0.25$ (Winner)

Thus, for the example shown above, although zone 1 turns out to be a winner for the connectivity measure, it is a loser for the accessibility measure. Zone 2 proves to be a winner for both measures.

4. Application: A Case Study with SE Trains in India

The passenger rail system in India is a critical component of the country's transportation infrastructure, playing a pivotal role in connecting diverse regions and fostering economic development. India's railway network has undergone significant expansion

across the nation, boasting a total route length of 67,415 km in 2019. In the fiscal year 2018–19, the Indian Railways documented nearly 8.5 billion originating passengers, signifying individuals whose journeys commenced with the train. The passenger-kilometers for suburban and suburban regions were approximately 1.2 trillion during this period [49].

Our focus in this case study is particularly on the superfast express (SE) trains of the Indian Railways. The SE trains are the long-distance premium services offered by the Indian Railways, known for higher speed and better amenities. These trains constitute almost 10% of the total trains (both suburban and non-suburban) operated by the Indian Railways [50,51] and with additional superfast charges levied over regular mail/express train charges (etrain.info, 2021 [52]). Further, these trains make fewer stops compared to their counterparts (i.e., other mail/express trains), allowing them to cover the same distance in a significantly shorter time span. However, there is a growing concern about the potential inequality in the distribution and accessibility of these SE trains and the services they provide across various zones of the Indian Railways [53].

Presently, India manages an extensive network of around 1200 SE trains, which includes renowned services such as Rajdhani Express, Duronto Express, Shatabdi Express, and various other superfast trains (etrain.info [52]). As reported by Indian Railways [12], these trains consistently maintain average speeds ranging from 85 to 95 km/hr. The average distance of travel by various SE trains ranges from 290 km to 1930 km [54] and has impacts on the sixteen zones of the Indian Railways, as indicated on the map presented in Figure 3. Note that the seventeenth zone, the Konkan Railway, was left out of this analysis as there were no available data on investment by the Indian Railways for the years of analysis.

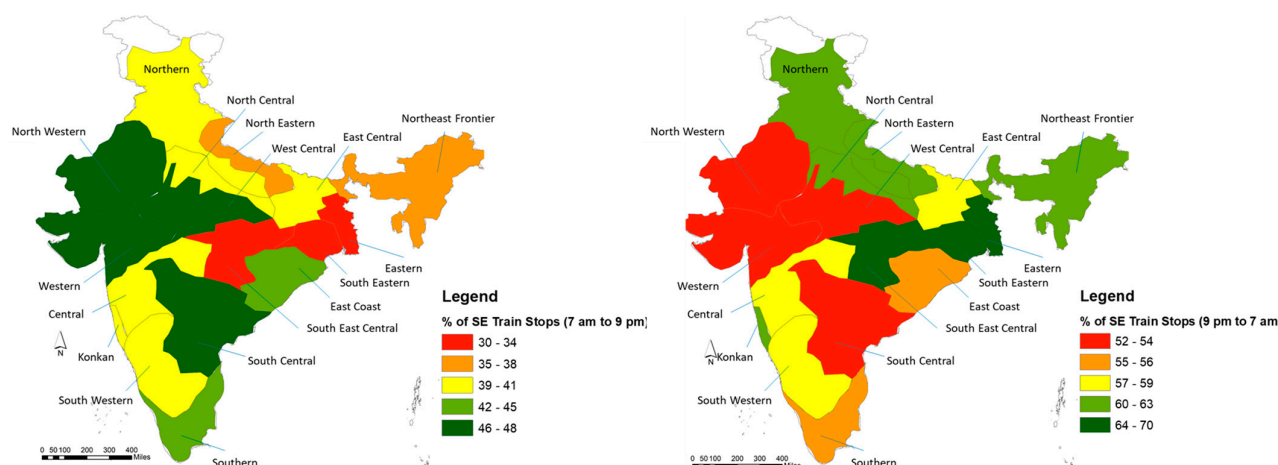


Figure 3. Percentage of SE train stops (at stations) across the zones.

Additionally, the three zones of Eastern, Southeastern, and Southeast Central were found to have the largest percentage (about 64–70%) of SE train stops during the 9 p.m.–7 a.m. time-frame. The Western, Northwestern, and South Central zones have the lowest percentage of about 52–54% SE train stops during the same travel period. Other details for all other zones are presented in the map of Figure 3.

The zones represent economically significant areas of operation for the Indian Railways, with SE trains and the network connecting the diverse socio-economic landscapes of the country within these seventeen zones. The operations of the Indian Railways in these zones continue seamlessly day and night [55]. However, our focus is only on those stations that are stops of SE trains. Thus, we study the impacts of inequality using the methodology presented in the earlier sections of this paper only for these stations for the two timeframes: 7 a.m.–9 p.m. (daytime as favorable) and 9 p.m.–7 a.m. (nighttime as unfavorable).

The subsequent section details the data collection efforts, some key findings, and outlines the assumptions made in this case study.

Data Collection: Travel Time, Population, and Investment

The spatial data of the SE train stations (as stops) and the rail network of the Indian Railways were obtained from various online sources (including [Mapcruzin.com](https://www.mapcruzin.com) [56,57]). The analysis was conducted with SE train arrival/departure data at 1620 stations with 1200 SE trains running per year (whether daily, weekly, or otherwise) and collected from 2016 to 2020. The stations served as a stop to at least one SE train operated by the Indian Railways.

Based on the data from the train timetable, the average number of SE trains that stop per station was 30 [50]. The historical data on arrival and departure were gathered from Cleartrip.com for the years analyzed. The data also contained information on delays of all the SE trains used to compute travel times between stations. The impedance needed for accessibility calculations was computed by adding the usual train travel time between stations and arrival/departure delays of the trains for 2016–2020. Thus, both the delay data and travel times in the timetables were used. There was an average delay of almost 3 h per day per zone of the Indian Railways for the years 2016 to 2020. The impedance thus obtained was used in the accessibility calculation for each station. The network with stations and rail tracks was used in connectivity calculations. The arrival and departure times were corroborated by various websites and sources that record Indian rail delays at stations, including the National Train Enquiry System of the Indian Railways [58].

Further, each station was mapped to a city it served. The city population data were aggregated to the spatial scale of the zone by extrapolating the city population from the year 2011 population data obtained from the Census Bureau of India [59]. The population data required for calculating the accessibility of stations (and zones) for the five years 2016 to 2020 were obtained by assuming an average population growth of 1% per year in India from the World Bank [60].

The zones of the Indian Railways were constructed on a geographical information system (GIS) software with the railway zone map from MapsofIndia [61], and the spatial map thus produced corroborated with the official webpage of various zones from the Indian Railways [57].

Equations (1) and (2) were utilized to calculate connectivity and accessibility for each station, respectively. Subsequently, the resulting values for each station were normalized against the highest connectivity and accessibility value among all stations across all zones. The average of these normalized values was then computed for the stations within each zone.

It was observed that the total connectivity of stations for the 7 a.m.–9 p.m. period is the largest for the Southern Railway zone. For the 9 p.m.–7 a.m. period, the Northern Railway has the largest total connectivity of the stations. The Northern Railway also has the largest total accessibility for the stations during the 7 a.m.–9 p.m. period, while the South Central Railway has the largest total accessibility for the stations during the 9 p.m.–7 a.m. period.

The values for connectivity, accessibility, and disparity remained relatively consistent across the zones in previous years. For instance, in 2020, as shown in Figure 4, the West Central Railway zone had the highest normalized connectivity during the 7 a.m.–9 p.m. timeframe, while the North Central Railway had the highest normalized connectivity during the 9 p.m.–7 a.m. timeframe. The Northern Railway zone had the greatest accessibility during the 7 a.m.–9 p.m. period, and the Eastern Railway zone led in accessibility during the 9 p.m.–7 a.m. period, as illustrated in Figure 5 for 2020.

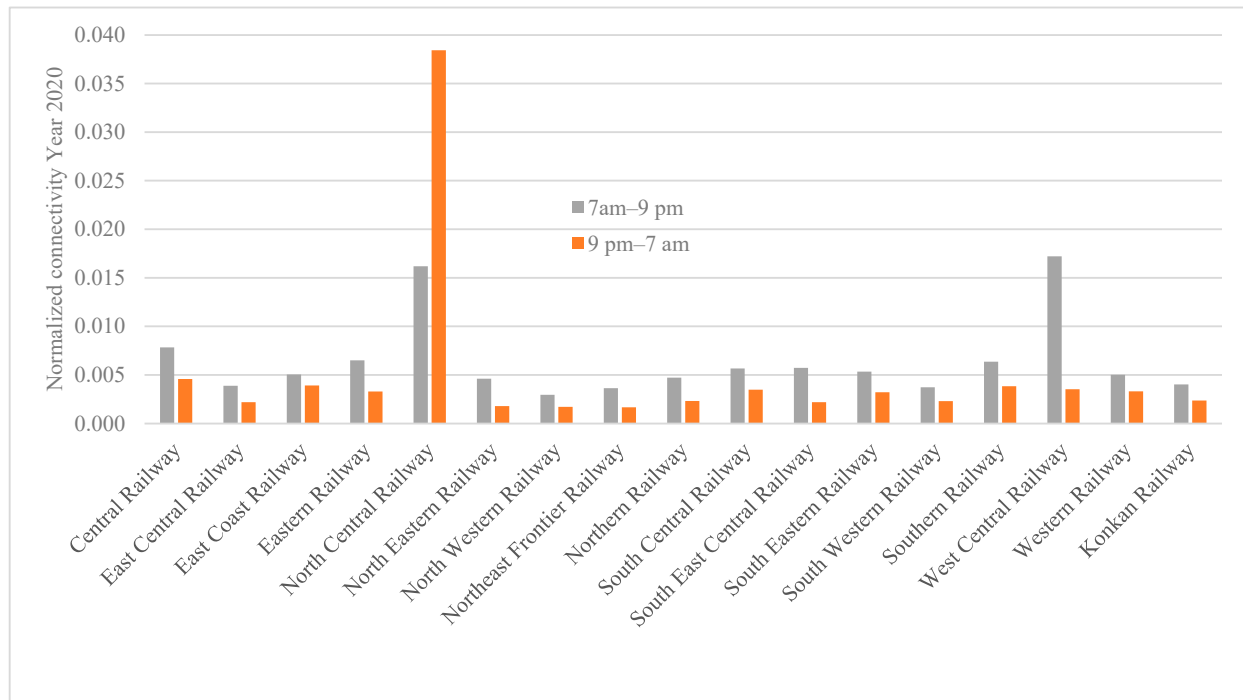


Figure 4. Year 2020 normalized connectivity for zones.

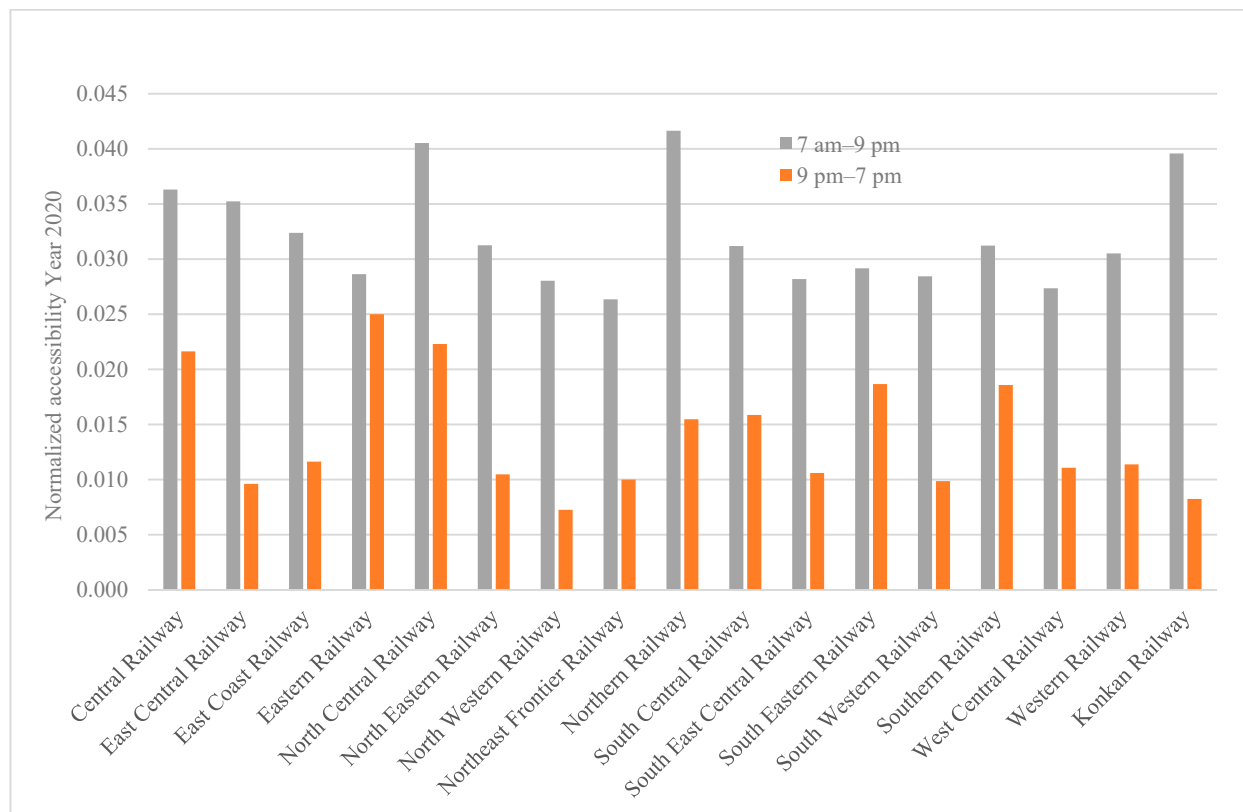


Figure 5. Year 2020 normalized accessibility for zones.

Theil's T index, previously discussed, was used to measure inequalities in both connectivity and accessibility. For example, in 2020, as shown in Figure 6, the South Central Railway zone exhibited the highest connectivity-based inequality in both the 7 a.m.–9 p.m. and 9 p.m.–7 a.m. periods, while the Southeast Central Railway zone had the highest accessibility-based inequality across both timeframes.

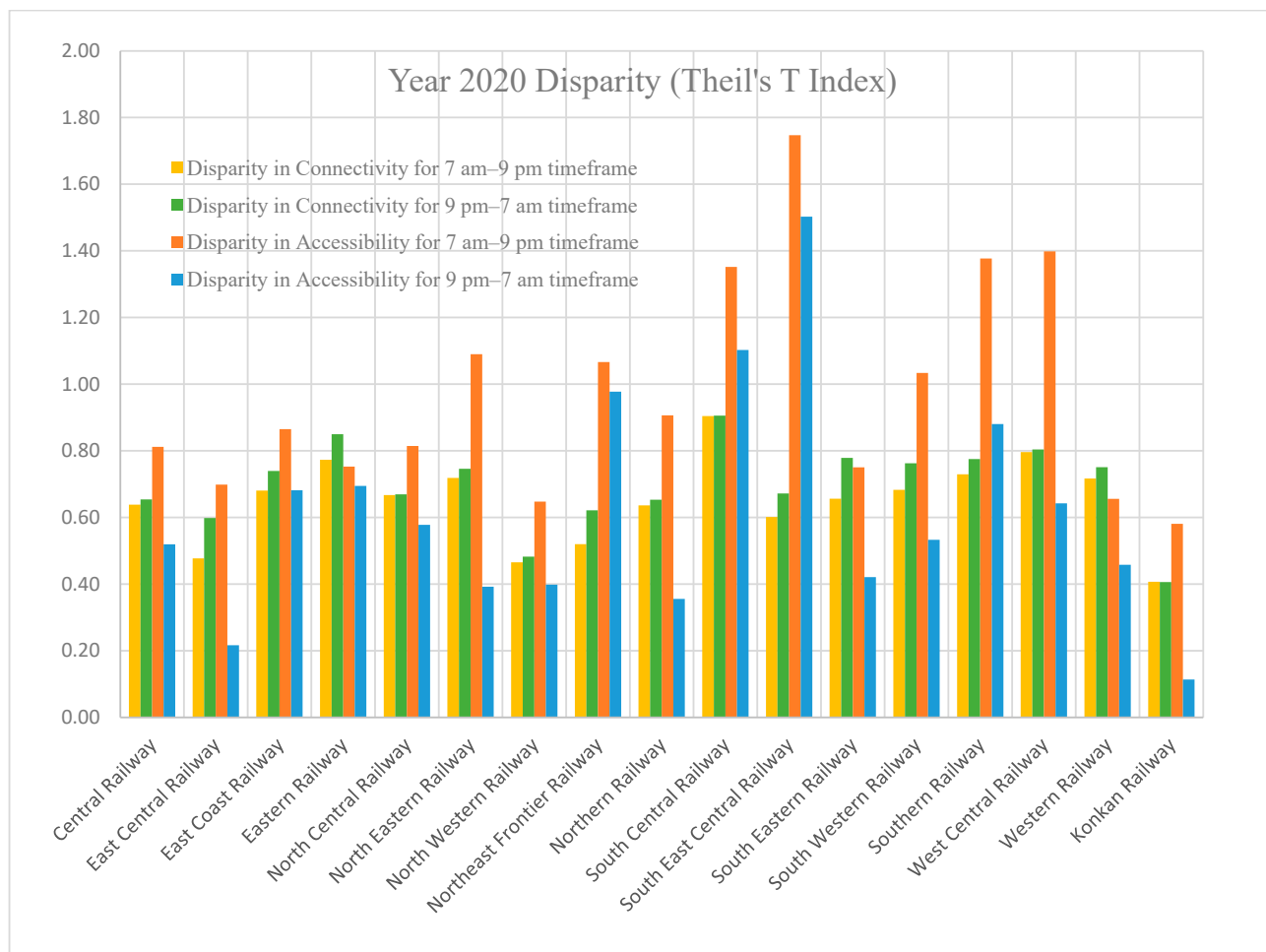


Figure 6. Year 2020 disparity with connectivity and accessibility measures.

The Indian Railways operates in cities with diverse economic backgrounds, each receiving varying levels of investment in their respective zones [31]. These cities, with their associated stations, are unevenly served by the SE trains of the Indian Railways. Figure 7 illustrates the fluctuating investment levels across zones from 2016 to 2020, consistently increasing within specific zones (data source: Indian Railways [62]). The absence of data for the Konkan Railway zone excludes its past investment information from the chart.

According to data from the Indian Railway budget, investments in each zone correlated with the passenger rail revenue generated during the same period [62]. Thus, we assumed that an increase in investment brought about an immediate increase in passenger revenue.

As per the Indian Railways publicly available financial data through budget reports, there is no distinct investment noted for the SE trains. The same rail network infrastructure, such as tracks, signal systems, etc., is used by both the SE trains and other mail/express trains of the Indian Railways. Financial reports of Indian Railways indicate that these investments primarily contribute to total working expenses, encompassing repairs and

maintenance of permanent way and works, carriages and wagons, and operating expenses such as rolling stock, equipment, traffic, and fuel [62].

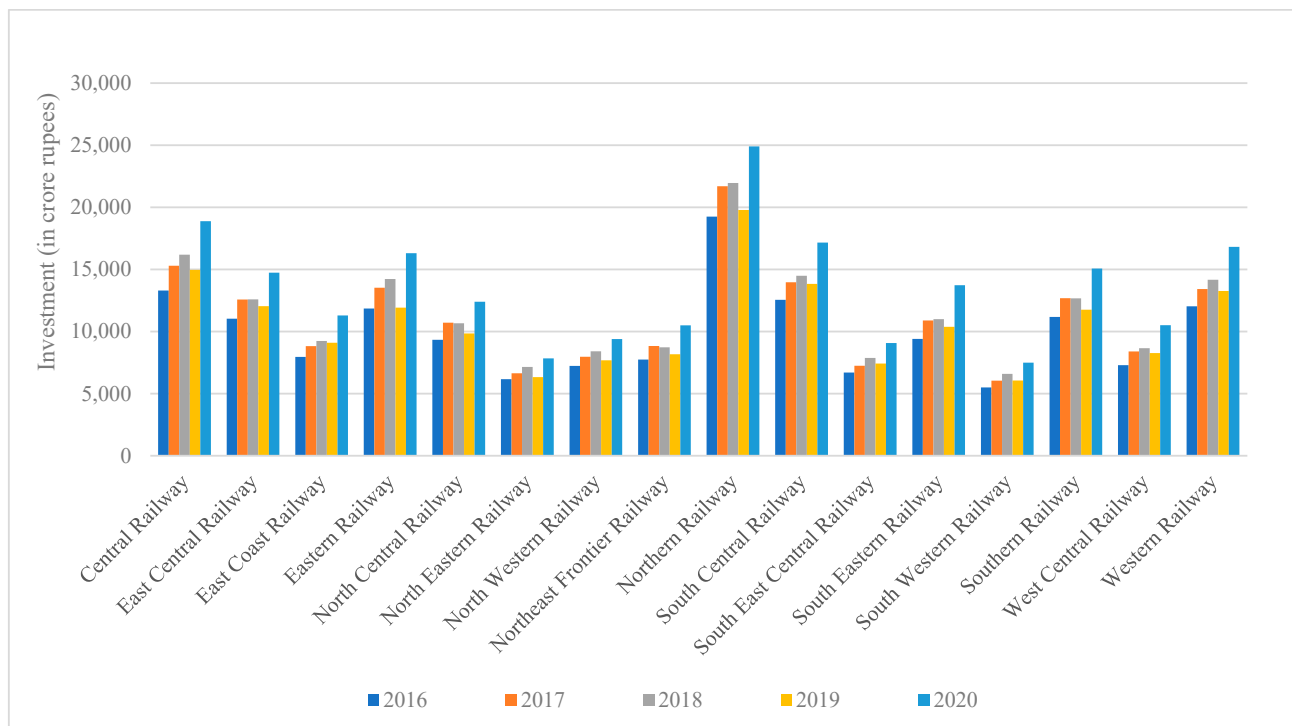


Figure 7. Zonal investment by the Indian Railways during 2016–2020.

5. Results and Discussion

We use the inequality-to-investment ratio defined earlier in Equation (5), supplemented by the evaluation rule of Table 1 presented in the earlier section, to determine zones as winners and losers.

Zones as winners and losers were determined across four analysis periods: 2016–2017, 2017–2018, 2018–2019, and 2019–2020. For the data on the sixteen zones analyzed for five years, eighty data points were obtained if considering connectivity or accessibility for each of the two timeframes, 7 a.m.–9 p.m. and 9 p.m.–7 a.m. The findings are summarized in Figure 8, with the values of the inequality-to-investment ratio presented in each cell.

Please note that in Figure 8, for simplification, the following year ranges are used: 2016–2017 represents 2016–2017, 2017–2018 represents 2017–2018, 2018–2019 represents 2018–2019, and 2019–2020 represents 2019–2020.

The analysis of the connectivity case for the 7 a.m.–9 p.m. timeframe showed that most of the zones were observed to be losers across the four analyzed periods when assessed for connectivity. The South East Central zone was a loser zone, except in 2018–2019, where it was neutral.

For the travel timeframe of 9 p.m.–7 a.m. for connectivity, most of the zones were also noted to be losers, and the number of zones as losers was the highest for the 2016–2017 period. The Northeast Frontier Railway and South East Central Railway zones were considered losers across all four periods.

Accessibility for the 7 a.m.–9 p.m. timeframe showed that almost all of the zones (except North Central, North Western, Northern, and Western) were losers during the 2016–2017 period and neutral during 2018–2019, and the majority of the zones were winners during the periods of 2017–2018 and 2019–2020.

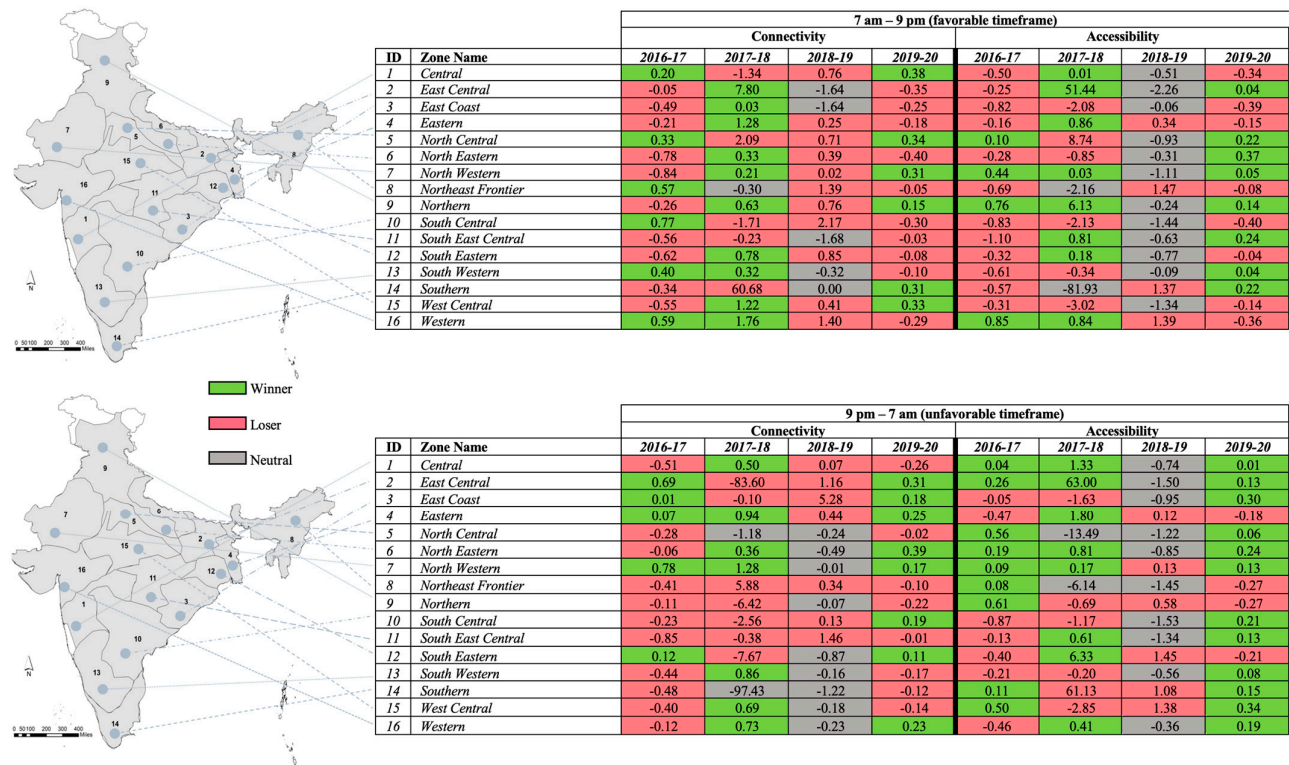


Figure 8. Zones as winner, loser, or neutral across the four periods for the two timeframes (value in cell is the inequality-to-investment ratio).

For accessibility during the 9 p.m.–7 a.m. timeframe, the majority of the zones were winners for the 2019–2020 period. Almost half the zones were winners or losers for the 2016–2017 and 2017–2018 periods. While zones are either neutral or losers for the 2018–2019 period, almost all the zones except Eastern, Northeast Frontier, South Central, and South Western were winners for the 2019–2020 period.

In summary, the findings indicate that most zones experienced an increase in inequality over this timeframe for connectivity, year after year. It can also be interpreted that the zone's connectivity, measured by the number of trains and their delays during the two timeframes, was insufficient to alleviate inequality throughout the analyzed periods.

Overall, for both connectivity and accessibility for 2018–2019 and for the two travel timeframes, none of the zones were winners. This indicates that an increase in investment over the years from 2016 to 2020 did not decrease inequality from 2018 to 2019, though other factors might have influenced this inequality increase that needs further investigation.

Comparing connectivity and accessibility, for the 7 a.m.–9 p.m. timeframe, the highest number of total ten zones repeat status as combined winners, losers, and neutral for each 2016–2017 and 2019–2020 periods. This is evident from the compilation shown in Table 8. For the same timeframe, the highest total of six zones continued as winners from the 2017–2018 period, and eight zones were repeat losers from the 2016–2017 period.

The largest number of zones with repeat status was twelve for the two timeframes (7 a.m.–9 p.m. and 9 p.m.–7 a.m.) for the accessibility measure for the period 2017–2018.

Based on the compilations shown in Table 8, it is evident that there is inconsistency in the number of zones as winners, losers, or neutral across the timeframes, periods, and the two measures. Similar to the year ranges in Figure 8, both Table 8 and the charts in Figure 9 use the following year ranges: 2016–2017 represents 2016–2017, 2017–2018 represents 2017–2018, 2018–2019 represents 2018–2019, and 2019–2020 represents 2019–2020.

Table 8. Frequency of zones repeating their status as winners, losers, or neutral across periods.

		Connectivity (7 a.m.–9 p.m.)				Accessibility (7 a.m.–9 p.m.)				Connectivity (9 p.m.–7 a.m.)				Accessibility (9 p.m.–7 a.m.)			
		2016– 2017	2017– 2018	2018– 2019	2019– 2020	2016– 2017	2017– 2018	2018– 2019	2019– 2020	2016– 2017	2017– 2018	2018– 2019	2019– 2020	2016– 2017	2017– 2018	2018– 2019	2019– 2020
Connectivity (7 a.m.–9 p.m.)	2016–2017					10(2,8,0)				5(0,5,0)							
	2017–2018						9(6,2,1)				8(6,2,0)						
	2018–2019							7(0,3,4)				6(0,4,2)					
	2019–2020								10(4,6,0)				4(1,3,0)				
Accessibility (7 a.m.–9 p.m.)	2016–2017	10(2,8,0)												9(3,6,0)			
	2017–2018		9(6,2,1)												12(7,4,1)		
	2018–2019			7(0,3,4)												10(0,2,8)	
	2019–2020				10(4,6,0)												10(7,3,0)
Connectivity (9 p.m.–7 a.m.)	2016–2017	5(0,5,0)												6(2,4,0)			
	2017–2018		8(6,2,0)												9(5,3,1)		
	2018–2019			6(0,4,2)												5(0,1,4)	
	2019–2020				4(1,3,0)												8(6,2,0)
Accessibility (9 p.m.–7 a.m.)	2016–2017					9(3,6,0)				6(2,4,0)							
	2017–2018						12(7,4,1)				9(5,3,1)						
	2018–2019							10(0,2,8)				5(0,1,4)					
	2019–2020								10(7,3,0)				8(6,2,0)				

a(b,c,d) stands for: a = number of zones repeating status. b = number of zones repeating status as winner zones. c = number of zones repeating status as loser zones. d = number of zones repeating status as neutral zones.

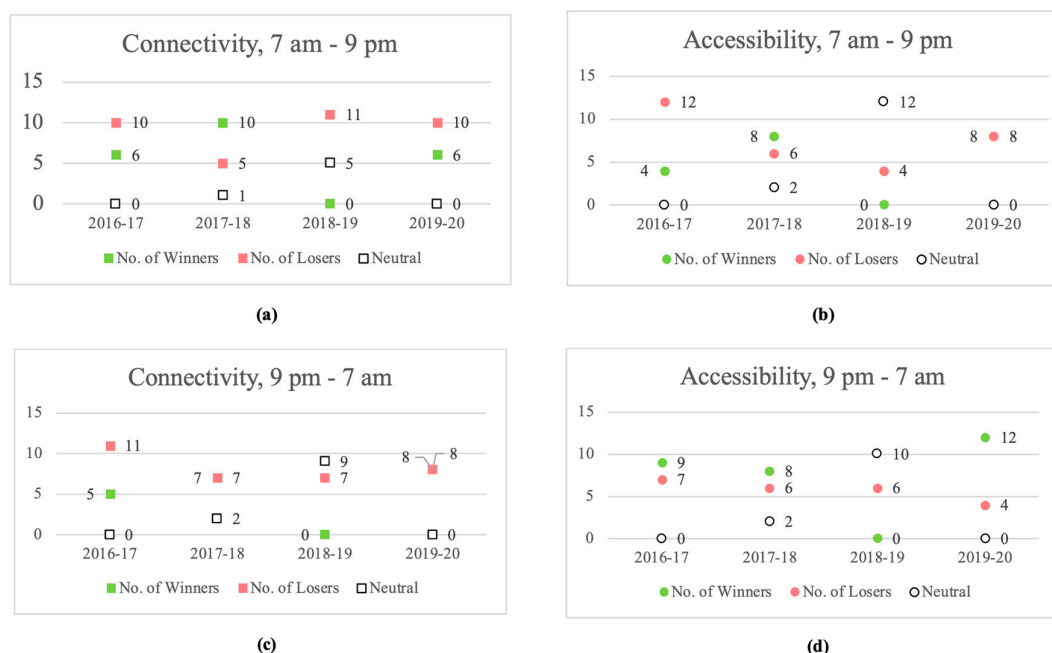


Figure 9. Count of zones as winners and losers (a) connectivity, 7 a.m.–9 p.m. (b) accessibility, 7 a.m.–9 p.m. (c) connectivity, 9 p.m.–7 a.m. (d) accessibility, 9 p.m.–7 a.m.

Essentially, the comprehensive analysis of the results presented in Table 8 yields two crucial insights. Firstly, the count of zones identified as repeat winners, losers, or neutral varies depending on the measures (be it connectivity or accessibility) and the considered timeframes. Secondly, none of the four periods consistently exhibits the same distribution of winners, losers, or neutral zones across both measures and timeframes. This variability underscores the disparities among zones and highlights the complexity involved in deducing guidance for investments, resource allocation, and the introduction or scheduling of new SE trains.

For both connectivity and accessibility, the number of loser zones was higher than the number of winners for the 7 a.m.–9 p.m. timeframe for 2016–2017. See Figure 9. On the other hand, during this same travel timeframe, the number of zones as winners was higher than the number of loser zones during 2017–2018 for connectivity and accessibility.

It was found for the 9 p.m.–7 a.m. timeframe for connectivity that the number of loser zones was higher than the number of winner zones compared to accessibility; this was observed for 2016–2017, 2017–2018, and 2019–2020. Therefore, if accessibility was considered the basis of analysis, the disparity among zones seemed to have reduced, possibly due to service improvements of SE trains. This could have occurred due to their reduced delays or other means of improving efficiency and reliability through upgrades in the rail infrastructure with investments by the Indian Railways for the respective zones.

An aggregate level of analysis was conducted using the odds ratio, signifying the performance persistency of zones as winners and losers from one analyzed period to another. A ratio greater than 1 meant that a zone that was a loser in one period had remained a loser during the next period with a 50% probability, thus promoting inequality. On the contrary, the ratios below 1 meant that a zone that was a winner in a period was a loser in the next period with a probability above 50%, which also promoted inequality.

With reference to the information above, odd ratios are computed and compiled as shown in Tables 9 and 10. It was observed that for the 9 p.m.–7 a.m. timeframe for connectivity, a zone that was a loser zone continued to be a loser zone for the periods 2017–2018 to 2018–2019 and 2018–2019 to 2019–2020 for the 7 a.m.–9 p.m. timeframe. For accessibility, however, zones that were losers continued to be losers for all the periods when

considering the 9 p.m.–7 a.m. timeframe. Therefore, these findings indicate that disparities among zones continued with a 50% probability when considering the number of loser zones presented in Figure 8, in which case, for accessibility, a significant number of loser zones continued to be the loser from 2016 to 2020 during 9 p.m.–7 a.m. timeframe.

Table 9. Cross-product ratios for connectivity.

Timeframe	2016–2017 to 2017–2018	2017–2018 to 2018–2019	2018–2019 to 2019–2020
9 p.m.–7 a.m.	0.58	2.20	0.78
7 a.m.–9 p.m.	0.21	0.47	1.18

Table 10. Cross-product ratios for accessibility.

Timeframe	2016–2017 to 2017–2018	2017–2018 to 2018–2019	2018–2019 to 2019–2020
9 p.m.–7 a.m.	1.11	1.01	1.2
7 a.m.–9 p.m.	2.33	0.20	2.33

Exclusively for the observation from the period 2016–2017 to 2017–2018, the ratios were less than 1 for connectivity and higher than 1 for accessibility. If connectivity was used to assess inequality, a zone that is a loser had transformed into a winner zone from 2016–2017 to 2017–2018 with a 50% probability, thereby improving inequality. On the other hand, with an assessment based on accessibility, a zone had remained a loser zone from 2016–2017 to 2017–2018, aggravating the disparity. Thus, the choice of the performance measure, whether connectivity or accessibility, matters in determining the success of policy-related interventions (such as investments) in improving zone disparities.

6. Conclusions

In this research, a framework for evaluating inequality is presented for investment received for zones within which passenger rails operate. The inequality is advertently created when trains stop during unfavorable timeframes, dictated by the trains' arrivals and departures schedules, such as during nighttime when the station lacks first-/last-mile connectivity or safety considerations around the station are important. Using an inequality-to-investment ratio developed, winners and losers over a given period of time are determined.

A case study was carried out with the superfast express (SE) trains of the Indian Railways for the favorable (7 a.m.–9 p.m.) and unfavorable (9 p.m.–7 a.m.) timeframes of the passenger travel. It was found that zone investments, when accounting for inequalities, created a variable number of zones as winners, losers, or neutral across the timeframes, evaluation periods, and the two measures: connectivity and accessibility. All the zones were either losers or neutrals during the 2018–2019 period for either of the two timeframes and for connectivity and accessibility measures. Specific outputs showed that for accessibility, the 9 p.m.–7 a.m. timeframe saw more winners than losers during both the 2016–2017 and 2019–2020 periods. A very high number of zones were losers than winners for accessibility during the 2016–2017 period for the 7 a.m.–9 p.m. period.

Therefore, for the investments that did not diminish the inequality of a zone and inadvertently created a loser zone, it can be inferred that investments must be increased or reconsidered to increase connectivity and accessibility for specific zones.

It is further suggested that any possible schedule changes in the SE train in the Indian Railways zones that are marginalized (less strategically served) be made to address their connectivity and accessibility during 'unfavorable' travel hours of 9 p.m.–7 a.m. Furthermore, appropriate investments and provisions for first-/last-mile transport around stations during unfavorable timeframes should be made for zones to reduce their inequalities compared to other zones. Otherwise, the zones already receiving significant investment

and being well connected and accessible with passenger rails will continue to experience economic prosperity. In contrast, zones with SE train arrivals and departure schedules mostly during ‘unfavorable’ times might not experience the desired economic growth by SE trains connecting to India’s megacities.

The case study results and interpretations of the outcome are mainly applicable to the zones of the Indian Railways analyzed, as the investment data were only available and utilized at this spatial scale. However, the framework presented in this research is generic because of its simple approach and could be applied to evaluate inequalities for any entity, whether city, state, or region receiving investment or government-led funds. Therefore, using the inequality evaluation framework presented in this paper to identify the winning and losing entities through the simplified approach proposed could be very useful. It would provide first-hand information to agencies for infrastructure development, scheduling, and resource allocation, ultimately creating a more balanced and sustainable railway system.

Our future research will focus on applying the framework outlined in this paper to evaluate the inequality across the entire Indian passenger rail mobility system, encompassing SE trains and all other mail/express trains operated by Indian Railways.

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Exhibit SR-4

Connectivity Evaluations

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Connectivity evaluations of Indian Railways zones with superfast express (SE) trains

Shailesh Chandra^{a,*}, Sundaravalli Narayanaswami^b, R. Thirumaleswara Naik^c, Vivek Mishra^d

^aAssociate Professor, California State University Long Beach, USA

^bAssociate Professor, Indian Institute of Management Ahmedabad, India

^cSenior Scientific Officer, Indian Institute of Science Bangalore, India

^dGraduate Student, California State University Long Beach, USA

Abstract

A high percentage of rail ridership in India is facilitated by the superfast express (SE) trains that operate across and within the 17 zones of the Indian Railways. While large cities with SE train stations in zones have some form of first/last-mile transport connectivity and considered safe to access during nighttime or early wee hours, several other stations do not have this privilege. Therefore, considering the 9 pm – 7 am period as not so friendly hours of travel, and the 7 am – 9 pm period as the desired hours of travel, analysis shows that the inequalities in centrality and accessibility exist between the zones more than between the stations within a zone for these two periods. Further, it was found that the two zones of the Northern Railway and the Southern Railway have the largest percentage of stations with deterioration in centrality for the 7 am – 9 pm period compared to the 24-hour period. If evaluated for accessibility, there were five zones of the South Western Railway, Southern Railway, East Coast Railway, South Eastern Railway, and Northern Railway that would experience deterioration in accessibility for the same desired hours of travel. It was also found that the investment made by the Indian Railways across the 17 zones between 2016 - 2020 resulted in positively impacting the total employment growth in the zones, and it was significant. The mean accessibility of stations in a zone, served by SE trains, had a significant positive impact on the zonal job opportunities during the 7 am – 9 pm period. The mean accessibility for the 9 pm – 7 am period showed a significant negative impact on job opportunities. These findings suggest that the Indian Railways should continue investing and improving zonal accessibility with new SE trains, particularly during the 7 am – 9 pm period, to spur economic growth in the zones. In addition, such investments with new SE trains should be appropriately planned to reduce disparity among zones.

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Keywords: Indian Railways, inequality, centrality, accessibility, first/last mile, rail.

1. Introduction and Background

Passenger rail connects spatially far apart places and plays a vital role in attenuating regional disparities (Fraszczyk et al., 2016). Rail as a transit system can address social equity issues and justice by allowing transport access to those who are poor or in more disadvantaged positions (Lu, 2018). Rail has been known to reduce per capita travel costs and

*Corresponding author. Tel.: +1-562-985-4654

E-mail address: shailesh.chandra@csulb.edu

boost economic growth both in developed nations such as the United States of America (Litman, 2015) and developing nations such as India (Pradhan and Bagchi, 2013). Numerous studies underline the economic development benefits of rail transportation. Economic development can assume various forms, including business growth, employment, and income increase. Rail attempts to achieve these forms due to the connectivity and access it facilitates between two large cities (Litman, 2015). However, this is not always the case. Past studies on economic agglomeration on a spatial scale indicate that an increase in regional disparities can result from an increase in infrastructure investments and connectivity – this occurs when the more prominent cities can now draw on increased resources from the new connected smaller cities (Fujita and Thisse, 1996). Thus, connectivity between rich and poor regions may not always work out to be as mutually beneficial (Liu et al., 2020).

The superfast express (SE) trains in India are a unique candidate for study. They connect the country's vast socio-economic landscapes and operate day and night (Ravi, 2019). Some SE trains are always crowded and marred with delays but are quite popular in India (Kumar et al. 2018; India.com, 2020). However, very little is known on the quality of connectivity provided by the SE trains when compared for day versus night service (i.e., the odd travel hours). This is important as policy decisions on investments in rail network expansions/improvements can occur accordingly to minimize regional disparities through SE train connectivity. Recent studies have focused on understanding the regional disparity (or integration) and economic impacts of the rail to a great extent – however, in the context of high-speed rails (Liu et al., 2020; Chandra and Vadali, 2014). The same for the SE trains has seldom been carried out for the role they may play (or maybe not?) in improving the socio-economic disparities in developing nations like India.

This paper evaluates the service provided by the SE trains across the 17 zones of the Indian Railways using a new formula for station connectivity evaluations for the two periods - the standard 24-hour period and the odd travel hour period (involving nighttime to early wee hours). The analysis is carried out to understand a zone's contribution to job opportunities from the Indian Railways investment from 2016 to 2020.

The rest of the paper is structured as follows: Section 2 provides a review of pertinent literature. Section 3 presents the methodology adopted for SE train service network connectivity and procedure for evaluating the spillover effects of investment and accessibility for zones. Section 4 presents the analysis and findings on connectivity and zonal disparities. Section 5 discusses investment policy and other connectivity implications in zones, and section 5 concludes the research.

2. Literature Review

Respective rail authorities of various nations have undertaken improvements through investments in passenger rail with the primary goal of providing alternatives to highway travel and improving mobility (Wenner et al., 2020; Murakami and Cervero; Sitharamaraju et al., 2020). However, these improvements should always be in-sync with passenger expectations and perceptions of the rail system's quality of service (such as reduced delays and comfort) (Johnson, 2020; Choudhary et al., 2021). Eliminating delays by increasing rail speeds is a common step to improving service that can have varied impacts on intra- and inter-agglomeration mobility (Romashina, 2021). One of the motivations for improving rail service is to attract riders to the transit and sustain economic growth for a region (Chandra and Mazin, 2020).

Recently various surrogates of economic growth “potential” have been studied in the literature for rail transportation – but those that are scholarly have been limited mainly to high-speed rail. For example, the growth potential used in the context of internal accessibility, also known as ‘self-potential’, is significant for regions or zones with high employment and population served by the rail – again for high-speed trains (Chandra and Vadali, 2014). Despite the best efforts, we could not find a single study that is comprehensive enough to discuss the economic growth potential from the semi-high speed and express trains with lower speeds than the high-speed rail.

Similarly, on the other extreme, literature exists that assesses passenger rails' economic potential, such as light rail with a lower speed than semi-high speed and express trains. In one example study, Fan et al. (2012) found that accessibility increased among low-wage workers compared to medium and high-wage workers after implementing the light rail line service. Furthermore, the accessibility gains were spatially predominant in areas that exhibited the rail connected with bus routes, indicating that last-mile connectivity with bus service plays a vital role in rail success.

2.1. Connectivity Measures in Practice

Graph theory has been widely used in devising measures for transit connectivity. For example, Kaplan et al. (2014) used transit connectivity that considered various measures, including in-vehicle time, access/egress times, waiting time, service reliability, frequency, and ‘seamless’ transfers along multi-modal paths. Other studies have considered a host of connectivity measures such as node connectivity and line connectivity (Mishra et al., 2012). However, none of these and other studies that exist are targeted to assess service quality with specific time of the day considerations of the service accounting for first/last mile connectivity convenience in rail transportation.

The importance of rail connectivity in facilitating a satisfactory level of performance and service has been well acknowledged. For example, a study by Zhu et al. (2018) highlighted the significant contribution of train service in connectivity for 19 of the 20 top-ranking domestic routes in China, and Xu et al. (2019) noted rail connectivity to stations as critical to performance assessment of rail transportation.

One of the popular measures of network connectivity is centrality, generally used in the field of network theory, e.g., in physical networks, computer science, epidemiology etc. (Martínez et al., 2003; Liu et al., 2005; Bell et al., 1999; Junker et al., 2006; Guimerà et al., 2005). Although other network connectivity measures are also used in public transportation, centrality has only recently been proposed (Mishra et al., 2012). To measure centrality – often called the topological feature of a network – the “accessibility” measure has also been used (Wangtu et al., 2018; Liu et al., 2020). The premise is that focusing only on the topology of a network isn’t sufficient and passenger flow has to be considered in describing the appropriate connectivity measure for a network (Sun et al., 2018). In the context of rail, most studies we came across analyzed the impact of high-speed rail in identifying the winner or loser cities with accessibility (Moyano et al., 2018). However, minimal understanding exists on the centrality or accessibility impacts of SE trains on the cities and regions they serve.

Accessibility in public transport has been studied in forms, such as location and potential (Geurs and van Wee, 2004). The location-based accessibility is the simplest form expressed as travel time, distance, or cost requirements to access a destination (Karou and Hull, 2014). The potential-based accessibility measure accounts for the transportation network’s ability to facilitate access to a population or workforce (Saghapour et al., 2016; Chandra and Vadali, 2013). Most research adopts either of these two forms of accessibility for rail.

2.2. Social Equity and Disparity with Rail

Transit accessibility as a connectivity measure is usually used as a yardstick to measure transport equity (Ermagun and Tilahun, 2020). Spatial impact analysis can be used to evaluate equity implications derived from changes to the distribution of accessibility values in the specific urban agglomerations of the rail projects (Monzón, 2013). Proximity to population centers, the existence of other rail stations, and the quality of the overall transportation system impact these implications. With the service offered to high-density populations, rail services with high speeds are usually recommended for cities that serve as regional core (Martin, 1997). In contrast, towns outside such high-speed rail service suffer spatial disparities but benefit from SE trains by providing transport accessibility within lower-density areas (Kim and Sultana, 2015). Studies show that population groups have been at the helm of equity-related studies resulting from accessibility, i.e., accessibility has been studied in terms of its distribution among population groups by assessing the spatial equity of transport infrastructure projects (Bröcker et al., 2010).

Transportation equity is often variedly related and labeled by names such as justice and fairness (Bröcker et al., 2004; Monzón et al., 2013). An equitable transit system is often considered attractive to a more extensive society and for a sustained level of ridership. One of the efforts in research has been to measure public transport equity from a graph-theoretical approach (Welch and Mishra, 2013). In this paper, we adopt a similar approach to identify inequalities in connectivity across zones demarcated by the rail networks of the Indian Railways.

Policies that drive infrastructure investments in a city should reduce inequalities, which motivates disparity analysis (Liu et al., 2020). Connectivity as centrality and accessibility provided by rail transportation brings about economic changes, and it often requires policy interventions to distribute the economic developments across cities connected via stations appropriately. Three commonly used measures can be deployed for disparity evaluation, and these include dispersion indices such as coefficient of variation (Gutierrez, 2001), Gini coefficient based on Lorenz curve (Kim, 2000), and entropy index such as the Thiel’s index (Chen and Haynes, 2017).

The SE trains of the Indian Railways connect cities of varied population sizes and thus, facilitate the distribution of economic activities. This also results in uneven gains in growth for cities, thereby creating inequalities or disparities among them. Thus, an assessment of disparities helps determine appropriate policies needed to support cities to prosper economically and not be left behind from taking advantage of the benefits resulting from rail connectivity.

2.3. Rail Transportation in India

India's first documented rail operation began in 1853 (IR History, 2020), and since then, the railway network has expanded throughout the country, with 67,415 kilometers of route lengths in 2019. During 2018-19, the Indian Railways had recorded almost 8.5 billion passengers originating (i.e., passengers with their trip beginning with the train). The passenger-kilometers for suburban and sub-urban regions have been around 1.2 trillion (Ministry of Railways, 2020). The upper-class segment constituted about 179 million passengers originating and 1.26 billion passenger-kilometers. The cars in this class are primarily two-tiered sleeping or seating chairs found in most long-distance SE trains. There are currently around 1,200 such trains operating in India, including the Rajdhani Express, Duronto Express, Shatabdi Express, and other superfast trains of the Indian Railways (etrain.info, 2020). The average speeds of these trains are 85 - 95 km/hr (Indian Railways, 2020). Twenty-four pairs of Rajdhani Express trains operate in India and connect New Delhi, the capital of India, to other state capital cities of the country (Indianrailinfo, 2020). About 25 pairs of operational Duronto Express trains are known for their non-stop services between India's major cities and state capitals (etrain.info, 2020). The Duronto Express trains and the Rajdhani Express trains are similar in terms of fare and priority in operations. The Shatabdi Express trains are also superfast trains, and there are currently 25 of these operating between major cities – out of those, 15 trains originate from New Delhi (goibibo.com, 2020). These trains are mostly day-trains, i.e., they complete their journey on the same day after leaving their originating stations. Superfast trains have a higher number of halts, and the fare is lower than the three express trains of Shatabdi, Rajdhani, and Duronto. As per a report from the Indian Railway Year Book 2018-19, although the upper-class segment had a tiny percentage of share for passenger-kilometers, the revenue in earnings per passenger-kilometer was the highest from the upper-class (Indian Railways, 2020). There is an estimated investment of US\$ 715 billion through various projects in Indian railways by 2030 (IBEF, 2021), as well as investment from privately operated train services such as the Delhi-Lucknow Tejas Express (PTI, 2019).

In this paper, we examine the regional disparity in connectivity of the zones that divide the Indian rail service network of the Indian Railways. The seventeen Indian Railway zones are studied for connectivity using degree centrality and accessibility - both employed as indicators of economic potential. The zones studied are namely, the Central Railway, East Central Railway, East Coast Railway, Eastern Railway, North Central Railway, North Eastern Railway, North Western Railway, Northeast Frontier Railway, Northern Railway, South Central Railway, South East Central Railway, South Eastern Railway, South Western Railway, Southern Railway, West Central Railway, Western Railway, and the Konkan Railway. These zones are assumed to be economically significant regions of activity for the Indian Railways resulting from 1.2 trillion passenger-kilometers across SE train stations in India.

3. Methodology

A generic form of connectivity using degree centrality and accessibility - both employed as indicators of economic potential – have been proposed in this paper. However, a basic understanding of the rail transportation network has been developed first and discussed below.

3.1. Network Topology

The transportation network for rail can be divided into two parts – the physical network consisting of the rail tracks and the service network that comprises trains that service passengers across the network. While the rail tracks represent a fixed form of connectivity offered across stations, the service network indicates where passenger flows occur. In this context, a general form of rail infrastructure and service network can be defined based on the work by Li and Rong (2020). The rail network can be represented as a graph, $G = (V, T, S)$ where V is the stations of SE trains, T is the links or rail lines, with travel time information, used by SE trains. Subsequent details of the networks are discussed below.

3.1.1. Infrastructure Network

The infrastructure network is formed by enumerating all existing links of graph G with V stations and T links. In other words, the infrastructure network is formed by links between any two stations of the rail network connected by tracks. Fig. 1 shows the infrastructure network created by six stations A to F. The infrastructure network is independent of the number of trains serving these stations.

3.1.2. Travel Time Network

The travel time network represents travel time as the weight for each link between stations served by a train (see Fig. 1). It is assumed that this network has equal travel time in each direction for a link. The general representation of the travel time of a link is T_{ij}^k where k stands for the train number operating between stations i and j .

3.1.3. Service Network

All the links among stations V that are served by train(s) form the service network. The S links of the SE train service network are divided into parts S_I and S_{II} - representing Part-I and Part-II service networks (see Fig. 1). The Part-I service network is formed when the departure time of train k at the origin station i or the train k 's arrival time at destination j is within the desired travel periods. On the other hand, in Part-II, the arrival or the departure time of the train k at the origin station i (or the destination station j) is during the odd hours of travel that may not be convenient to passengers. Thus, travel in the Part-II service network is *not* considered friendly from a passenger point of view. For example, under Part-II, a nighttime or early morning hour travel (as odd travel times) might not be perceived safe by a passenger due to dark or a lack of first/last mile connectivity to stations.

Therefore, based on the above narrative, for a dummy variable, say $t_{i,j}^k = 1$ as in Part-I, it would indicate train k that serves the link connecting station i to j has an arrival or departure time at i (or at j) that is during the friendly travel periods. By default, $t_{i,j}^k = 0$ in Part-II service network.

From the network throughput perspective, entities managing the above-discussed networks aim to utilize as many links from the infrastructure network with as many trains as possible to create more extensive service networks. Policies that allow such efforts are generally encouraged, although operational budget constraints limit those efforts.

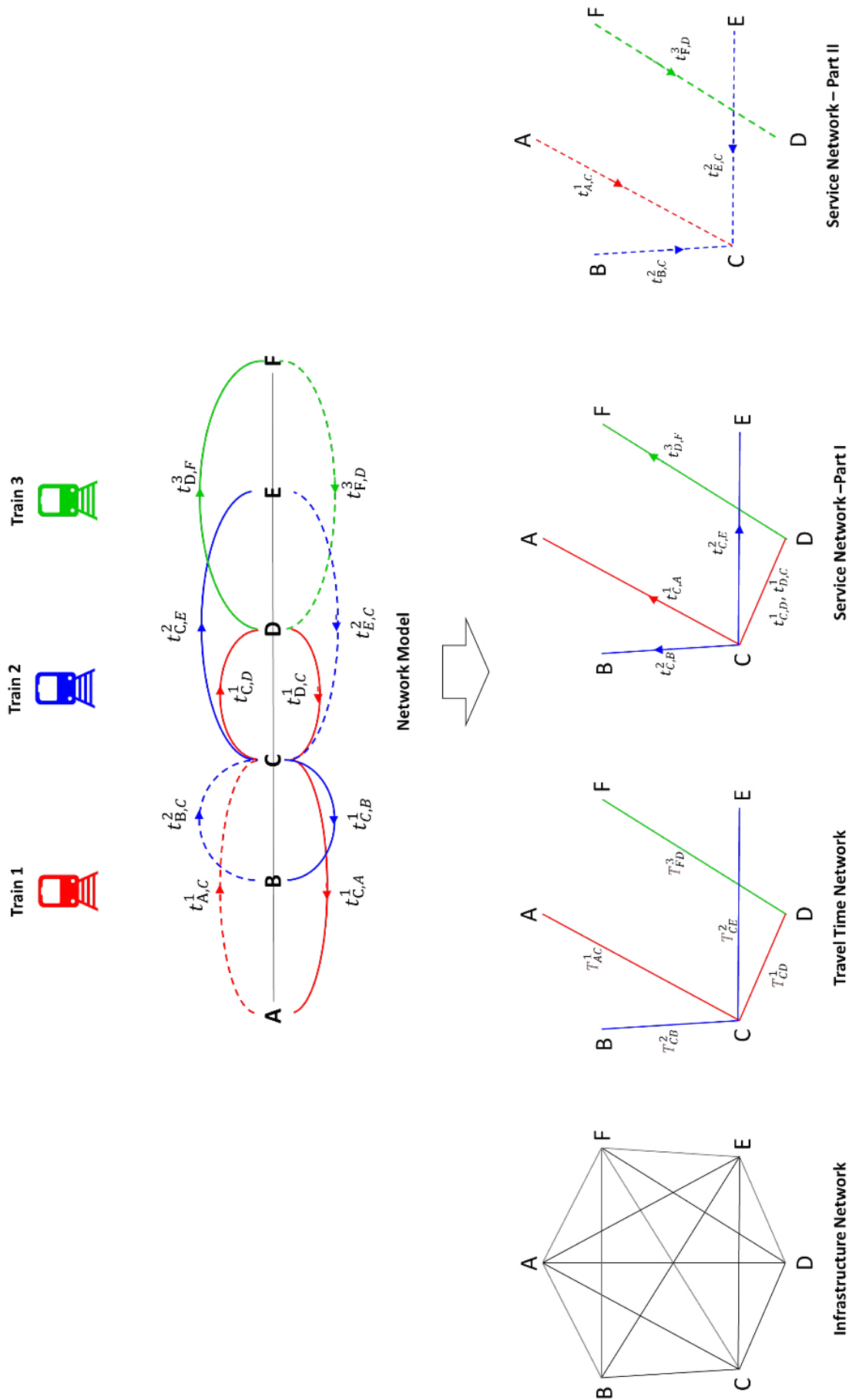


Figure 1: Network representations of the three-train service example

3.2. Connectivity

A formulation for connectivity is proposed for the two service networks discussed above. We use centrality and accessibility as the primary measures of connectivity of the service network.

3.2.1. Centrality

A modified form of “degree centrality” measure has been used in this research. The measure serves as the indicator of connectivity for stations and extends for zones that contain these stations. During late night to early morning wee travel hours, travel by rail becomes unfriendly especially due to lack of an adequate first/last mile transport connectivity or safety concerns of rail passengers at dark. Therefore, an appropriate degree centrality formula (C_i) of a station i is required to be developed for Part II railway service network shown in Fig. 1 and is expressed as:

$$C_i = \sum_{i \neq j \in N} t_{i,j,a}^k w_{ij}^k + \sum_{i \neq j \in N} t_{j,i,d}^k w_{ji}^k \quad (1)$$

where,

$$t_{i,j,a}^k = \begin{cases} 1, & \text{if a train } k \text{ departs station } i \text{ during desirable / friendly periods to reach station } j \\ 0, & \text{otherwise} \end{cases}$$

$$t_{j,i,d}^k = \begin{cases} 1, & \text{if a train } k \text{ departs station } j \text{ to arrive at station } i \text{ during desirable / friendly periods} \\ 0, & \text{otherwise} \end{cases}$$

$$w_{ij}^k = 1, \text{ if there exists a train } k \text{ from station } i \text{ to station } j, \text{ otherwise } 0.$$

$$w_{ji}^k = 1, \text{ if there exists a train } k \text{ from station } j \text{ to station } i, \text{ otherwise } 0.$$

3.2.2. Accessibility

Accessibility is often used as an indicator for determining economic potential (Chandra and Vadali, 2014), and in this research, it is regarded as the attraction potential of a railway station – because the higher the accessibility, the higher is the attraction of passengers towards the station. Thus, the following formulation for accessibility of station i , A_i , is proposed:

$$A_i = \sum_{k=1}^K \sum_{j=1}^J \frac{\delta_{i,j,k} P_j}{I_{i,j,k}} \quad (2)$$

where,

P_j = population of the city/town that has the railway station j ,

J = total number of stations,

K = total number of trains being analyzed, and

$I_{i,j,k}$ = impedance (travel time, distance etc.) between station i and station j with train k , and

a dummy variable $\delta_{i,j,k} = \begin{cases} 1 & x < T_{i,j,k} < y \\ 0 & \text{otherwise} \end{cases}$. The time $T_{i,j,k}$ is the arrival time of train k from station i to station j .

If the train k arrives at station j anytime between desired/friendly time intervals of x and y , the dummy variable $\delta_{i,j,k}$ assumes the value of 1; otherwise, it is 0. The times x and y for a station based on the safety and convenience of passenger arrival/departure at a railway station. Convenience could include first/last mile transport to access the station and safety is ensured during particular time intervals of travel to access the station.

As an example, if x is 7 (i.e. 7 am) and y is 21 (i.e. 9 pm), a train k that arrives at station i from station j between any time after 7 am and 9 pm, the dummy variable $\delta_{i,j,k}$ is 1 and the train contributes to increasing the

accessibility of station i . For this same example, if the train k departs from station j and arrives at station i after 9 pm and before 7 am of the next day, the dummy variable $\delta_{i,j,k}$ is 0 and the train k would not contribute to the accessibility of station i . Stations that have a high number of SE train stops (arrival or departure) during the unfriendly travel periods makes those stations not so attractive for passenger travel. This could have a negative impact on the economic growth potential of the cities which the stations served. All the links among stations V that are served by train(s) form the service network. The S links of the SE train service network are divided into parts SI and SII - representing Part-I and Part-II service networks (see Fig. 1). The Part-I service network is formed when the departure time of train k at the origin station i or the train k 's arrival time at destination j is within the desired travel periods. On the other hand, in Part-II, the arrival or the departure time of the train k at the origin station i (or the destination station j) is during the odd hours of travel that may not be convenient to passengers. Thus, travel in the Part-II service network is not considered friendly from a passenger point of view. For example, under Part-II, a nighttime or early morning hour travel (as odd travel times) might not be perceived safe by a passenger due to dark or a lack of first/last mile connectivity to stations.

Therefore, based on the above narrative, for a dummy variable, say $\delta = 1$ as in Part-I, it would indicate train k that serves the link connecting station i to j has an arrival or departure time at i (or at j) that is during the friendly travel periods. By default, $\delta = 0$ in Part-II service network.

From the network throughput perspective, entities managing the above-discussed networks aim to utilize as many links from the infrastructure network with as many trains as possible to create more extensive service networks. Policies that allow such efforts are generally encouraged, although operational budget constraints limit those efforts.

3.2.3. Improvement or Deterioration in Connectivity

A methodology has been developed for assessing the zonal improvement (or deterioration) in connectivity (through centrality and accessibility measures). The procedure involves focusing on connectivity changes for the stations of the zone by considering the significance of train arrival/departure at the station during the friendly (or odd) travel hours compared to the 24-hour travel time scenario.

Improvement or deterioration of the SE train stations in a zone can also facilitate understanding of how the zone ranks in passenger service with respect to other zones of the Indian Railways. For example, if most of the stations in a zone show improvement in connectivity for the friendly travel time periods (compared to the 24-hour period), then the zone will have an overall improvement in the connectivity as well. Same reasoning holds for the deterioration in connectivity of a zone. Therefore, we present next the formulation to determine improvement/deterioration in centrality and accessibility for SE train stations.

We define the improvement in centrality (I_{C_i}) of a station when $I_{C_i} = (L_{i,p_{C,24-hr}} - L_{i,p_{C,d}}) > 0$

where,

$L_{i,p_{C,24-hr}}$ = centrality of station i positioned at $p_{C,24-hr}$ in ranking obtained by sorting in a descending order the centrality values of all the SE train stations in that zone for the 24-hr travel time case

$L_{i,p_{C,d}}$ = centrality of station i positioned at $p_{C,d}$ in ranking obtained by sorting in a descending order the centrality values of all the SE train stations in that zone for the friendly travel time case.

Similarly, we have, for the accessibility (I_{A_i}) of a station when $I_{A_i} = (L_{i,p_{A,24-hr}} - L_{i,p_{A,d}}) > 0$ for the '24-hr' period and friendly travel time considerations,

$L_{i,p_{A,24-hr}}$ = accessibility of station i positioned at $p_{A,24-hr}$ in ranking obtained by sorting in a descending order the accessibility values of all the SE train stations in that zone for the 24-hr travel time case, and

$L_{i,p_{A,d}}$ = accessibility of station i positioned at $p_{A,d}$ in ranking obtained by sorting in a descending order the accessibility values of all the SE train stations in that zone for the daytime travel time case.

Similarly, the deterioration in centrality (D_{C_i}) and accessibility (D_{A_i}) of a station i occurs with the respective inequalities $D_{C_i} = (L_{i,p_{C,24-hr}} - L_{i,p_{C,d}}) < 0$ and $D_{A_i} = (L_{i,p_{A,24-hr}} - L_{i,p_{A,d}}) < 0$.

An example of the improvement and deterioration calculation for connectivity (centrality) with five hypothetical station examples has been shown in Table 1. In the example shown, it is observed that the stations A and

B exhibit improvement in ranking while stations C and D show deterioration in ranking when considering comparisons of the rankings for the Travel Period - I versus the Travel Period - II. Similar calculations can be done for station showing improvement or deterioration in accessibility rankings. The zonal improvement (or deterioration) in centrality or accessibility rankings from I to II is calculated by summing up the number of stations with improvements (or deterioration), multiplying it by hundred, and dividing the product by the number of stations in that zone. Note that the sum of the percentage improvement and deterioration in centrality for zone 1 do not add to 100% because station C did not experience any change in centrality ranking.

Table 1: Illustration of improvement and deterioration calculation with five-station examples.

Station	Zone Number	Centrality Values		Rankings*		Improvement/ Deterioration in Ranking (I to II)	Zonal Changes	
		Travel Period - I	Travel Period – II	Station Ranking Travel Period - I	Station Ranking Travel Period – II		Improvement %	Deterioration %
A	1	10	50	5	1	Improvement (from 5 to 1)	66.7%	0
B	1	20	40	4	2	Improvement (from 4 to 2)		
C	1	30	30	3	3	No change (from 3 to 3)		
D	2	40	20	2	4	Deterioration (from 2 to 4)	0	100%
E	2	50	10	1	5	Deterioration (from 1 to 5)		

* Note: The higher the centrality, the better is the ranking.

3.3. Spatial Disparity with Rail

The spatial disparity can reveal gaps in SHSE connectivity and to further justify and guide future zonal-level investment policies. Zones that have benefitted or not benefitted with SE train service during friendly (or odd) hours of travel periods are identified with the disparity analysis. This is particularly important for the Indian Railways which serves number of cities (through stations) that are economically diverse and attracting varying levels of zonal investments (IBEF, 2022). These cities also lie within their respective zones that are unevenly/unequally served by the SE trains of the Indian Railways.

Zonal disparity brought about by differences in services to stations of cities could impact the regional economic health of India. A disparity analysis could reflect any modifications, expansions, or adjustments needed in operating SE train, and to expand service connectivity among the cities within zones of the Indian Railways. The analysis could further entail identifying those cities with zones that can be made attractive with improvements in connectivity of SE trains rather than expansion in rail infrastructure.

Existing measures that can be used to evaluate spatial disparity include – dispersion indices, Lorenz curve indices, and entropy indices (Liu et al., 2020). The first two noted indices namely, the dispersion indices and Lorenz curve, cannot be decomposed to capture “between” zone and “among” zone disparities as required to understand the interactions of disparities at the zonal level. However, the Theil’s T index (an entropy index) is a decomposable measure of economic inequality and can be disintegrated into population groups, income sources or other dimensions (Cowell, 2006). In literature, the index has been used as a measure for ‘between-group’ and ‘within-group’ disparities (Liu et al., 2020), and hence, in the context of examining disparities between and among zones (as groups) of the Indian Railways in this paper, the use of this index fits the purpose.

With reference to the modified definitions of variables to the original Theil T index (Theil, 1967), the following is the formula for the index as defined below:

$$T = \frac{1}{n} \sum_{i=1}^n \frac{x_i}{\mu} \ln \left(\frac{x_i}{\mu} \right) \quad (3)$$

where,

n = number of stations from all the zones

x_i = measure of the economic potential of a station i

μ = mean economic potential measure for all n stations

The between-zone and the with-in zone inequalities are defined as $T_B = \sum_{j=1}^m S_j T_j$ and $T_W = \sum_{j=1}^m S_j \ln \left(\frac{\bar{x}_j}{\mu} \right)$

, respectively, where $S_j = \frac{n_j}{n} \frac{\bar{x}_j}{\mu}$, m being the number of zones, n_j is the number of stations in zone j , T_j is the

Theil's T index of zone j , and $\bar{x}_j$ is the average centrality measure of zone j .

3.4. Spillover Effects of Rail Investment

The benefits of investments in rail infrastructure of a specific zone are often not limited to within that zone itself. The benefits could spatially spread across the other neighboring zones due to connectivity resulting in a spatial autocorrelation among the zones for economic opportunities. A positive autocorrelation for similarities and negative autocorrelation for dissimilarities. The autocorrelation can be described through the proximity established using a weight matrix considering the contingency distance between two zones (Radzinski and Dziecielski, 2021). To identify a positive or negative spatial correlation, Moran's I could be used with a weight matrix as one of the inputs (Shi et al., 2020). The Moran's I varies between -1 (no autocorrelation) and 1 (autocorrelation). Once the spatial autocorrelation is established, an ordinary least squares regression (OLS) with Lagrangian Multiplier (LM) test can be used to determine the choice of a suitable spatial model relating the dependent and independent variables.

In spatial econometrics, the Spatial Durbin Model (SDM) has often been considered for empirical analysis to capture spatial spillovers of transportation investments and has the following formulation (LeSage and Pace, 2009; Yu et al., 2013):

$$y = \rho W y + X \beta + W X \theta + \alpha l_n + \varepsilon \quad (4)$$

where,

y is the output (e.g., job opportunities), ρ is the coefficient of spatial autocorrelation, along with α and β as parameters that need to be estimated, and ε is the error term. In this research, we have X as a vector of two independent variables – i) the investment in the local currency of the zone, and ii) the service quality of the rail transportation expressed as the zonal accessibility for the friendly and odd-hour travel times.

The SDM in Eq. 4 includes the spatially lagged y and lagged X as the independent variables to account for spillover effects. W is the spatially weighted binary contiguity matrix, which assumes that zones that share a common boundary can only impact each other's employment opportunities.

With basic transformations of terms in Eq. (4), the direct and indirect impacts can be derived by rearranging the spatial weight matrix as follows:

$$y = (I_n - \rho W)^{-1} (X \beta + W X \theta + \alpha l_n + \varepsilon) \quad (5)$$

$$\Rightarrow y = \sum_{r=1}^k S_r(W) x_r + V(W) l_n \alpha + V(W) \varepsilon$$

where,

$$S_r(W) = V(W) (I_n \beta_r + W \theta_r), V(W) = (I_n - \rho W)^{-1} \text{ and, therefore,}$$

$$\begin{bmatrix} y_1 \\ y_2 \\ \vdots \\ y_n \end{bmatrix} = \sum_{r=1}^k \begin{bmatrix} S_r(W)_{11} & S_r(W)_{12} & \vdots & S_r(W)_{1n} \\ S_r(W)_{21} & S_r(W)_{22} & \vdots & S_r(W)_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ S_r(W)_{n1} & \cdots & \cdots & S_r(W)_{nn} \end{bmatrix} \begin{bmatrix} x_{1r} \\ x_{2r} \\ \vdots \\ x_{nr} \end{bmatrix} + V(W)l_n\alpha + V(W)\varepsilon \quad (6)$$

The average direct impacts are equal to the average of the diagonal elements of the matrix $S_r(W)$, which means that for an observation i the average impact of the dependent variables to independent variables is the coefficient of the regression represented by the average $S_r(W)_{ii}$. The average total impact is equal to the average of sum of all the rows (or columns) elements of the matrix $S_r(W)$. The average total impacts from a given row in Eq. (6) is the average total change in the dependent variable for that row for equal amount change in the independent variables across all n observations. The average indirect impacts (which represent the spillover effects) are equal to the average total impact minus the average direct impact. The Maximum Likelihood (ML) procedures can be adopted to eliminate endogeneity in estimating the parameters of the spatial model in Eq. (6) (Yu et al., 2013). LeSage's code for econometric analysis in MATLAB can be deployed for this purpose (LeSage, 2021). Thus, parameter estimates can be obtained for all three effects – direct, indirect, and total – to appropriately understand the impacts of the independent variables. A special case of SDM results when $\theta = 0$ and is called the spatial autoregression (SAR) model, while a spatially lagged X (SLX) model is obtained when $\rho = 0$ in the SDM.

4. Analysis

An analysis has been carried out to understand the inequality in connectivity and impacts of investments in the seventeen zones of the Indian Railways.

Data needed for assessing centrality and accessibility measures for connectivity were the spatial data of the SE train stations in India and the rail network of the Indian Railways. These were obtained from various online sources (including Mapcrizin.com, 2021; Indian Railways, 2021). The 17 zones of the Indian Railways were constructed on a geographical information system (GIS) software with the railway zonal map from the MapsofIndia (MapsofIndia, 2021) and corroborated with official webpage of various zones from the Indian Railways (Indian Railways, 2021d).

For the year 2020, there were 1620 railway stations identified in India that served as a stop to at least one SE train operated by the Indian Railways. The city population data of the year 2011 obtained from the Census Bureau of India were used to denote population that was assumed to have access to station (2011 Census Data, 2020). The chart in Fig. 2 shows the variation of a zone's population to the number of SE train stations. The trendline indicates somewhat correlated data indicating that each railway zone has an adequate number of stations for the population that it serves with the semi-high speed and express trains. However, how well a zone with stations is relatively situated to other zones in providing equal opportunities to the passengers for semi-high speed and express train travel can only be understood with a disparity analysis – as discussed later in this paper.

The map in Fig. 3 shows the percentage of stops of SE trains across various zones of the Indian railway network for the unfriendly travel periods of 9 pm – 7 am: during these hours travel by rail becomes unfriendly especially due to lack of an adequate first/last mile transport connectivity and safety concerns of rail passengers at dark, especially in developing nations like India (Chandrasekhar, 2020). It is assumed that higher the percentage for a zone, the more unfriendly that zone is for 9 pm – 7 am travel period.

It is observed that the three zones of Eastern Railway, South Eastern Railway, and South East Central Railway have the largest percentage of about 64-70% of SE train stops during the 9 pm – 7 am travel periods. The Western Railway, North Western Railway, and the South Central Railway have the lowest percentage of about 52-54% SE train stops during the same travel periods. The map in Fig. 3 also shows the spatial location of the top-three stations (cities) with the highest frequency of stops during the unfriendly travel period. Some of the major stations are Kanpur Central, Vijayawada Junction, Jhansi, etc., as shown in Fig. 3.

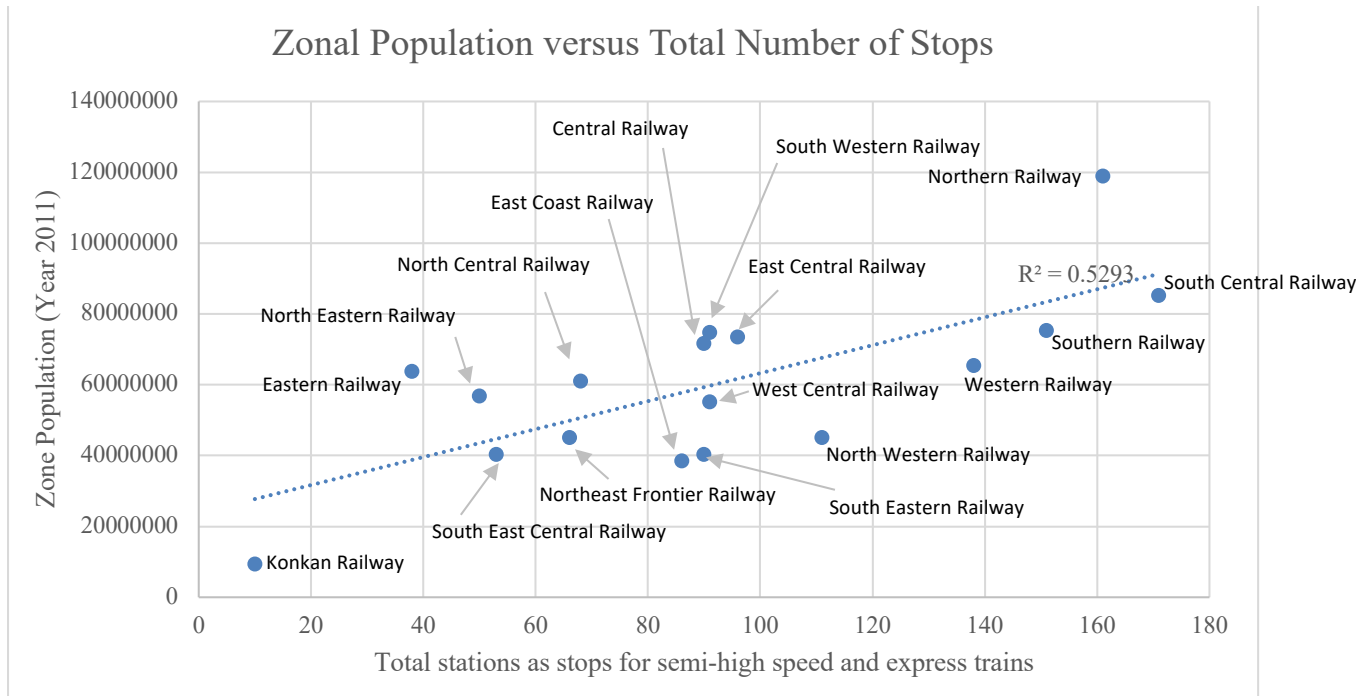


Figure 2: Variation in total number of stations versus population in the 17 zones.

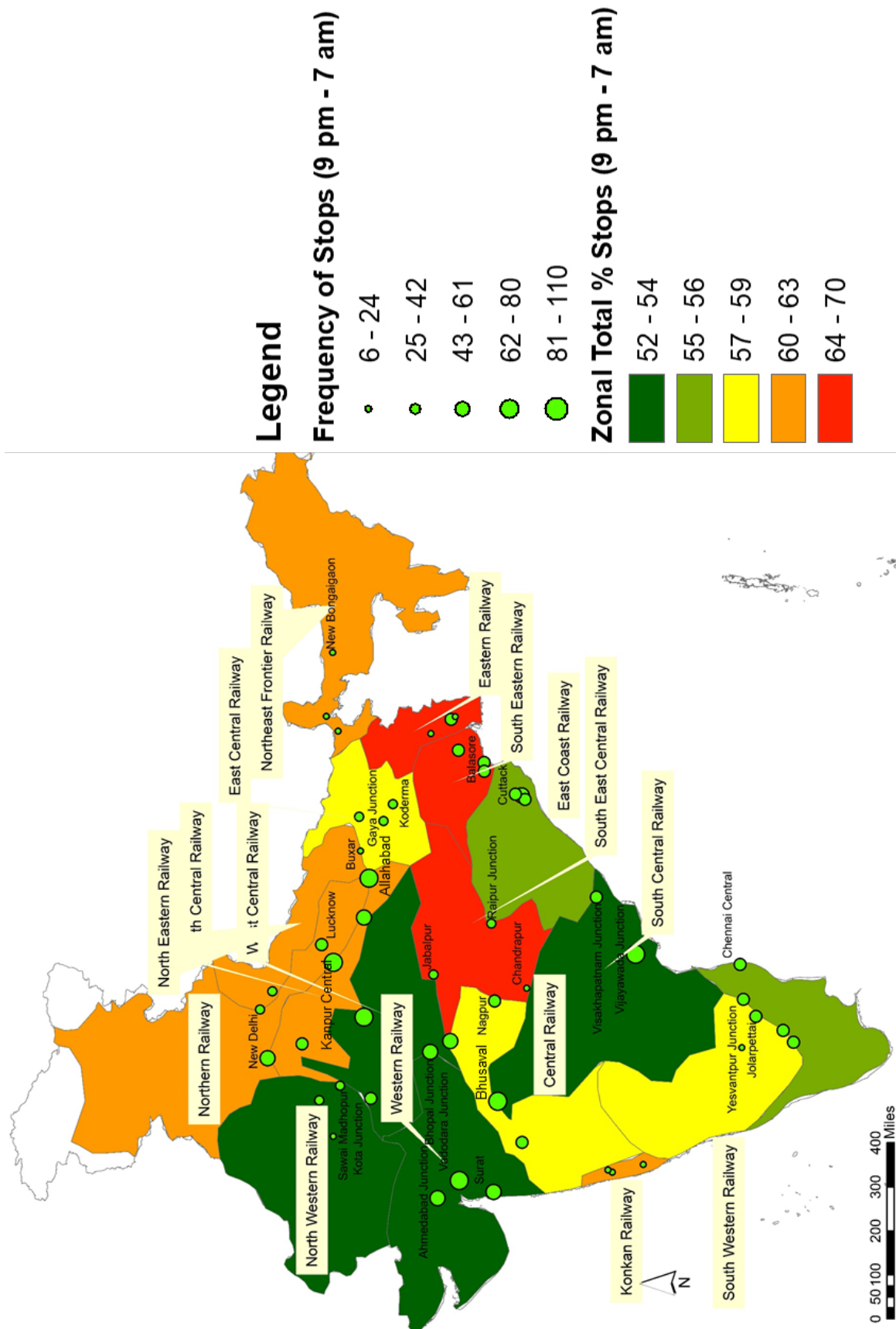


Figure 3: Frequency of train stops between 9 pm – 7 am (nighttime) across stations and zones

4.1. Centrality and Accessibility Outcomes

The percentage of stations with improvement (and deterioration) in centralities (and accessibilities) were evaluated for the seventeen zones. The evaluations were for the two connectivity measures for the 7 am – 9 pm period relative to the connectivity in the 24-hour period with the 9 pm – 7 am travel period considered to be unfriendly for passengers. Therefore, the purpose of comparing the 24-hour period with the 7 am – 9 pm period is to get a comparison between the overall connectivity for the 24-hour period and the actual connectivity that is offered to the rail passengers during the time period 7 am – 9 pm considered as friendly. This would highlight zones attracting passengers during the friendly 7 am – 9 pm travel periods. Further, the significance of identifying these zones is that the policy decisions geared to increase zone connectivity by introducing a new SE train service could consider any centrality or accessibility impacts during friendly 7 am – 9 pm travel periods of passengers.

The population data required for the calculation of accessibility of stations (and zones) for the five-year period 2016 to 2020 were obtained by assuming an average population growth of 1% per year of India from the World Bank and using the 2011 Census year population data as the base year (The World Bank, 2022).

The degree centrality and accessibility computed for each station were spatially mapped to the station's zone. From the maps in Figs. 4 and 5, it is observed that in the south, the zones of the South Central Railway, South Western Railway, and Southern Railway and, in the north, the zones of the Northern Railway and the North Central Railway - would have the highest percentage of stations with improvement in centrality for the 7 am – 9 pm travel period relative to 24-hour period centrality. This percentage improvement would be in the range of 44 to 50%. The lowest percentage of stations with improvement in centrality would be in the Eastern Railway and the Konkan Railway zones of around 20-24%.

The Konkan Railway zone stations of about 20-27% will be the lowest centrality deterioration for the 7 am – 9 pm period compared to the 24-hour period. On the other hand, several zones with a high percentage of stations would deteriorate in centrality for the 7 am – 9 pm period compared to the 24-hour period. These zones include the Western Railway and North Eastern Railway, and the others, with about 49-55% of stations - as shown in the map of Fig. 4 with the red color shade. This means that connectivity seems to reduce in these zones, with the high percentage of stations noticing a drop in centrality during the friendly 7 am – 9 pm travel periods.

Comparing the two maps in Fig. 4, it is observed that the two zones of the Northern Railway and the Southern Railway have the most significant percentage of stations that would experience changes (improvement and deterioration simultaneously) in centrality for the friendly 7 am – 9 pm travel periods compared to the 24-hour period. This observation could be due to a significantly high number of SE trains that terminate or originate at the New Delhi railway station in the Northern Railway zone and the Chennai Central railway station in the Southern Railway zone.

For the accessibility case, the North Western Railway, Western Railway, and South East Central Railway zones would have the highest percentage of stations experiencing improvement for the friendly 7 am – 9 pm period (see Fig. 5). These percentages would be in the range of 50-52%. The zones of the East Central Railway, Eastern Railway, Konkan Railway, and Southern Railway would have the lowest percentage of stations with improvement in accessibility for the same travel periods. The largest percentage of stations experiencing deterioration in accessibility for this travel period would be for the South Western Railway, Southern Railway, East Coast Railway, South Eastern Railway, and Northern Railway. The percentage would be around 53-55%.

The Eastern Railway and the Konkan Railway would have the lowest percentage of stations (around 40-43%) that would experience deterioration in accessibility. The Southern Railway zone, among all the other zones, would have the lowest percentage of stations with accessibilities that would deteriorate and improve in the 7 am – 9 pm period compared to the entire 24-hour period. The total percentage of stations that would exhibit changes (improvement and deterioration) in connectivity for a zone does not add to 100%. Some of the stations in any zone would not change in centrality or accessibility for the 7 am – 9 pm travel periods.

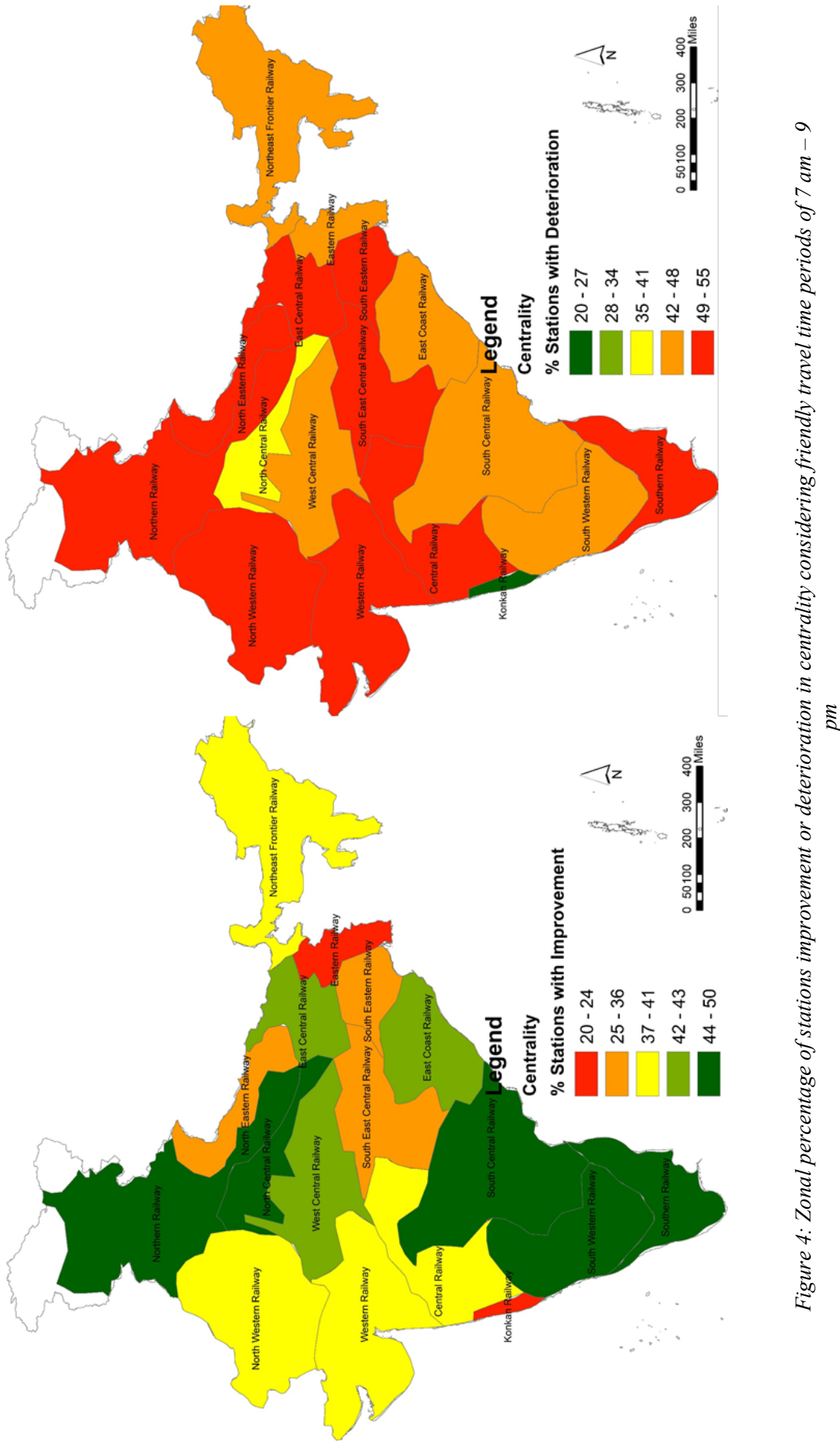


Figure 4: Zonal percentage of stations improvement or deterioration in centrality considering friendly travel time periods of 7 am – 9 pm

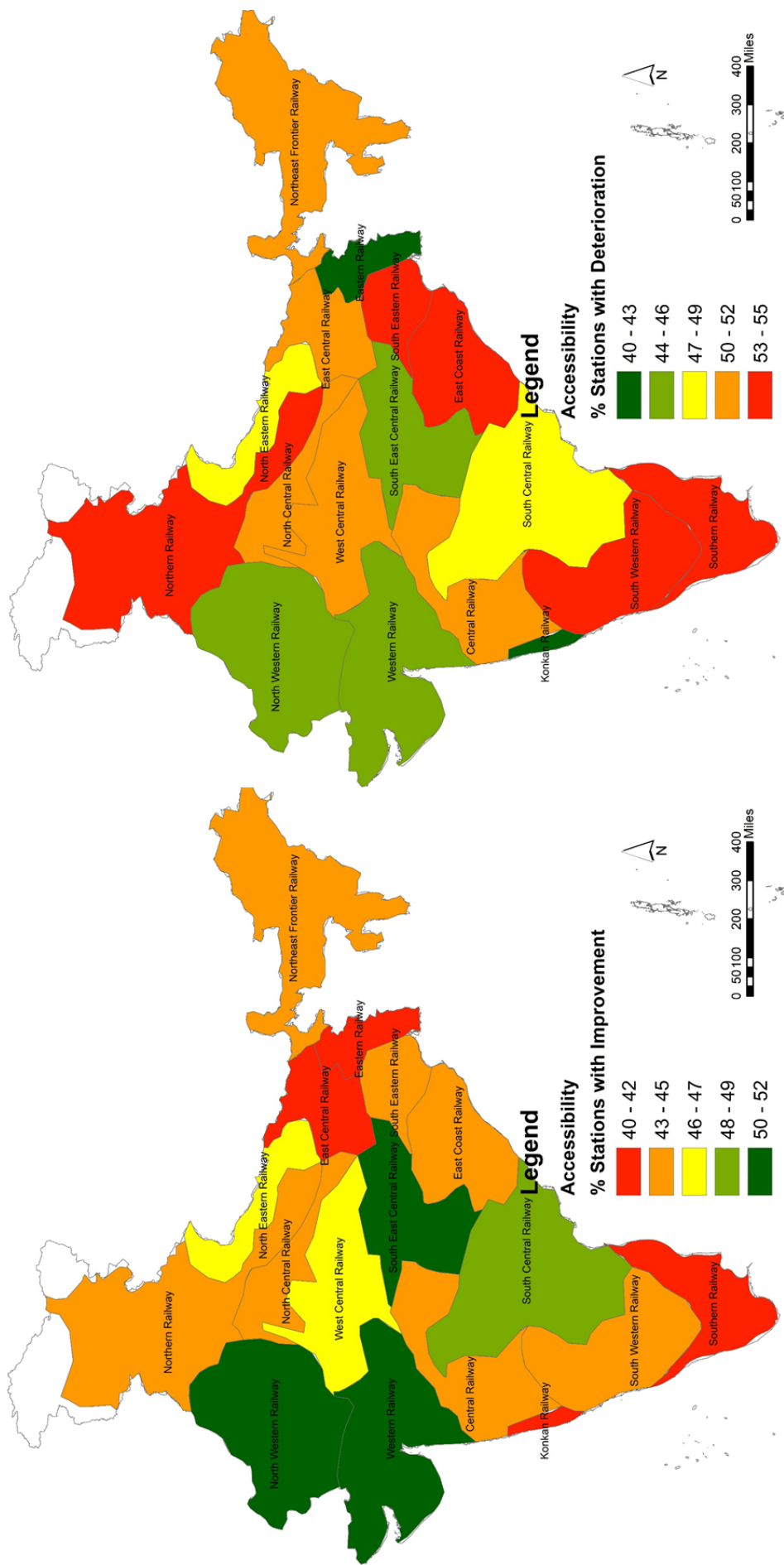


Figure 5: Zonal percentage of stations improvement or deterioration in accessibility considering friendly travel time periods of 7 am – 9 pm

The mean centrality and accessibility of stations in a zone were normalized as percentage with respect to the highest zonal centrality and accessibility and presented in the maps of Figs. 6 and 7, respectively.

It was observed that the Central Railway exhibits the largest normalized percentage centrality compared to the other sixteen zones for both the 24-hour period and 7 am – 9 pm period (see map in Fig. 6) - which could be because several trains arrive or depart at the stations of this zone than at the stations in the other zones and as corroborated with the Indian Railways timetable for the SE trains and an independent website on running status of trains (Running Status, 2021; Indian Railways, 2021^e). Thus, on average, a station in the Central Railway zone is more connected than the stations in other zones. The Northeast Frontier Railway and the North Western Railway zones have the lowest normalized centrality for the 24-hour period and the 7 am – 9 pm period, respectively. Thus, it appears that the stations in these two zones have the lowest SE train connectivity compared to the other zones in those two respective time periods. It is also noted that there was an almost equal distribution of normalized centralities among majority of the zones for the two time periods. This could indicate that stations in most zones would do not experience changes in centrality (connectivity) for the two time periods.

Based on the findings as illustrated using the map in Fig. 7, the zone of North Central Railway has the most significant normalized accessibility for both the 24 hour period and the 7 am – 9 pm period. The high accessibility value for North Central Railway zone is attributed to an increased number of SE trains for the two analyzed time periods. This is also supplemented by the proximity of the stations in these two zones to cities in which they lie with a high population, thus eventually contributing to large values of average accessibility.

Therefore, stations in the North Central Railway zone have an increased number of SE train stops than the other stations in other zones both during the 24-hour period and the 7 am - 9 pm period.

It is observed that several zones including the South Western Railway, East Coast Railway, Western Railway, and the North Frontier Railway have decreased in SE train accessibility from the 24- hour period to the 7 am – 9 pm period. This could be expected because the number of train stops are reduced during the 7 am – 9 pm period compared to the total 24-hour period. But this also indicates that a significant number of SE train stops are during the unfriendly 9 pm – 7 am period.

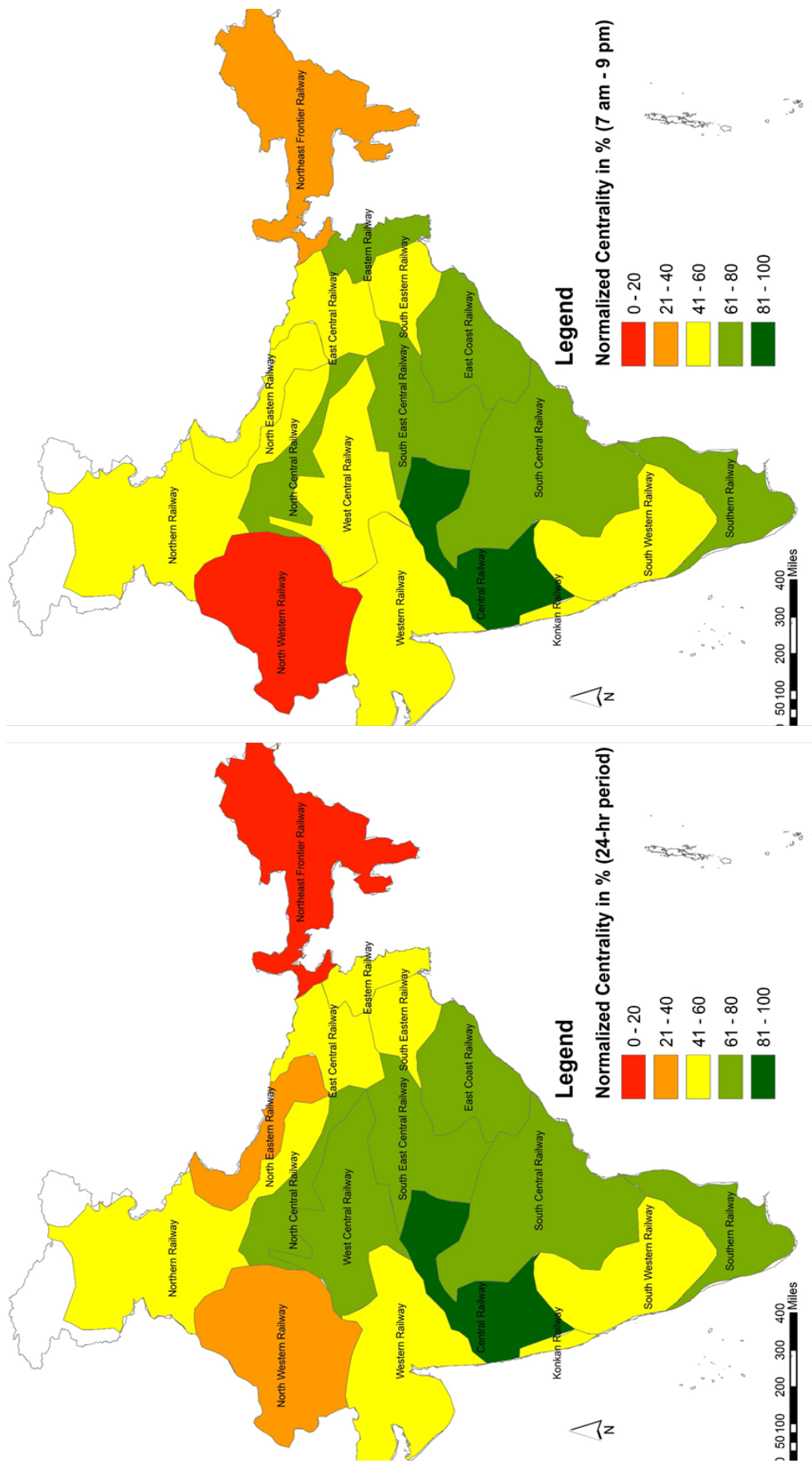


Figure 6: Normalized centrality in percentage for zones for the 24-hr and the 7 am – 9 pm periods

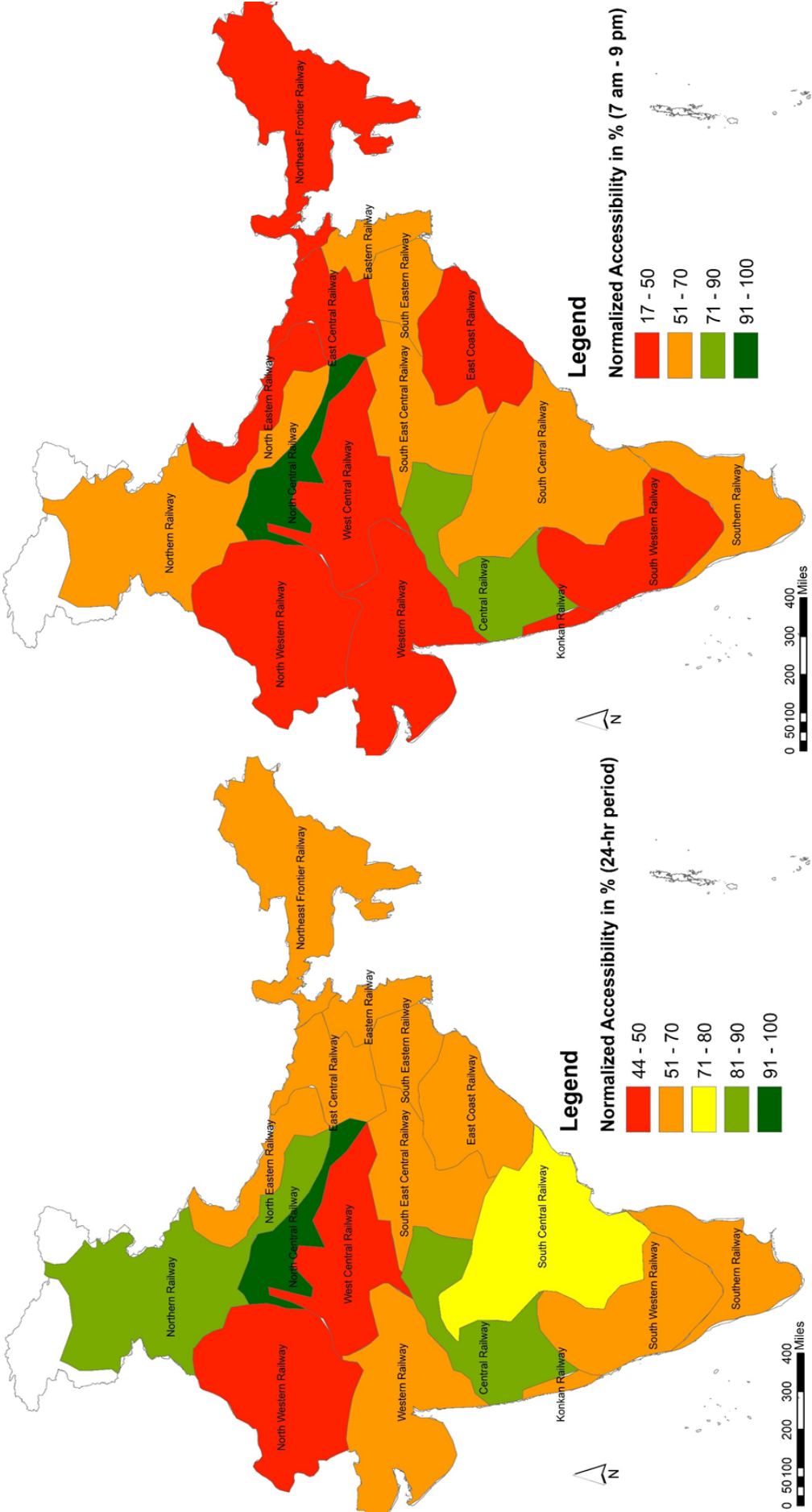


Figure 7: Normalized accessibility in percentage for the 24-hr and the 7 am – 9 pm periods

4.2. Disparity Analysis

The disparity analysis showed that if only the degree centrality is considered in evaluating inequality in connectivity among zones, the South Eastern Railway and the Eastern Railway have the largest Theil's T index (equal to 1 each) for the 24-hour period (see Fig. 8). Thus, these two zones are the least connected with SE trains for the 24-hour period considered. This also suggests that if the 24-hour period of SE trains is considered, the stations in these two zones could offer a reduced economic opportunity to the population they serve, compared to those in other zones.

For the friendly 7 am – 9 pm travel periods, the South Central Railway zone has the largest inequality in centrality (Theil's T index equal to 0.9) as compared to other zones of the Indian Railways. Thus, indicating that the scheduled stops of SE trains between 7 am to 9 pm at the stations of this zone is not much of a utility with reduced connectivity and minimal opportunities for economic growth in this time period. In comparison to the other zones, the Konkan Railway with the lowest indices for centrality for the 24-hour period (equal to 0.41) and the 7 am – 9 pm travel period (equal to 0.4) has the highest potential for economic growth with SE train service that its stations receive.

Based on a similar analysis with accessibility, the zones of the South East Central Railway and the Southern Railway exhibit the largest inequality for the 24-hour travel period with T index equal to 0.62 and 0.59, respectively. This shows that the population in these two zones has an overall reduced opportunity for economic growth than other zones for the 24-hour period with the SE train connectivity. For the 7 am – 9 pm travel period, with Theil's T index of 1.75, the South East Central Railway zone (compared to the other zones of the Indian Railways) also has the least opportunity for economic growth with the SE train connectivity.

The zone of Konkan Railway, compared to other zones, has the smallest Theil's T index equal to 0.09 and 0.58 for the 24-hour and the 7 am – 9 pm travel periods, respectively, and thus, has the largest potential for economic growth with SE train connectivity.

With the decomposition analysis for the 'between-zone' inequality and 'within' zone inequality for the two periods presented in the chart of Fig. 9, it is evident that there is a large disparity in the centrality of stations between different zones (with Theil's T index equal to 0.82 for the 24-hour period and 0.71 for the 7 am – 9 pm travel period) than within the zones (with Theil's T index equal to 0.01 for both the periods). Therefore, for the two time periods the connectivity provided by SE trains is highly unequal between zones. Therefore, while this poses an economic advantage for some zones, other zones do not have the same advantage because of unequal opportunity.

When analyzed for disparity using accessibility, the between-zone inequality is higher for the service times 7 am – 9 pm travel period (with Theil's T index equal to 1.03) as compared to the 24-hour travel period (with Theil's T index equal to 0.36). However, the index for these periods is higher than the Theil's T index of 0.03 for the 24-hour period, and the index of 0.09 for the 7 am – 9 pm travel period for the 'within' zone inequality of accessibility. This indicates that the economic opportunities from travel with SE trains considering accessibility (as a surrogate of connectivity) are unequal for the population with-in the zones compared to between zones for the two time periods.

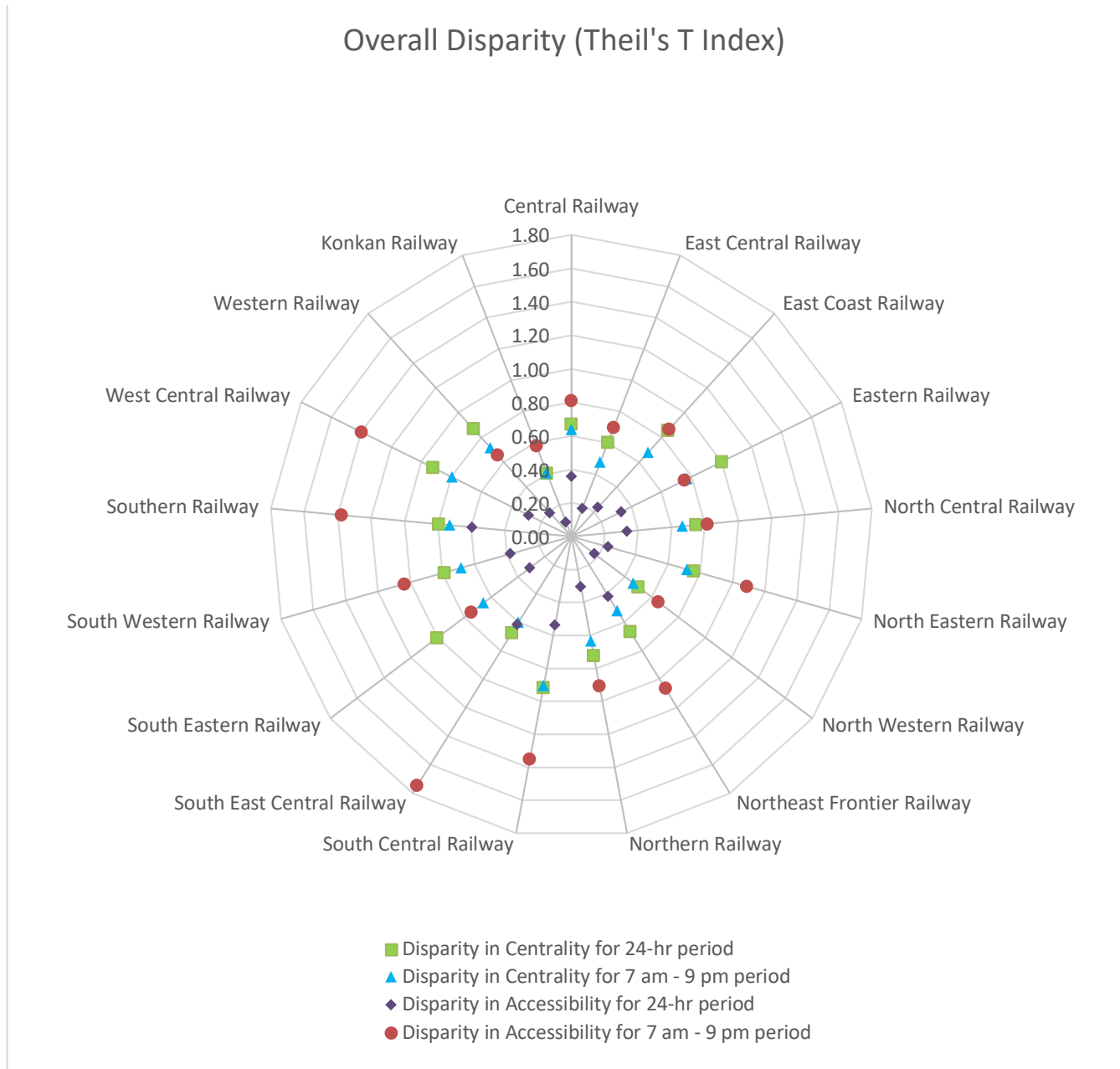


Figure 8: Overall disparity outcomes for with centrality and accessibility measures

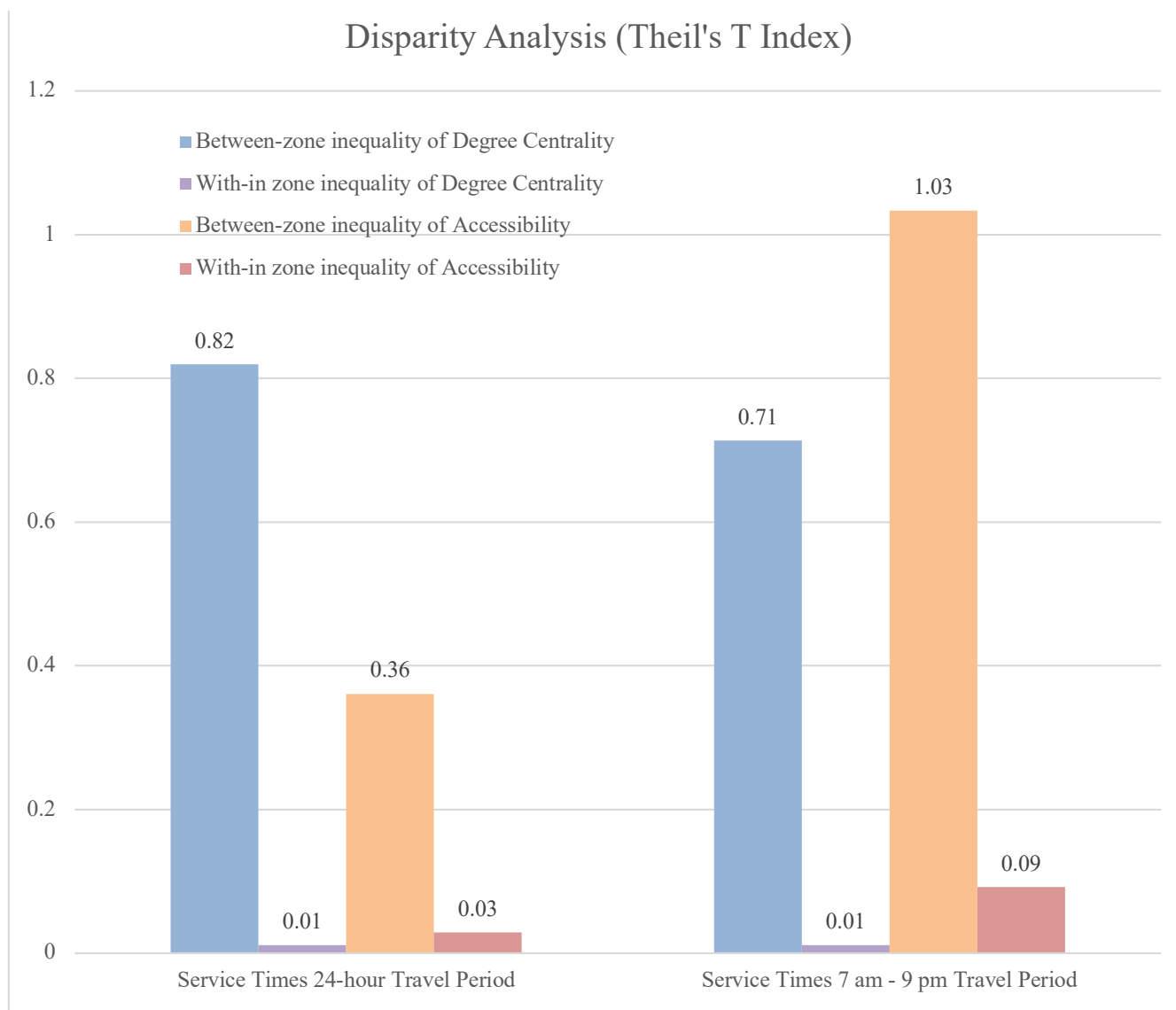


Figure 9: Between-zone and with-in zone disparity analysis

5. Investment Policy and Other Connectivity Implications

The Indian Railways provide direct funding for the operation and maintenance of rail infrastructure and services. One of the outcomes that need investigation relates to how these investments have been effective in the zones' economic growth, which would alleviate disparities that currently exist between them. Therefore, we discuss the impact of investment by the Indian Railways on employment in zones in the past five years. This discussion is necessary to see if the efforts in investment by the Indian Railways have facilitated increased employment in zones. Further, we also discuss how station accessibility with SE train arrival/departure delays in a zone during friendly and unfriendly travel periods are also causal to employment impacts in the past five years.

5.1. Investment in Indian Railways

The data on investment in passenger rail by the Indian Railways were gathered from the Indian Railways webpage (Indian Railways, 2021c). The chart in Fig. 10 shows that the investment has been varying over the last five years

(from 2016 to 2020) across the zones but consistently increasing for a given zone. Data from Indian Railway budget show that the level of investment in each zone has been proportional to passenger rail revenue generated in a zone over the same period - although a major percentage of the revenue generated for Indian Railways is from goods transportation. These investments are primarily accounted for as total working expenses and include repairs and maintenance of permanent way and works, carriages and wagons, operating expenses (rolling stock and equipment, traffic, and fuel), staff welfare and amenities, etc. (Indian Railways, 2021a). While some of the zones in recent years have a higher than 100% operating ratio, the ratio is lower than 100% for zones such as East Coast Railway and North Central Railway – which indicates that not all zones are profitable for the Indian Railways. However, very little is known if investment in rail changes economic growth or impacts job opportunities in any way within the zones.



Figure 10: Annual zonal investments by the Indian Railways (Source: Chandra and Narayanaswami, 2024)

5.2. Train delays

Accessibility (compared to centrality) can be considered a suitable indicator of SE train connectivity for the Indian Railway zones. This is because accessibility is sensitive to the city population served by the SE train stations and impedance (travel time). Therefore, further insights were drawn on the impact of zonal accessibility (through SE trains) on the zones' economic growth and the investment undertaken by the Indian Railways.

We specifically focus on accessibility impacts for the two time periods of analysis: 7 am – 9 pm, and 9 pm – 7 am. While the 7 am – 9 pm travel period is considered friendly to passengers, 9 pm – 7 am is deemed unfriendly due to a lack of first/last mile connectivity or safety reasons.

For these two analysis periods, yearly SE trains running status gathered from Cleartrip.com from 2016 to 2020 showed the trains' impunctuality (delays) at various stations of the zones. The variation in the annual delays of these trains at the zonal level is shown in Figs. 11 and 12.

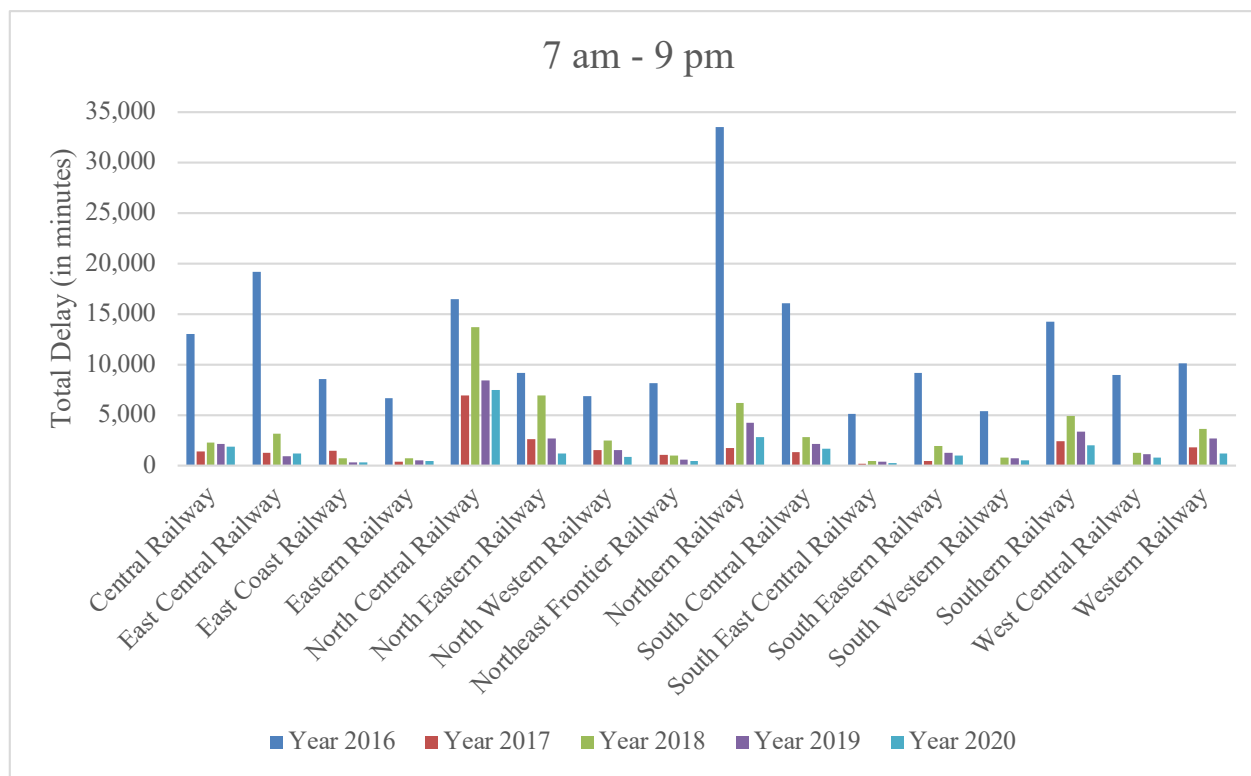


Figure 11: Annual delay variation for SE trains across zones for train stops for the 7 am – 9 pm period

The total annual SE train delay for 2016 was the highest for the Northern Railway zone for the 7 am – 9 pm period, exceeding 30,000 minutes. Other zones in that year had a higher delay than in the other years. The zones of the East Central Railway, East Coast Railway, North Eastern Railway, North Western Railway, Southern Railway, and Western Railway had the lowest delays in 2020. The rest of the zones experienced the lowest delays in 2017 for the 7 am – 9 pm period.

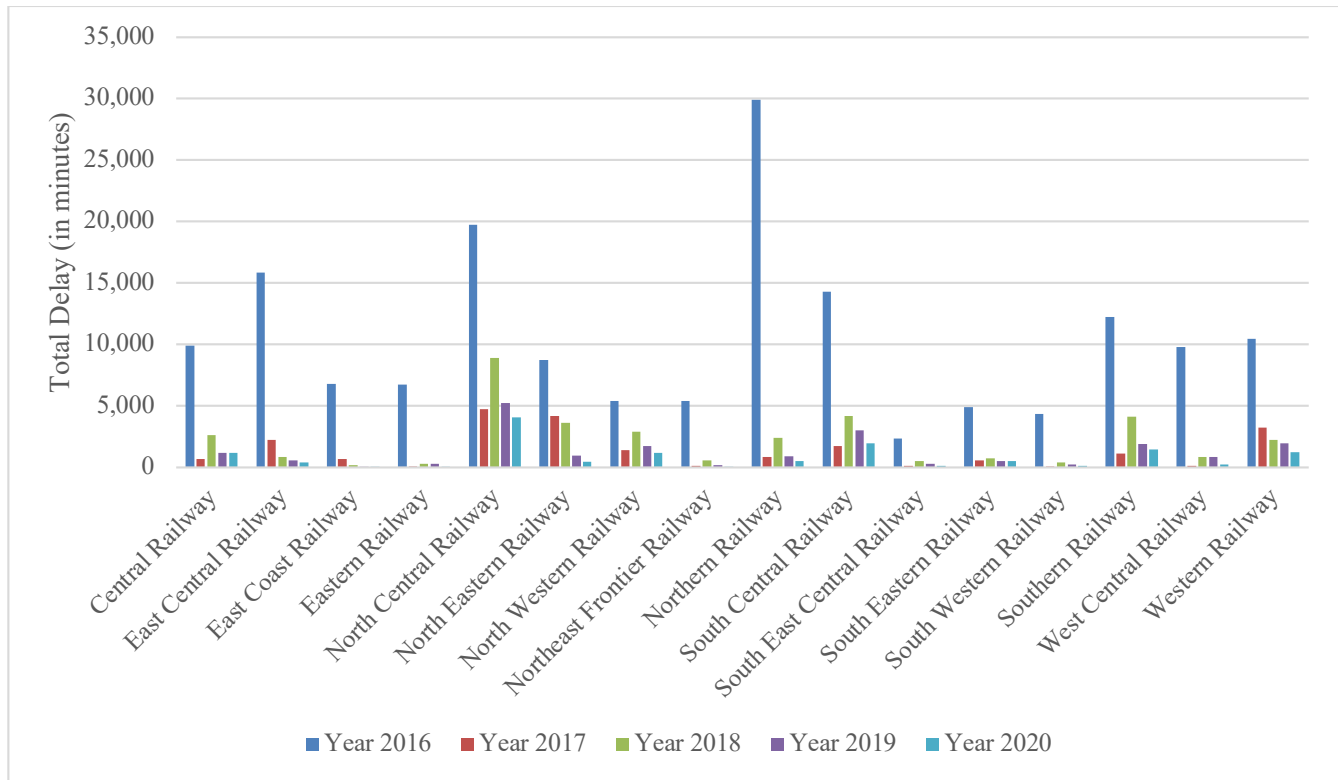


Figure 12: Annual delay variation for SE trains across zones for train stops from 9 pm – 7 am

For the 9 pm – 7 am period, the total annual delays were the highest in 2016 for all the zones, with the Northern Railway zone experiencing the largest delay (about 30,000 minutes) compared to the other zones. The zones with the lowest delays in 2020 were the East Central Railway, East Coast Railway, North Eastern Railway, North Western Railway, South Eastern Railway, South Western Railway, and Western Railway. The rest of the zones had the least delays in 2017 from 9 pm – 7 am.

Accessibility for each station was calculated by considering the impedance as the sum of the actual origin-destination travel time of SE trains and the same origin-destination pair delay. Thus, variations in the zonal accessibilities using stations' city population were calculated for 2016-2020. The population for each of the years from 2016 to 2020 was calculated using the base census year population of 2011 and extrapolating with a population growth rate of 1% obtained from the World Bank (as noted in the previous section). A zone's accessibility equal to the mean station accessibility in a zone was calculated for each of the five years for 2016-2020 for the two periods: 7 am – 9 pm, and 9 pm – 7 am. Thus, 80 data points for the five years (2016-2020) were obtained on accessibility for each period for the sixteen zones (excluding the zone of Konkan Railway, which did not have information on investment by the Indian Railways).

5.3. Spillover effects

Spillover effects of investment and SE train service connectivity on employment across zones were analyzed. Data on employment (job opportunities created) from 2016 to 2020 for zones were used as the dependent variable, while the investment and the zonal mean accessibility were used as the independent variables. Each Indian Railways zone is made up of several districts, that have information on total workers (indicative of the total employment) as gathered from the Office of the Registrar General & Census Commissioner, India for the year 2011 (Census – 2011, 2021) and these were aggregated to yield employment data. Further extrapolation was carried out using employment growth rate from 2012-2019 (Trading Economics, 2021) and the five-year employment data from 2016 to 2020 were prepared.

The distances among centroid of zones varied significantly. Our focus was on investigating spatial impacts on job opportunities among zones that were immediate neighbors to each other. We chose to consider a queen-type

measure of accounting two zones as neighbors if there was any common share of their boundary required to calculate spatial autocorrelation. GeoDA, a popular open-source spatial analysis software, was used to calculate the Moran's I and conduct the LM tests (GeoDA, 2021). In our study, the value of Moran's I was positive and significant at 1% for the employment distribution (as job opportunities) of zones. LM test output indicated that a spatial lag model could be used for further analysis for the job opportunities (used as the dependent variable) and the zonal investment and accessibility with odd hours of travel used as the two independent variables. The LM test suggested the suitability of a spatial autoregressive (SAR) model for our case with total zone employment (y) as the dependent variable.

Findings from analysis show that the impact of investment on job opportunities in zones are significant at 1% level with estimates of 43.206. Limiting the findings just to the coefficients estimated for investments by the Indian Railways does not solely reflect their marginal effects on job opportunities; therefore, assessing its average direct and indirect effects become necessary. The estimations are based on the concept of spillover effects introduced in the earlier section.

The investment had positive direct and indirect effects. The estimates are found to be 54.392 (at 1% significance level) and 115.318 (at 5% significance level) for the direct and indirect effects (spillover effects), respectively. The total effect was 169.711 (at 5% significance level). The above findings are summarized in Table 2. Thus, the policy implications of the findings are that the Indian Railways could continue to invest in sustaining job opportunities in the 17 zones.

The mean accessibility of zones during the 7 am – 9 pm period positively impacted the job opportunities with weight equal to 0.020, the direct impact weight being 0.025, the indirect impact weight being 0.054 – all at 10% significance level. However, the weight of the total impact of mean accessibility on job opportunities was not significant. Therefore, based on the analysis, there is some evidence that a zone's job opportunities by SE train connectivity during the friendly 7 am – 9 pm travel periods are impacted by the neighboring zones' accessibilities with the direct and indirect effects.

The mean accessibility of zones during the 9 pm – 7 am period is significant at a 5% significance level with a (negative) weight of -0.033. This shows that with the increase in the 9 pm – 7 am accessibility, there have been a reduction in the zonal employment. The direct impact weight of the accessibility in this period being a -0.043 at 5% significance level indicates that this accessibility has contributed to a decrease in job opportunities in a zone. Further, the indirect effects (spillover effects) of 9 pm – 7 am accessibility also lower job opportunities in neighboring zones. The indirect impact is -0.091 at 5% significance level, and the total impact weight is -0.133 at 10% significance level. These findings indicate that the accessibility during this period have been detrimental to the job opportunities in the zones. Thus, policies that would increase the 9 pm – 7 am accessibility provided by the Indian Railways could increase job opportunities in the zones.

The spatial autocorrelation coefficient (ρ) is found to be 0.723 and spatial dependence for job opportunities is observed at 1% significance level. This indicates that a zone is affected by job opportunities in neighboring zones.

Table 2: Estimation results of SAR for the Indian Railway zones

Variable	Weight	Direct effect	Indirect effect	Total effect
Investment	43.206*** (11.762)	54.392*** (8.448)	115.318** (2.289)	169.711*** (3.043)
Mean Accessibility (7 am – 9 pm)	0.020* (1.357)	0.025* (1.344)	0.054* (1.099)	0.079 (1.195)
Mean Accessibility (9 pm – 7 am)	-0.033** (-2.310)	-0.043** (-2.261)	-0.091** (-1.573)	-0.133* (-1.806)
ρ	0.723*** (9.449)			
R^2	0.9911			

Note: The numbers in parenthesis denote t-statistics

* Significance level 10%

** Significance level 5%

*** Significance level 1%

Recent thrust of the Indian Railways has been on soliciting private participation in rail connectivity and capacity augmentation projects (Indian Railways, 2021b). The underlying policy goals of these efforts include supplementing government investment in rail projects by private capital flows and matching the transport capacity needs with the expected GDP growth of the nation. Indian Railways anticipates maximizing land use and spurring economic activities around stations in future with assistance from private entities under a public-private partnership (PPP) model (BusinessLine²⁰²¹). However, complete privatization of the Indian Railways has been forewarned and call for a check on productive efficiency, particularly ensuring an equity-allocation (Kuriakose and Gupta, 2021).

PPPs in projects of the Indian Railways have been evolving over last several years in India, including operating partnerships with state governments and some PPPs in the past have been discontinued as well (Raghuram and Gangwar, 2010). All these efforts indicate Indian Railways is keeping a foreword approach to revitalizing poorly performing elements of the passenger rail operations in India.

To enhance connectivity across the zones, the Indian Railways had introduced over 800 new passenger trains in the last five years, out of which 67 new express trains were rolled into service in 2014-2015 (Economic Times, 2022a). Currently, there are two fully operational semi-high speed Vande Bharat trains that terminate and originate at New Delhi railway station, which is in the national capital of India and lies in the most populated zone of the Northern Railway (IRCTC, 2022; 2011 Census Data, 2020). As per the Union Budget, there are also plans to introduce 400 new Vande Bharat semi high-speed trains between other major stations/cities of India in the next three years (Economic Times, 2022b).

6. Concluding Remarks

This research investigated the impact of SE train connectivity on seventeen zones of the Indian Railways. Degree centrality and accessibility were used as the two measures of connectivity. Further, evaluations were carried out to assess the inequality among zones and understand how the past five years of accessibility variations and investment policy of the Indian Railways have played a critical role in economic growth (in terms of job opportunities) of the zones.

In this paper, the analyses focused on two time periods: a 24-hour period and the 7 am – 9 pm period. The comparative analysis between the two periods highlighted the SE train connectivity during the 7 am – 9 pm period relative to the daily connectivity of the 24-hour period. The 9 pm – 7 am travel period is considered unfriendly for rail passenger travel in India due to the lack of adequate first/last mile connectivity and safety concerns in accessing rail stations at night or early wee hours of travel. Therefore, the effective connectivity from the SE trains would be during the 7 am – 9 pm travel period.

An analysis carried out with the early year 2020 SE train connectivity of the Indian Railways showed a more substantial disparity in their connectivity among its seventeen zones than within a zone for the two periods. The Theil's T index for the between-zone disparity was much higher than the within-zone disparity for both centrality and accessibility measures – indicating that the inequalities in connectivity exist between zones more than between the stations (as cities) in a zone.

The South Central Railway zone has the largest inequality in centrality as compared to other zones of the Indian Railways for the 7 am – 9 pm period, and the two zones of the South East Central Railway and the Southern Railway exhibit the same for the 24-hour travel period. The Konkan Railway ranks low in disparity for both 7 am – 9 pm period and the 24-hour period. For accessibility, the South East Central Railway zone exhibits the largest inequality for both travel periods. For both periods, it was found that the Central Railway has the largest mean centrality, and the North Central Railway zone has the largest mean accessibility of all the stations in the respective zones.

Positive spillover effects of SE train accessibility on job opportunities in zones are observed during the 7 am – 9 pm period, which suggests that policies that increase the zonal connectivity through new SE trains could be promoted by the Indian Railways. This would mean that a zone lagging behind other zones in economic growth could have an increased SE train connectivity – particularly to those with high employment. At the same time, although SE train accessibility could increase during the 9 pm – 7 am period, due to a negative spillover effect, SE train connectivity may not add to the increase in employment at the zonal level. In this research, it was also observed the positive impact of Indian Railways investment in various zones and the subsequent increase in employment. This indicates that for

sustained economic growth in zones, the Indian Railways should continue to invest in the rail network infrastructure (including tracks, rolling stocks, etc.).

In summary, two important suggestions are made based on the research findings on SE train connectivity - first, the Indian Railways should focus on identifying zones for investments that have the most considerable disparity in centrality and accessibility as these two are indicators of the economic potential of a zone, and second, to enhance the service quality offered by the SE trains, their halts at critical stations for passenger boardings/alighting during nighttime and early morning hours should be facilitated with increased first/last mile connectivity and enhanced safety through streetlights leading to the stations. In the absence of these steps, the economically prosperous zones will continue to advance in employment growth, resulting in increased disparity with marginalized/less strategically served zones of the Indian Railways.

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